MONTHLY NOTICES

OF THE

ROYAL ASTRONOMICAL SOCIETY.

VOL. LXXXIII. FEBRUARY 9, 1923. No. 4

ANNUAL GENERAL MEETING.

Professor EDDINGTON, F.R.S., PRESIDENT, in the Chair.

The Minutes of the preceding Annual General Meeting were read and confirmed.

The Report of the Auditors of the Treasurer's accounts for the year 1922 was read, and is given on p. 224.

The Annual Report of the Council was partly read.

The Address was delivered by the President, after which the Gold Medal was handed to Mr. Post Wheeler, Counsellor to the American Embassy, for transmission to Professor A. A. Michelson, to whom the Medal had been awarded for his application of the Interferometer to astronomical measurements.

The President having appointed the Scrutineers, the Society proceeded to the ballot for Officers and Council for the ensuing year. The names of those elected are given on p. 319.

The thanks of the Meeting were given to the retiring Members of Council, and also to the Auditors of the Treasurer's accounts and to the Scrutineers of the ballot.

Dinsmore Alter, Professor of Physics and Astronomy, University of Kansas, Lawrence, Kansas, U.S.A.;

A. Frederick Collins, 6 and 7 Tavistock Place, London W.C. 1;

Howard H. Cox, 38 Braemar Avenue, Wood Green, London, N. 22;

Philip Dilk Gilbert, B.A., Weymouth College, Weymouth, and Stratton, Swindon, Wilts;

George Lawrence Harrison, 88 Boulevarde, Hull;

L. C. Martin, D.Sc., Lecturer on Optics at the Imperial College of Science and Technology, South Kensington, London, S.W. 7;

A. M. Mosharrafa, B.Sc., The Manor House, Alphington, Exeter;

Lt.-Com. Harold Leo Quick, R.A.N., Navy Office, Melbourne, Australia; and

John Shearer, B.A., B.Sc., Observer, Carnegie Institution's Magnetic Observatory, Watheroo, Western Australia,

were balloted for and duly elected Fellows of the Society.

The following candidates were proposed for election as Fellows of the Society, the names of the proposers from personal knowledge being appended :—

Arthur Crossfield Davies, Justice of the Peace for Co. Down, Lenaderg House, Banbridge, Co. Down, Ireland (proposed by W. R. Kelly);

David Carder Davies, 19 Ashtead Road, London, E. 5 (proposed by R. Inwards);

Albert Downing C. Hooper, 42 High Street, Maldon, Essex (proposed by A. Berry);

Carl Emil Luplau Janssen, Director of Urania Observatory, Dr. Olgas Vej, 25, Copenhagen F, Denmark (proposed by T. E. R. Phillips); and

Lieut. William Henry Plummer, F.R.G.S., R.N.R., 19 Hawthorne Avenue, Swansea (proposed by C. W. Franklin).

Report of the Council to the Hundred and Third Annual General Meeting of the Society.

Progress and Present State of the Society.

	Patron	Honorary Member	Fellows		Associates	Total
			Compounders.	Annual Subscribers		
1921 December 31	1	1	228	527	49	806
Since elected	...	...	+6	+100	...	...
Deceased	...	...	−8	− 8	−2	...
Resigned	...	...	...	− 8	...	...
Since compounded	...	...	...	...	...	...
Removed from list	...	...	...	− 6	...	...
Restored to list	...	...	...	+ 1	...	...
1922 December 31	1	1	226	606	47	881

(Five Associates are also Fellows, and are therefore counted twice in the above table.)

Colonel Stratton's Account as Treasurer of the Royal

RECEIVED.

	£	s.	d.	£	s.	d.
Balances, 1922 January 1:—						
Cheque outstanding 1922 Jan. 1	3	3	0			
In hand of Assistant Secretary on Petty Cash Account	8	19	8			
				12	2	8
Received on account of Dividends:—						
£1,250 Metropolitan 3-per-cent. Stock	37	10	0			
£2,000 Swansea Corporation 3½-per-cent. Stock ...	70	0	0			
£3,500 East India Railway 3-per-cent. Debenture Stock	77	8	9			
£3,500 London and North-Western Railway 3-per-cent. Debenture Stock	74	16	3			
£4,000 Midland Railway 2½-per-cent. Debenture Stock	71	5	0			
£1,000 Lancashire and Yorkshire Railway 3-per-cent. Preference Stock (half-year's interest)	10	10	0			
£750 London and North-Western Railway 4-per-cent. Consolidated Preference Stock (half-year's interest)	10	17	6			
£2,400 Gas Light and Coke Co. 3-per-cent. Debenture Stock	51	6	0			
£2,000 Commercial Gas Co. 3-per-cent. Debenture Stock	44	2	6			
£3,852, 12s. 6d. War Loan Inscribed 5-per-cent. Stock	192	12	6			
£542, 18s. 2d. India 3½-per-cent. Stock	19	0	0			
Interest on Deposit Account (including £13, 7s. 10d. due to the Special Fund)	15	11	6			
				675	0	0
Received on account of Subscriptions:—						
Arrears	194	16	0			
Annual Contributions for 1922	863	2	0			
,, ,, in advance 1923–25	14	14	0			
Admission Fees	191	2	0			
First Contributions	154	7	0			
				1418	1	0
Compositions				189	0	0
Sale of Publications:—						
At Society's Rooms (including £25, 16s. 0d. received in advance for Centenary Volume)	61	17	11			
At Wheldon & Wesley's £201 13 6						
Less 10 per cent. discount ... 20 3 4						
	181	10	2			
Sale of Waste Paper	1	15	8			
Herschel's Works: half share, received from the Royal Society	1	0	0			
Franklin-Adams Charts	200	0	0			
				446	3	9
Sales of Photographs				64	11	6
Income Tax refunded				141	12	0
London Mathematical Society, for use of Rooms, etc. ...				20	0	0
Received from Fellows for Centenary Dinner Tickets ...				108	3	0
,, ,, use of Telephone				0	7	10
Contributions to Special Fund				1389	4	6
Cheque outstanding, 1922 December 31				13	6	8
				£4,477	12	11

Examined and found correct.

(Signed) R. T. CULLEN.
W. GEORGE LINGHAM.
FRANK ROBBINS.

1923 *January* 26.

Astronomical Society, from 1922 *January* 1 *to December* 31.

PAID.

	£ s. d.	£ s. d.
Debit Balance at Bankers', 1922 Jan. 1		8 10 9
Due to Assistant Secretary 1922 Jan. 1, on account of Library Fund		18 5 4
Assistant Secretary: Salary, etc., to 1922 October 16 ...	239 15 11	
Extra clerical assistance	35 6 0	
Clerk: Salary 1921 December–1922 December	173 6 8	
,, Bonus	25 0 0	
Porter and wife: Wages	172 14 0	
		646 2 7
House Duty	2 12 6	
Fire Insurance	10 13 0	
Insurance of Stock at Printers'	1 18 3	
Insurance of Portrait of Sir D. Gill	0 9 0	
Accident Insurance	1 6 2	
Servants' Insurance Stamps	1 17 6	
		18 16 5
Printing, etc.:—		
Monthly Notices, 9 numbers	863 5 1	
Geophysical Supplement	18 16 0	
Miscellaneous	99 17 9	
Photo-engraving	49 18 11	
		1031 17 9
Reproduction of Photographs for sale		61 8 3
Purchase of Books for Library		54 3 2
Expenses of Meetings of R.A.S.	22 3 5	
Expenses of Geophysical Discussions	6 2 2	
Expenses of Centenary Celebration	222 3 9	
		250 9 4
Telephone: Subscription, Deposit, etc.	11 1 3	
Registration of Telegraphic Address	2 0 0	
Postage, etc. (including postage of *Monthly Notices*) ...	106 13 2	
Carriage on Franklin-Adams Charts	15 0 0	
Carriage of Parcels	0 18 7	
Stationery and Office expenses	25 2 9	
		160 15 9
House expenses	15 7 2	
Coal, Gas, Electric Light, etc.	138 4 0	
Fittings and Repairs (including new Fire Hose £61) ...	88 15 6	
Sundries	11 8 3	
		253 14 11
Lee and Janson Fund Grant		9 0 0
Assistant Secretary for work for London Mathematical Society in 1921	10 0 0	
Expenses of Meetings of London Mathematical Society ...	6 4 2	
		16 4 2
Cheque Book and Bankers' Charges on Cheques	1 3 1	
Deficiency on French Money Order	0 0 7	
		1 3 8
Balances 1922 December 31:—		
Cheques outstanding, 1922, December 31	9 5 0	
In hand of Assistant Secretary on Petty Cash Account	3 17 0	
In hand of Assistant Secretary on account of Library Committee	26 15 9	
At Bankers', on Current Account, as per Pass Book (including Lee and Janson Fund, £36, 5s. 4d.; Hannah Jackson Fund, £75, 4s. 5d.)	257 3 1	
At Bankers', on Deposit Account (including Special Fund, £1389, 4s. 6d.)	1650 0 0	
		1947 0 10
		£4,477 12 11

Report of the Auditors.

We, the undersigned, appointed Auditors by the Fellows assembled in ordinary meeting 1922 December, now report that we have examined the Treasurer's account of receipts and expenditure for the calendar year 1922, and have found and certified the same to be correct.

At the close of the year the two cash balances in hand of the acting assistant Secretary together with the balance on current account at the Bank, plus cheques not credited and minus one cheque drawn but not presented, amounted to £283, 14*s.* 2*d.* In addition £1650 was on deposit at 1 per cent.

The year 1922 will be memorable by reason of the very substantial improvement in the state of the Society's financial position—a change brought about by the genius of the late Treasurer (Col. E. H. Grove-Hills, deceased), who, though suffering from a painful and incurable malady, had the energy to raise a special fund (£1389, 4*s.* 6*d.* actually received) besides increasing our Fellowship list in a most remarkable degree. He was instrumental too in securing the increase of £125 in the income from investments which arises from the gift of £2500 War Stock 1929–47 from Sir Charles Parsons as a memorial of the late Earl of Rosse. These three benefits joined with the fact that it has not been necessary to incur the expense of a volume of *Memoirs* will lighten the task of the quite recently appointed Finance Committee, to which body your Auditors have submitted some forty-one recommendations or criticisms.

The Railways Act of 1921 has made one alteration in the list of investments. The Lancashire and Yorkshire Railway having been absorbed in the London and North Western Railway Co., our £1000 3 per cent. Preference Stock has become £750 4 per cent. Consolidated Preference Stock in the latter company. This change had its origin in the wisdom of the Legislature, and the Society had no option but submission.

The tabular statement of the investments shows improvement—depreciation is about £1600 less, and the interest earned is 3*s.* 11*d.* higher at £3, 6*s.* 7*d.* per cent.

Interest on deposit averaged £1, 9*s.* 3*d.* per cent. per annum.

Income Tax deducted at the source equalled £53, 18*s.* 6*d.* as far as April 5, and £77, 15*s.* 6*d.* for the latter part of the year. The former amount has been recovered and this added to £87, 13*s.* 6*d.* received on account of 1921 makes the sum £141, 12*s.* shown in the accounts.

Concerning the library, we note that of the six books reported a year ago as missing one has been found, namely:—

Namur, A., *Tables de Logarithmes à 12 décimales.* 8vo. Bruxelles, 1877,

but on the other hand—

Cooper, Edward J., *Cometic Orbits.* 8vo. Dublin, 1852,

cannot be found.

No application for the renewal of the loan of instrument No. 133 (3¼-inch Ross equatorial refractor) has been received, as required by the regulations, and we have reason to doubt whether the instrument is in use.

A list has been laid on the table of the names of those Fellows who are in arrear for sums due at the last Annual General Meeting of the Society, with the amount due against each Fellow's name.

The Society's holdings in three railway and two gas companies have been certified by the Bank, where the respective deeds are lodged for safe custody, and the four stocks inscribed at the Bank of England have been verified in the usual manner.

Lastly, we wish to express our admiration of the neatness and the accuracy with which our late assistant Secretary always presented his account. For painstaking care and devotion to the interest of the Society, he will not be surpassed easily, if at all.

(Signed) R. T. Cullen.
W. George Lingham.
Frank Robbins.

1923 *February* 9.

Amount of Stock.	Description.	When Acquired.	Original Value.	Cost per cent.	Value at 1922 Dec. 29.	Yield per cent. of Col. 4.	Total Interest.
£ s. d.			£ s. d.		£	£ s. d.	£ s. d.
1250 0 0	Metropolitan Cons., 3	1883–90	1254 8 0	100·352	963	2 19 9	37 10 0
2000 0 0	Swansea Corp., 3½	1905–9	2136 10 3	106·825	1460	3 5 6	70 0 0
3500 0 0	East India Rlwy., 3	1899–1909	3576 17 6	102·196	1873	2 18 8	105 0 0
3500 0 0	Lond. N.W. Deb., 3	1899–1908	3871 8 0	110·623	2205	2 14 3	105 0 0
4000 0 0	Midland Deb., 2½	1899–1901	3750 5 3	93·757	2080	2 13 4	100 0 0
750 0 0	Lond. N.W. Cons. Pref., 4	1903–10	867 11 0	115·673	600	3 9 2	30 0 0
2400 0 0	Gas Light Coke Co. Deb., 3	1899–1913	2328 2 10	97·006	1452	3 1 10	72 0 0
2000 0 0	Comm. Gas Deb., 3	1899–1910	1899 0 9	94·952	1150	3 3 2	60 0 0
19,400 0 0	Sub-totals and means		19,684 3 7	101·465	11,783	2 18 10	579 10 0
3,852 12 6	War Stock, 1929–47, 5	1915–22	3,782 7 3	98·176	3,852	5 1 10	192 12 6
542 18 2	India Stock, 3½	1920	300 0 0	55·258	342	6 6 8	19 0 0
23,795 10 8	Totals and means.		23,766 10 10	99·878	15,977	3 6 7	791 2 6
					Depreciation 7,789, say 33 per cent.		

Gifts to the General Funds of the Society.

The John Lee Gift: In 1836 John Lee, Esq., LL.D., gave the Advowson of the living of Hartwell; in 1844 he gave the Advowson of the living of Stone, Bucks. In 1879 these Advowsons were repurchased by his heir, Edward Lee, Esq., for £700.

The Lawson Bequest (1856): Henry Lawson, Esq., bequeathed to the Society the sum of £200, free of legacy duty.

The Carrington Bequest (1876): R. C. Carrington, Esq., F.R.S., bequeathed to the Society £2,000 Consols.

The Lambert Bequest (1877): C. J. Lambert, Esq., presented to the Society the sum of £500, free of legacy duty, being part of a sum of money bequeathed by his father, Charles Lambert, Esq., for scientific purposes.

The Knott Gift (1899): Mrs. George Knott presented to the Society the sum of £250, to defray the cost of printing *Memoirs*, vol. lii., containing the late Mr. Knott's Observations of Variable Stars.

The Peek Bequest (1904): Sir C. E. Peek bequeathed to the Society the sum of £200, as a contribution towards the expense of printing the Rousdon Variable Star Observations (*Memoirs*, vol. lv., part 1).

The McClean Bequest (1905): Frank McClean, Esq., LL.D., F.R.S., bequeathed to the Society the sum of £2,000, free of legacy duty.

The Farrar Bequest (1906): The Rev. A. S. Farrar, D.D., bequeathed to the Society the sum of £100, free of legacy duty.

The Brook Gift (1907): C. L. Brook, Esq., gave one-third of the cost of printing Pogson's Variable Star Observations (*Memoirs*, vol. lviii.).

The Gill Bequest (1914): Sir David Gill, K.C.B., F.R.S., bequeathed to the Society the sum of £250, in aid of Astronomical Research, in grateful remembrance of a like sum paid out of the funds of the Society in aid of his expedition to Ascension in the year 1877. In accordance with his wishes this bequest was expended to defray part of the cost of publishing vol. lxiii. of the *Memoirs*, containing the Theory of the Four Great Satellites of Jupiter.

The Parsons Gift (1922): The Hon. Sir Charles Parsons, K.C.B., F.R.S., gave to the Society £2,500 War Stock (1929–47), in memory of his father, the 3rd Earl of Rosse, who made the 6-foot Reflector, and who died 1867 October 31.

The Grove-Hills Bequest (1922): Colonel E. H. Grove-Hills, C.M.G., F.R.S., bequeathed to the Society, free of legacy duty, his collection of early Astronomical and Mathematical books up to the year 1700, together with a sum of £250 to cover the cost of the removal thereof to the Society's rooms.

Trust Funds.

The Lee and Janson Fund: In 1834 John Lee, Esq., LL.D., gave £100, and in 1876 T. C. Janson, Esq., bequeathed £200, the interest to be given to the widow or orphan of a deceased Fellow. These amounts now stand in £334, 10*s.* 9*d.* East Indian Railway 3-per-cent. Debenture Stock.

	£	*s.*	*d.*		£	*s.*	*d.*
1922 Jan. 1. Balance of Fund	35	4	7	1922. Grant from Fund ...	9	0	0
1922. Dividends, East Indian Railway ...	10	0	9	1922 Dec. 31. Balance of Fund	36	5	4
	£45	5	4		£45	5	4

The Turnor Fund and the Horrocks Memorial Fund: In 1853 the Rev. Charles Turnor, M.A., bequeathed a sum of £500, which now stands in £464, 18*s.* East Indian Railway 3-per-cent. Debenture Stock. In 1876 the Society received a sum of £38, 15*s.*, the unexpended balance of a fund collected for the purpose of erecting a memorial in Westminster Abbey to Jeremiah Horrocks: to this the Society added the sum of £55, 15*s.*, and the whole now stands in £103, 6*s.* East Indian Railway 3-per-cent. Debenture Stock. The interest on these funds is to be used in the purchase of books for the Library.

	£	*s.*	*d.*		£	*s.*	*d.*
1922. Dividends, East Indian Railway ...	17	1	0	1922. Purchase of Books ...	14	0	4
				1922 Jan. 1. Due from Fund	3	0	8
	£17	1	0		£17	1	0

The Harry Watson Fund: In 1920 Mrs. Watson presented to the Society the sum of £300 as a memorial to her late husband, Lieut.-Col. Harry James Watson, F.R.A.S., the interest on this sum to be used for the purchase of books for the Library, such books to be indicated by a special book-plate. This gift now stands invested in £542, 18*s.* 2*d.* India 3½-per-cent. stock.

	£	*s.*	*d.*		£	*s.*	*d.*
1922 Jan. 1. Balance of Fund	18	10	0	1922. Purchase of Books ...	37	10	0
1922. Dividends, India 3½ per cent.	19	0	0				
	£37	10	0		£37	10	0

The Hannah Jackson (née Gwilt) Fund: In 1861 Mrs. Hannah Jackson (*née* Gwilt) gave £300 Consols, which now stands in £309, 18*s.* 6*d.* East Indian Railway 3-per-cent. Debenture Stock; the interest to be given in medals or other awards in accordance with the terms of the Trust.

	£	*s.*	*d.*		£	*s.*	*d.*
1922 Jan. 1. Balance of Fund	65	18	5	1922 Dec. 31. Balance of Fund	75	4	5
1922. Dividends, East Indian Railway ...	9	6	0				
	£75	4	5		£75	4	5

The Grove-Hills Special Fund: As already reported to the Society, the late Treasurer collected by his private efforts, with the approval of the Council, a Special Fund to meet the immediate financial difficulties of the Society. He aimed at securing £1,800, and at present the Fund stands at £1,389, 4*s.* 6*d.* The Council are hopeful that the Fund will reach the figure originally aimed at, and they will be able, during the interval, while the Fund is available to meet present expenditure, to frame a financial policy for the future. The following is the list of donors to this Fund:—

C. G. Abbot.
H. B. Adames.
A. C. Allen.
C. Anthony.

E. E. Barnard.
A. Berry.
S. Bolton.
J. G. Bower.
C. H. Brewitt-Taylor.
W. Briggs.
S. Brodetsky.
C. L. Brook.
A. N. Brown.
E. W. Brown.
J. S. Browne.
G. Byfield.

A. Y. G. Campbell.
W. W. Campbell.
E. Carpmael.
C. J. P. Cave.
F. Clements.
Sir C. Close.
Miss V. M. Common.
A. E. Conrady.
R. E. S. Cooper.
A. C. D. Crommelin.
J. Crouch.

F. Dannreuther.
C. G. Darwin.
P. L. H. Davis.
R. E. De Lury.
W. Doberck.
J. L. E. Dreyer.
J. E. Drower.
W. G. Duffield.
F. C. Dumat.
Sir F. W. Dyson.

A. S. Eddington.
T. E. Espin.
L. Evans.
T. L. Evans.
J. Evershed.

S. Fellows.
Sir A. M. Field.
A. H. Fison.
Sir G. S. Forbes.
D. R. Fotheringham.
J. K. Fotheringham.
A. Fowler.
Miss A. Fry.

W. H. St. Q. Gage.
W. T. Gayfer.
S. B. Gaythorpe.
W. B. Gibbs.
J. W. Gifford.
H. B. Goodwin.
W. du G. Gray.
Sir R. Gregory.
E. H. Grove-Hills.
H. H. Gruning.

W. Hall.
G. H. Hardy.
I. B. Hart.
W. Heath.
G. A. Hemsalech.
P. H. Hepburn.
Miss F. Herschel.
E. R. Hicks.
N. Holden.
H. P. Hollis.
T. H. L. Hony.
J. Hough.
S. S. Hough.
W. M. Hunt.

R. T. A. Innes.
R. Inwards.
Lord Iveagh.

H. F. Jackson.
J. H. Jeans.
R. Ll. Jones.

H. Knox-Shaw.

H. Lamb.
J. R. Leeson.
Sir G. P. Lenox-Conyngham.
A. E. Levin.
W. J. Lewis.
A. F. Lindemann.
W. G. Lingham.
J. E. Littlewood.
W. J. S. Lockyer.
A. W. Long.
F. W. Longbottom.
A. W. Lowe.

G. T. McCaw.
F. K. McClean.
W. M. McClean.
D. MacLean.
R. Malcolm.
E. E. Markwick.
J. W. Meares.
C. J. Merfield.
T. R. Merton.
E. A. Milne.
M. de Miremont.
G. Morham.

F. W. Nash.
A. N. Neate.
H. F. Newall.
A. M. Newbegin.

Sir C. Parsons.
C. Pendlebury.
C. D. Perrine.
J. S. Plaskett.
A. J. Potts.
E. Prentice.
T. Preston-Battersby.
J. Proudman.

Lord Rayleigh.
W. H. Rees.
F. Robbins.

Sir D. Salomons.
R. A. Sampson.
F. Schlesinger.
Sir A. Schuster.
F. H. Seares.
A. L. Selby.
C. B. Shaw.
J. F. Shepherd.
J. B. Smith.
J. H. Smith.
R. N. Stevens.
J. J. Steward.
Stonyhurst College.
F. J. M. Stratton.

E. O. Tancock.
J. Tatlock.
A. Taylor.
H. Thomson.
H. G. Tomkins.
E. Tourner.
H. S. Toy.
J. Tripplin.
H. H. Turner.

A. J. Walker.
F. W. Walker.
G. N. Watson.
E. Weldon.
C. J. Westland.
E. W. White.
E. T. Whitelow
R. Wilding.
H. E. Wood.
W. M. Worssell.
W. H. Wright.

A. E. Young.
T. E. Young.

Assets and Present Property of the Society, 1923 January 1.

	£ s. d.	£ s. d.
Balances, 1922 December 31 :—		
Cheques outstanding	9 5 0	
In hand of Assistant Secretary on Petty Cash Account	3 17 0	
In hand of Assistant Secretary on Account of Library Committee	26 15 9	
At Bankers, on Current Account	257 3 1	
,, ,, on Deposit	1650 0 0	
		1947 0 10
Due on account of Subscriptions :—		
13 Contributions of 3 year's standing	81 18 0	
31 ,, 2 ,, ,, ...	130 4 0	
72 ,, 1 ,, ,, ...	151 4 0	
Other sums	3 1 0	
	366 7 0	
Less paid in advance	14 14 0	
		351 13 0
Due from Wheldon & Wesley for sales of Publications during the last quarter of 1922	34 3 0	
Due for Publications sold	5 19 0	
		40 2 0
		2338 15 10

£1,250 Metropolitan 3-per-cent. Stock.

£2,000 Swansea Corporation 3½-per-cent. Stock.

£3,500 East Indian Railway 3-per-cent. Debenture Stock.

£3,500 London and North-Western Railway 3-per-cent. Debenture Stock.

£4,000 Midland Railway 2½-per-cent. Debenture Stock.

£750 London and North-Western 4 per-cent. Consolidated Preference Stock.

£2,400 Gas Light and Coke Co. 3-per-cent. Debenture Stock.

£2,000 Commercial Gas Co. 3-per-cent. Debenture Stock.

£3,852, 12s. 6d. War Loan 5-per-cent. Inscribed Stock.

£542, 18s. 2d. India 3½-per-cent. Stock.

Astronomical and other Manuscripts, Books, Prints, Photographs, and Instruments.

Furniture, etc.

Stock of Publications and Photographs.

Three Gold Medals.

Publications of the Society.

During the past year vol. lxxxii. of the *Monthly Notices* has been published, also *Geophysical Supplement to the Monthly Notices*, No. 1.

Geophysical Discussions and Publications.

Meetings for the discussion of geophysical subjects have been held in the rooms of the Society since 1917. At first these meetings were arranged by the British Association, but since 1919 they have been in the charge of a committee, eight members of which are appointed by the Society, and one each by the Royal Meteorological, Royal Geographical, Geological, and Physical Societies, and one by the British Astronomical Association. During the past year discussions have been held on the Depth of Earthquake Foci, the Maintenance of the Earth's Electric Charge, a 21-minute Periodicity in Earthquakes, and the Projection for the International Aeronautical Map.

Publication of geophysical papers in special supplements to the *Monthly Notices* has been initiated during the past year, and two parts have already appeared. These publications are not intended to include only papers read at the discussions, nor to include the whole of these, but to increase the accessibility of original work in all branches of geophysics. H. J.

The Assistant Secretary.

Owing to the vacancy in the post of Assistant Secretary caused by the lamented death of Mr. William Henry Wesley, who served the Society with devotion for forty-seven years, the Council has taken into consideration the choice of a successor. They have chosen Miss Kathleen Williams, who had assisted Mr. Wesley for nearly three years previous to his death. Her appointment dates from the Anniversary Meeting.

Obituary.

The Council regret that they have to record the loss by death of the following Fellows and Associates, and of the Assistant Secretary, during the past year :—

Fellows :—Henry Bridger Adames.
William Henry Mahoney Christie.
Frederick John Eld.
* Frank Flowers.
Edmond Herbert Grove-Hills (Treasurer).
George Stewart Hitchcock.
† Carlton John Lambert.
John George Lincoln.
Robert Pearce.
Andrew Simons.
Charles Michie Smith.
Edward John Stutter.
Clement Jennings Taylor.
George Lyon Tupman.
Peter Hately Waddell.
James Waterhouse.
Associates :—Fritz Cohn.
Jacobus Cornelius Kapteyn.
Assistant Secretary :—William Henry Wesley.

Henry Bridger Adames was born at Chichester, 1882 October 3, and was educated at the Prebendal School in that city.

He became interested in astronomy at a very early age. On leaving school he entered the Royal Observatory, Greenwich; while at the Observatory (from 1901 July to 1902 December) he worked in the Transit Circle Department (Reduction of Zenith Distances).

In 1903 he went to Canada and entered the Montreal office of the Canadian Pacific Railway. In 1909 he returned to England, being engaged, up to the time of his death, as an accountant.

He had many and varied literary and scientific interests; but devoted the greater part of his leisure to astronomy and meteorology.

In astronomy he gave his chief attention during recent years to the systematic observation of variable stars, using a 4-inch equatorial refractor. For many years he was a member of the Variable Star Section of the British Astronomical Association.

He became a Fellow of the Royal Meteorological Society in 1912, and served for a period on the Council of that Society.‡ He collaborated during the past eight years with Mr. J. E. Clark in the preparation of Annual Reports on the Phenological Observations in the British Islands, which were published by the Royal Meteorological Society.

* Died in 1920, but death not reported till 1922.
† Died in 1921, but death not reported till 1922.
‡ 1916–17 or 1917–18.

He was elected a Fellow of the Royal Geographical Society in 1913.

He married in 1908 Miss Queenie Moyle of Chichester, whom he had known from childhood.

His death occurred at Ilford on 1922 December 9.

He was elected a Fellow of the Society on 1902 June 13.

Sir W. H. M. Christie, K.C.B., F.R.S.—Early in the year the Society lost by the death of Sir William Christie one of its oldest Fellows, for he was at the time the ninth in the list in order of date of election. His unobtrusive manner and personality and the fact that he had taken little active part in the affairs of the Society during the last ten years of his life caused him to be unfamiliar to the general body of present Fellows; but it may be said with truth that during the last fifty years no person has had more influence directly or indirectly on British astronomy. For forty years he was an official astronomer, and for twenty-nine of them the titular head of astronomy in England.

William Henry Mahoney Christie was the third son and the eldest child of the second marriage of Samuel Hunter Christie, F.R.S., Professor of Mathematics in the Royal Military Academy at Woolwich from 1838 to 1854, the author of many papers on magnetism, for whom the invention of the method of the bridge in electrical measurements is claimed. The second son of Samuel Christie, James Robert, who was also F.R.S. and on the staff of the Royal Military Academy as First Mathematical Master, was a Fellow of our Society and the author of a book on Practical Astronomy and other works.* The family was originally Scotch, the father of S. H. Christie having come to London in the eighteenth century. The subject of this memoir, whose mother was of an Irish family, whence came his third name, was born on 1845 October 1, and was educated at King's College School, London, and Trinity College, Cambridge, going up to the University in 1864. He was elected a Foundation Scholar of the College at Easter 1866, and achieved the position of fourth wrangler in the Mathematical Tripos of 1868, when the first three places were filled by Fletcher Moulton, G. H. Darwin, and Charles Smith, afterwards Master of Sidney Sussex College.

Christie sat for the Sheepshanks Exhibition in 1866, but was not successful, the exhibition being awarded in that year to Charles Niven, afterwards Professor of Natural Philosophy in the University of Aberdeen. Airy was one of the three examiners, and on the paper that he set, which was entirely on practical astronomy, Christie was undoubtedly first—showing an intimate knowledge of the adjustment and use of instruments, especially with Greenwich methods. Airy says in his Report: † "Christie is methodical, clear, and practical," which were qualities that he retained throughout life. Another of the examiners wrote: "Christie is very clear and sensible in what he knows and a very satisfactory examinee." But he failed in the paper set by Professor Cayley, which was wholly on Planetary and Lunar theory,

* See *Monthly Notices*, 40, 188.

† These Reports are preserved in the Record Room at Greenwich, and have been inspected by permission of the Astronomer Royal.—H.P.H.

and after careful consideration the examiners decided in favour of Niven, because he was "the man of most knowledge."

After taking his degree Christie remained at Cambridge, occupied in teaching, and was elected a Fellow of his College in 1869 October. In 1870 June Mr. E. J. Stone, Chief Assistant at the Royal Observatory, Greenwich, was appointed to the position of H.M. Astronomer at the Cape of Good Hope, and left Greenwich in the following August. Airy selected Christie as Stone's successor, possibly because of his quality as shown in the Sheepshanks examination, and perhaps on account of his presumed hereditary talent. Christie joined the Staff of the Observatory in the autumn of that year, and quickly took up a position in the astronomical world. He joined our Society in 1871 January, was elected a Member of Council in 1872 February, and sat at the Council table in various capacities for forty years from that date, except in the year 1879–1880, when he stood off at his own wish. There is reason for thinking that Airy hoped Christie would take up the lunar theory as his special research, as it was Airy's at that time, but after making some study of Delaunay's papers and results, Christie abandoned the motion of the Moon as a line of work, and his early papers show that he was more interested in colour, spectroscopy, and optics generally. The first of these papers was, however, more of a mechanical nature, for it describes the now seemingly obvious paper-covered drum used until this day on the Greenwich Transit-Circle for recording the observation of declination of stars, invented by Christie in the year 1873. In that year a very important innovation in the work of the Royal Observatory was brought about by his influence, this being the addition of the systematic photography of the Solar surface, and the spectroscopic observation of the Sun and stars. The latter has not survived, but the recording of the positions of sunspots is one of the routine researches of the present day at Greenwich. Christie as Chief Assistant took occasional turns with the Transit-Circle and Altazimuth, the two principal instruments of the Observatory at that period, but examination of the papers communicated to the Society, indexed under the heading Royal Observatory, Greenwich, will show his special interest in the spectroscope, and in the latter half of the year 1877 a half-prism spectroscope, made on a new principle evolved by him, was brought into use. The instrument is described in the *Proceedings of the Royal Society*, No. 179, 1877, but briefly it may be said that the train is made up of half-prisms, each being the half of an isosceles prism cut in two by a plane perpendicular to the base, and gives an unusually large dispersion. The design has not been followed elsewhere, but the fact that it was constructed and brought into use is sufficient to show Christie's optical ingenuity and capability. The entries in the index to the *Monthly Notices* under his own name at this period are for the most part similarly of an optical nature. "On the Colour and Brightness of Stars as measured with a New Photometer" (1874 Jan.), is a paper which describes an instrument made for him by Browning, and includes a series of observations made with it. "On a New Form of Solar Eyepiece" (1876 January), relates to another instrument made to his design, whilst two papers on the appearance of Venus (1876 December and 1878 January) are contribu-

tions to a discussion then proceeding as to whether the solar light is reflected from Venus specularly; that is, as from a looking-glass, or whether it is scattered; and are based on actual observations of the planet made with polarising eyepiece and dark wedge. But though these were his favourite mental activities in these early years, it is not to be forgotten that he kept himself well in touch with the staple work of the Royal Observatory, as is shown by his discussion of the "Systematic Errors of the Greenwich North Polar Distances," to be found in vol. xlv of the *Memoirs*. He was during these ten years the trusted helper of Sir George Airy, conducting the routine of the Observatory with precision and skill, and held a place in our Society, which was then passing through a troublous period of internal faction, respected by all for his official position, his personal dignity, and his courtesy of manner. Not the least of his efforts outside his official work at this time was the establishment of the *Observatory Magazine*, which, with the encouragement of several friends, he initiated in the year 1877, and which has continued practically in the form that he devised under many successive Editors until the present time. Christie was sole Editor until 1881 October when, fearing lest this would interfere with his duties as Astronomer Royal, he offered the editorship to Mr. E. W. Maunder, and the two names appeared on the cover as joint Editors until the end of the year 1882, when Mr. Christie's connection with the *Magazine* ceased.

The retirement of Sir George Airy in 1881 August, and the consequent necessity for appointing a successor created considerable interest not only in astronomical circles but also among the larger public that keeps itself lightly in touch with scientific occurrences generally. The appointment of a new Astronomer Royal had not taken place for forty-six years, the name of Airy was fairly well known as that of a very learned man, and the circumstances of the science of astronomy had much changed, which all led to much speculation as to the next occupant of the exalted position. Many names were suggested, but a promotion from the staff apparently did not occur to people as a probable course, though it was obviously a suitable one, and the choice of the Crown —it is well known that Airy, Dr. Huggins, and Mr. Spottiswoode, then President of the Royal Society, were among the advisers—fell on Mr. Christie, who was unfeignedly surprised when he heard of his advancement. It was not an easy task to follow Airy in the superintendence of the Royal Observatory, for there was much that required alteration. Though the institution was rated as a shore department of the Admiralty, Airy took pride in keeping it distinct and different in its methods from the ordinary offices of the Civil Service. It was his pleasure to effect economies in small details and to rely on his own designs and ideas rather than to adopt the instruments and procedure of business establishments, the result being that in the year 1881 the visitor to Greenwich Observatory found himself in surroundings half a century out of date. It was a surprise to find that the excellent work on sun-spots and on earth magnetism emanated from wooden sheds, forty years old, and that the electrical batteries on which the transmission of the Greenwich time signal depended were in the

hands of a small tradesman of the town. It was such things as these that Christie was called upon to remedy, and he did it steadily and carefully. The deal desks at which the computers worked were replaced by mahogany, and the officers and workmen of the General Post Office were called upon to put on a satisfactory footing and take over the telegraph wires and electrical apparatus internal and external. The first purely astronomical improvement Christie made related to the collimators of the Transit-Circle, which formerly rested on solid brick piers north and south of the instrument, and blocked out the view, so that observation of stars by reflexion at a greater distance than 40° from the zenith was prevented. Early in the year 1882 the piers were cut away and the collimators mounted as they are now, on upright arms turning on centres below, allowing them to be swung on one side when not in use, to extend the range of that class of observation. The extension and elaborations of the sun-spot record next came under Christie's consideration. At the beginning of this work in 1873 the photographs on which it depended showed an image of the sun 4 inches in diameter, but means were now adopted to double that size; a micrometer was designed and made for the measurement of these larger plates, and the co-operation of other observatories was sought to make the daily record complete. The valuable Greenwich record of positions of sun-spots and faculæ that has now been carried on with such completeness for nearly fifty years, projected and designed as to its details by Christie, is not the least of the items that stand to his credit.

It is needless to follow in detail the many alterations in procedure, the many improvements in structure and furnishing of the Royal Observatory that happened in the early years of his tenure of office as Astronomer Royal, though it may be well to record the short existence of a low building covered by the dome now surmounting the south observatory, in which the Lassell telescope was housed. Christie, on behalf of the Admiralty, accepted this telescope, being the one with which Hyperion was discovered in 1848, as a gift when it was offered by the Misses Lassell in 1883, thinking that a reflector of 2 feet aperture would be well suited for observation of faint satellites and comets beyond the power of the instrumental means then available. The instrument was set up in this building made for it in the south ground and much energy was spent in its installation. But in the end it was found unsuitable for any work involving measurement because of the instability of the mounting, and its use was finally abandoned. The necessity for a large telescope, however, remained present in Christie's mind and led to the procuring of a refracting telescope of 28-inch aperture to replace the Merz 12¾-inch, that had been set up in the south-east tower of the Observatory about 1859. The planning of this objective was much to Christie's taste, for, as has been said, he had much optical talent. He consulted and had much correspondence with Sir George Stokes on the subject, and together they designed an objective of two lenses which, though normally suitable for visual observation, could by separation and reversal of one lens be made suitable for photography. The ingenuity shown in the design of the dome which covers this instrument calls for comment. The new telescope with object-glass of 28 feet focal

length is in the same building and on the mounting which formerly carried the smaller telescope, and it was necessary that the character of the dome should be enlarged, which was effected by giving it the shape of an involute of a circle, and the result to the eye is not inartistic. The construction of this object-glass, which was eventually made by the firm of Grubb from discs supplied by Messrs. Chance and mounted in 1893 July, was first mentioned in the Astronomer Royal's Report for 1886. In the same Report there is a foreshadowing of the addition of photography, other than solar, to the work of the Royal Observatory, for the suggestion is made that the Merz refractor might be adapted to photography by the addition of a correcting lens, but the suggestion was never realised; the needs of photography being otherwise provided for as will be shown later.

In the Report for the next year, 1887, the plans for the first enlargement and extension of the buildings of the Observatory, which formed such a prominent feature of Christie's control, were set out. The Admiralty had sanctioned and made provision in the estimates for a considerable alteration and extension of the rooms used as computing rooms, lower and upper, built by Bradley, which would take in his old quadrant pier, and on this would be placed the 6-inch Cooke refractor, for occasional observations and to carry the photoheliograph. The advantage of thus concentrating the work of the Observatory in the old building was pointed out, from which it appears that Christie at that time did not contemplate the splendid building, which is the main observatory to-day, in the south grounds. This came later, and was built piecemeal in the years 1891 to 1899, section being added to section as money was supplied for the purpose by the Treasury in response to the Astronomer Royal's tactful demands. It is curious to think that this grew out of an original scheme for a storehouse or museum to house various portable instruments that was to be attached to the Lassell building, mentioned in the Report for 1889. The scheme for placing the Cooke Equatorial on the upward extension of the quadrant pier, proposed in 1887, was also modified when the Astrographic Chart was initiated, and the telescope to be used for that work at Greenwich was set up in that place. In 1894 Sir Henry Thompson made his noble gift of £5000 to the Observatory to provide a photographic telescope, and the acceptance of this offer gave a reason for the destruction of the Lassell building erected in 1884 and the re-erection of the dome on the central tower of the new building—the Physical Observatory, as it was then called. The building of this went on steadily but slowly, and at the same time extensions, projected by Christie, were going on in other directions. The construction of a new Altazimuth or Universal Transit-Circle was proposed in 1892, the observations with which would be treated in an ingenious manner according to formulæ to be found in a paper by Christie in the *Monthly Notices* of May of that year (1892). This instrument was made by Messrs. Troughton & Simms, and the building to house it erected in 1894 and 1895. In the years immediately following this the Observatory domain was increased by the addition of an enclosure in the park on the east side of the Observatory, at a distance of between 300 and 400 yards, as the site for a Magnetic

Pavilion to contain the instruments used for determination of absolute values of the magnetic elements. This was completed and the instruments installed in 1898 September. It will be readily understood that this decade was a busy period in Christie's life. The consideration of details of these instruments and consultation with Government departments as to the new buildings, left little time for individual astronomical work. In one Report he complains that he could never find leisure to complete the optical calculations for a photographic corrector to be applied to large visual instruments, such as has been already mentioned, as he would like to have done. His time was largely occupied in drafting long letters, many of which were written in his own hand, to the instrument makers and to the Admiralty and Treasury. Simultaneously the normal work of the Observatory was going on. The observations with the Transit-Circle were increasing; the chronometer work was expanding, and the Astrographic Chart and Catalogue were proceeding, and though these things were largely in the hands of his Chief Assistant and Assistants, he exercised meticulous supervision over all, even to such small details as to the position of shelves to carry chronometers or the making of cupboards to hold photographs.

Besides, and perhaps beyond the material changes in building and equipment that Christie wrought, the changes in the staff and the manner of their selection must be mentioned, though the subject must be touched on lightly. In the early history of Greenwich Observatory the Assistants of the Astronomer Royal were undoubtedly men drawn to the work by natural inclination, content with small reward, and if one may judge from Maskelyne's treatment of the luckless Kinnebrook wholly at the mercy of their chief, who might discharge them for such a venial offence as a difference of personal equation. Their selection lay wholly with the Astronomer Royal until in the year 1872 the appointment of Assistants passed into the hands of the Civil Service Commissioners, who made selection in the usual way by examination, the salary offered and the examination being of such type as might attract men with university degrees of minor merit—but the conditions did not attract largely. On several occasions when appointments were offered it was difficult to secure candidates, and experience showed that the work at Greenwich, being largely of a mechanical nature, was efficiently done by those who had been apprenticed to the "trade," so that Christie reverted more or less to an earlier method of forming the staff, and temporary computers passed into a class of established computers, which has now become Assistants. A change was made also in connection with the higher office of Chief Assistant. Airy had from the beginning an Adjutant. First, the Rev. Robert Main, then Mr. E. J. Stone, and afterwards, as we have seen, Mr. Christie; gentlemen whom he chose for their mathematical attainments, nominally that they might help him if need be in his researches, or in any detail of the Observatory work in which such talent was required, but who actually helped him by taking responsible charge of the routine work of the establishment. Christie when holding the office had influenced Airy to establish a "Physical Department" represented by the sun-spot work, and with this the study of Earth-magnetism might fittingly be associated. Since the work of

the Observatory was then developing along two diverging lines it appeared to him that the office of Chief Assistant should be duplicated, and that there should be an officer of this class for each branch. The attitude of resistance of the staff to this, or rather to the method of appointment of the Chief Assistant, which is wholly in the hands of the Astronomer Royal, has been hinted at in another place (*Observatory*, **45**, 78), where it is contended that by selecting these officers from outside the staff, the science of astronomy is enriched by talent and genius which would otherwise find its way elsewhere, and the argument has much weight and is supported by experience. Christie carried his point in spite of some internal opposition in addition to the normal objection to increase of staff and expenditure from the controlling offices of the Admiralty and Treasury. Much of his time was spent on official interviews and correspondence on these matters, which, looking back from the present-day point of view, was certainly not wasted effort, noting the effect it has had on astronomy, though it brought him no personal reputation as an astronomer.

Though many items may be omitted in this brief sketch of the activities of Sir William Christie in these twenty-nine years, the part that he took in the establishment of Greenwich as the prime meridian of the World, and in the matter of the reform of the time-system generally should not be forgotten. He was the representative of Great Britain at the meeting of the International Geodetic Association at Rome in 1883 when Mr. Sandford Fleming's proposal for the introduction of a universal day based on the Greenwich meridian was discussed. He was not one of the representatives at the conference at Washington in the following year, where it was resolved to recommend the adoption of the meridian of Greenwich as the zero of longitude, and the Greenwich civil day, beginning at Greenwich midnight and reckoned from 0 to 24 hours, as the standard of time-reckoning, but he had given his support freely and whole-heartedly to these proposals and there is little doubt but that this had much effect. At his instigation the Board of Visitors in 1885 recommended almost unanimously that the day in the British Nautical Almanac should be arranged for the year 1891 to begin at Greenwich midnight so as to agree with civil reckoning. As we know this recommendation is not being carried out until nearly forty years after it was proposed, but Christie took the responsibility of making this unification of the day in certain departments of the observatory work at the beginning of 1885, and on January 1 of that year the hour hand of the public clock at the Observatory gate was put forward by twelve hours.

His co-operation was often sought for the determination of differences of longitude from Greenwich, and for this purpose a Transit Pavilion was built in 1891 in the front court on Bradley's meridian, which was used occasionally by members of his staff and by visiting astronomers and geodesists. The arc Greenwich-Paris was measured three times under his direction (the first time was before the erection of this building), the longitude of Montreal was determined in 1892 in co-operation with Professor McLeod of the M'Gill University, the Astronomer Royal consenting to take part in the work on condition that the longitude of

Waterville near Valentia in Ireland was determined, since this was near the western end of the European longitude arc on the parallel of 52° N. This was followed a few years later by determination of the longitude of Killorglin, whereby the fixation of the astronomical position of this western end for which Greenwich had taken the responsibility some forty years earlier was satisfactorily completed. Christie took no personal share in these observations, but merely organised. He was a good and careful observer judging by his work when he was chief assistant, but after he became Astronomer Royal he observed no more except on one special occasion of a solar eclipse in 1882 May, which was seen in partial phase from Greenwich, and on the later occasions when he went with members of his staff on eclipse expeditions: to Japan in 1896, when the sky was completely overcast during totality; to Sahdol in India in 1898; to Ovar in Portugal in 1900; and to Sfax in Tunisia in 1905. "He observes too much," he was once heard to say in half cynical comment on the methods of one who like himself was the chief of an important national observatory, which apparently expressed his views on the duty of an Astronomer Royal in this respect.

To close with some personal details, Christie married, in 1881, Violette Mary, daughter of Sir Alfred Hickman, Bart., a well-known ironmaster in the Midlands, and sometime member for Wolverhampton in the House of Commons. Mrs. Christie died in 1888, leaving two sons, the younger of whom died in early childhood, the elder, Capt. H. A. H. Christie (Yeomanry), Barrister-at-law, is a Fellow of our Society. He received the distinction of Companion of the Bath on the occasion of Queen Victoria's Diamond Jubilee in 1897 and Knighthood followed on 1904 November 9. The degree of Doctor of Science was conferred on him by Oxford University on 1902 June 24, and he was the recipient of honours from many foreign learned Societies. He was elected to Fellowship of the Royal Society in 1881, and served on the Council and as a Vice-President for various periods. He gave up his position as Astronomer Royal on his sixty-fifth birthday, 1910 October 1, and retired to a country home, first at Warlingham, and afterwards at Downe in Kent, keeping himself in touch, however, with the progress of astronomy. Yachting had been a recreation with him, and in the later years of his life in the winter months he made a voyage to some place of congenial climate. In February 1921 he visited Professor W. H. Pickering at Mandeville, Jamaica, and there had the opportunity of making some remarkable observations of the Moon that he communicated to the Society in April of that year. He seemed to be in his usual health—remembering his advanced age—on the day of the Meeting of the Society in January last, and spoke of a projected voyage to Morocco, for which he started on the following Friday, January 20. The manner of the end will be learned from the following extracts from the log of the P. and O. Steamship *Morea*:—

"January 22, 1922, 4^h 30^m p.m. Surgeon reported death of Sir William Christie, 1st saloon passenger. Cause of death, Gastric Hæmorrhage.

"January 23, 1922, 11^h 45^m, Slow. 11^h 46^m, Stopped, and committed

to the deep the body of the late Sir William Christie. $11^h\ 49^m$, Full ahead. Lat. 40° 3·5′ N. Long. 9° 20′ W."

He was elected a Fellow of the Society on 1871 January 13. He was Secretary from 1880 February until 1882 February, President 1888–1890.

An engraved portrait from a painting by Jacomb Hood hangs in the Meeting Room. H. P. H.

Edmond Herbert Grove-Hills, Colonel R.E., who died on October 2 at his residence, 1 Campden Hill, London, W., was born 1864 August 1. His father, Herbert A. Hills of High-Head Castle, Cumberland, had been Judge of the Court of Appeal in Egypt, and his mother was a daughter of Sir William Grove. Edmond Hills was educated at Winchester and at the Royal Military Academy, whence he passed out as senior cadet of his term in 1884 with a commission in the Royal Engineers. For the next two years he was at the School of Military Engineering at Chatham, and he went to Gibraltar in 1887, obtaining his Captaincy in 1893. Amid other scientific pursuits of his corps he became expert in photography ; and his photographs of the solar spectrum at the eclipse of 1893 attracted attention. He had gone with Sir Ed. Thorpe to Fundium, in Senegal, W. Africa, to observe the eclipse of the sun of April 16 in that year, and taking two slit-spectroscopes he obtained successful photographs of the spectrum of the prominence and corona, the former showing the hydrogen lines up to H_ξ. In the following year, 1894, he communicated, in the *Memoirs* of the Society, an account of a method for obtaining longitudes in survey work from photographs of the moon. His interests in solar physics developed, and he went to Japan for the eclipse in 1896, but cloudy weather prevented observation. In India in 1898 he was more fortunate, and obtained excellent photographs of the "flash" spectrum demonstrating that the chromospheric spectrum was not an exact bright reversal of the ordinary Fraunhofer spectrum. For many years he was secretary of the Joint Permanent Eclipse Committee. His spectroscopic equipment was lent by him to observers of subsequent eclipses, and was afterwards presented to the Cambridge University Observatory.

In 1896 Hills was appointed Instructor in Chemistry and Photography at the School of Military Engineering at Chatham, and held the post until 1899, when he took charge of the Geographical Section of the General Staff at the War Office, continuing the work throughout the Boer War. Under his influence the staff of the Geographical Section of the Intelligence Department of the War Office was enlarged.

His other main scientific interest was the science of geodesy, both in theory and in the field. He organised and improved several State surveys ; in time he became associated with Sir David Gill in his promotion of the triangulation of South Africa and in the scheme of a Cape to Cairo meridian measurement. He reported officially on the Canadian surveys ; and devoted a Presidential Address to the section of Geography at the British Association in 1908 to a discussion of correlation of the surveys of the Empire. Arising from local connec-

tions, he took up the office of Honorary Director of the Observatory of the University of Durham, and devoted much time to the design and testing of a new type of suspended zenith telescope, on which he reported at length to the R.A.S. He had antiquarian tastes in scientific, especially astronomical, history. He became Brigadier-General during the war in 1918; during his career he received recognition in official honour lists for public service more than once. He was elected a Fellow of the Royal Society in 1911; from 1913 to 1915 he was President of the Royal Astronomical Society. He had been Treasurer of the Society from 1905 to 1913, and in the last year of his life, on the resignation of Mr. Knobel, he again took up the duties of the post at a critical time.

In 1901 the Governments of Chile and Argentina, finding themselves at variance about the boundary between their countries, had drifted to the verge of war, but eventually agreed to submit their differences to the arbitration of King Edward. A commission was appointed with Sir Thomas Holdich as President and Major Hills as Secretary. The commission reported recommending the watershed of the Cordilleras as the boundary, and the quarrel was settled to the satisfaction of both countries.

He carried through a considerable undertaking in theoretical geodynamics in conjunction with Sir J. Larmor, which was reported to the R.A.S. in a paper of date 1906 November, and a supplement in 1915 May. The very irregular track of the Earth's pole had been traced for many years at the central bureau at Potsdam from international observations round a circle of latitude; and it was felt to be time to try to find out something about the causes that were disturbing it. Free from disturbance it would describe an Eulerian circle, of constant period which for an elastic earth could be found only from the observations. Chandler, departing boldly from Euler's value 306 days for a rigid Earth, estimated it at 428 days. If the successive displaced positions of the pole are pricked on a sheet rotating with this same period, they will form on it a relative orbit in which the velocity is the measure of the couple or torque acting at that instant to disturb the Earth's motion. A graph of the succession of torques was thus constructed in the hope that it would show marked annual or other definite features; the result, however, seemed quite irregular, and has not yet been interpreted. Later, in 1918 May, the Astronomer Royal communicated to the R.A.S. a thorough formal analysis of all the main observations, conducted on the basis of separation into an Eulerian and an annual component, neglecting the residue, which proved to be feasible. The Eulerian period was corrected from Chandler's value 428 to 432 days. It was found that the Eulerian component was a circular motion, as it ought to be, but of amplitude which nearly doubled from 1904 to 1912. The other annual part proved to be fairly constant and nearly circular.

His grandfather, Sir W. R. Grove, acquired fame in his early days by his book on the *Correlation of the Physical Forces*, which arose out of lectures delivered at the Royal Institution; more substantially as an experimental physicist, Grove's wet battery maintaining its practical

pre-eminence until displaced by the dynamo; but most fundamentally of all, by his invention of the gas battery, which in later days became the clue to a great development in pure thermodynamics as applied to the electric and other properties of dilute solutions. Later he was known during many years for quiet legal and administrative power, and as one of the most efficient judges of the time. Colonel Hills, who recently prefixed an additional surname Grove, on becoming heir to his uncle, General Sir Coleridge Grove, inherited the family gifts of a widely informed interest in scientific questions and an admirable faculty of easy administration, from which scientific bodies, mainly the R.A.S., the Royal Society, and the Royal Institution, derived much advantage. It was in close accord with family tradition that, for the last years of his life, he filled the important office of Honorary Secretary of the Royal Institution.

In 1914 he had started for Russia to observe the eclipse in August of that year; but on the outbreak of the war he hurried back to England and was appointed Assistant Chief Engineer on the Eastern Command. He became Colonel in 1914, and reached the temporary rank of Brigadier General and was awarded the C.B.E.

After the war he resumed his scientific activities and had a photographic instrument constructed for the Durham Observatory, to be used for the study of variation of latitude, the position of the vertical being given by suspension of the instrument. He was engaged in trials of the new instrument at the time of his death.

In his second term of office as Treasurer of the Society he threw himself with zeal and energy into the task of improving its financial position at a time when, chiefly by reason of the great increase in the cost of printing, the Society's activities were in great danger of being seriously impaired. By his efforts not only was a considerable sum of money obtained by voluntary contributions, but also the number of Fellows was largely increased. The memory of his deep interest in the welfare of the Society is perpetuated by his legacy of the very valuable collection of old astronomical works and his provision of a considerable sum for the suitable installation of the collection in the rooms of the Society.

Hills married in 1892, Juliet, the daughter of the late James Spencer-Bell, M.P., of Fawe Park, Keswick, and had two sons and a daughter. Both his sons, just grown up, of high promise and great attraction, volunteered for the dangerous service of the Air Corps, then tentatively feeling its way; and both gave their lives for the national cause. No symptom of the distress which so heavy a bereavement must have produced was ever allowed, while the war lasted, to appear in his family or in public.

In May 1922 he attended the meeting of the International Unions of Astronomy and Geophysics at Rome. There he was attacked by what appeared to be a temporary indisposition; but as the summer passed it became evident that his illness was of a very serious nature, and he died on 1922 October 2, mourned by a wide circle of friends.

He was elected a Fellow of the Society on 1893 December 8.

The Rev. George Stewart Hitchcock was the son of the late George R. Hitchcock, Esq., of Blackrock, County Dublin. He was educated at Trinity College, Dublin, where he gained the distinction of being first Hebrew prizeman and Downes prizeman in preaching. He afterwards graduated at London University in 1892. Taking Holy Orders, he held curacies at Bridport, and S. Paul's, Chatham. On resigning the latter he was appointed chaplain under the Medway Board of Guardians. Two years later—in 1898—he became Unitarian minister in Chatham, but unable to find the rest he longed for, he joined the Roman Catholic Church, and became a member of the Society of Jesus. Carrying on his studies at Rome, he greatly distinguished himself as a Biblical scholar and writer. Later on difficulties and controversies arose, and at the latter part of 1914 he resumed his former post as minister to the Unitarian congregation at Chatham.

As a scholar he possessed a profound knowledge of Hebrew, Aramaic, and Syriac, and he wrote a number of standard books on Biblical criticism and exegesis. He was also a writer of poems and hymns. Amongst other activities, he made great use of his wide knowledge of literature and philosophy, and conducted, at Chatham, classes in these subjects, to which large numbers were attracted. During the three winters preceding his death he had been lecturing for the Kent Education Committee on Logic, Psychology, and Economics. He died in Dublin on 1922 July 10.

He was elected a Fellow of the Society on 1921 Jan. 14.

Carlton John Lambert was born at Finsbury on 1844 August 22. In 1859 he entered the City of London School under Dr. Mortimer, and going thence to Pembroke College, Cambridge, he took his degree in 1867 as third wrangler. In 1868 he was elected a Fellow of Pembroke, and became mathematical lecturer there in the following year.

In 1871 he was appointed assistant master at Wellington College under Dr. Benson, afterwards Archbishop of Canterbury. In the same year he married Elizabeth Brown Brown, daughter of Alexander Young Brown of Edinburgh. In 1873 he moved to Brockley, on his appointment as Professor of Mathematics at the Royal Naval College, Greenwich, and whilst there several men who subsequently became famous passed through his hands, amongst others Admirals Jellicoe, Beatty, Sturdee, and Madden. After thirty-six years at the College he retired, at the age of 65, to Buckhurst Lodge, Westerham Hill.

In his private life he carried out a great deal of philanthropic work in his leisure moments. He founded the Greenwich Provident Dispensary, in which he took a deep interest right up to the time of his death. He also started and partly endowed the Cudham District Nursing Association, which renders much needed service in a very scattered district. The endowment fund of the Association is now known as the Carlton John Lambert Memorial Fund. He also gave many public lectures on popular and instructive subjects, the best of which was, perhaps, his favourite lecture on the Moon. His great hobby was photography, and particularly stereoscopic and natural-colour work, to which he devoted years of patient work.

Amongst the prizes and honours won by him during his career were the following:—the Beaufoy Medal, City of London School; the Gold Medal, Queen's prize, 1st for mechanical physics, South Kensington; the Silver Medal, Queen's prize, 2nd for experimental physics; the Adkins Memorial Medal; and the Gold Medal of the Chinese Order Pao Sing, 2nd class.

In 1877 he presented to the Society the sum of £500, details of which will be found in the list of gifts to the General Fund.

He died after a short illness on 1921 November 6, and was buried in Keston Churchyard, Kent. He is survived by his wife and one son, Leslie Carlton Lambert.

He was elected a Fellow of the Society on 1870 April 8.

Sir Robert Pearce was a son of Joseph Pearce, a Borough Accountant of Ipswich. He was educated at the Ipswich Grammar School, and began his professional life as an accountant. In 1865 he was admitted a solicitor and practised in Ipswich until he came to London in 1872, and entered the firm of Baylis, Pearce & Co. He married in 1880 Elizabeth Deane, who died in 1910, and by whom he has one daughter. In 1914 he married Margaret Exton.

From 1906 to 1918 he represented the Leek Division of Staffordshire in Parliament, except for a few months in 1910 when he was defeated.

Sir Robert was largely responsible for the passing of the Daylight Saving Bill, being the first to introduce the late Mr. William Willett's scheme into Parliament. He was knighted in 1916 in recognition of the work he had done in Parliament to promote this measure. He was also much interested in the problem of Calendar Reform, and introduced a bill into Parliament dealing with this matter and including a proposal for the fixing of Easter.

Sir Robert was associated with Cripplegate for fifty years, and acted as vestry clerk and clerk to the Cripplegate Charities, of which he compiled an exhaustive history. He was also solicitor to the City Parochial Foundation, and was one of the principal movers in bringing the parochial charities under the management of a single board.

Sir Robert wrote many beautiful sonnets, and was at one time a keen mountaineer.

He died on 1922 Sept. 29, in his 83rd year. A memorial service was held at the Church of St. Giles, Cripplegate, which the Lord Mayor and Lady Mayoress attended in state.

He was elected a Fellow of the Society on 1877 Jan. 12.

Professor Charles Michie Smith was born on 1854 July 13, at Keig, Aberdeen, and was educated at Aberdeen and Edinburgh. After taking his B.Sc. degree in 1876 he was appointed the same year Professor of Physics at the Christian College in Madras. Fifteen years later he succeeded Pogson as Government Astronomer in Madras. Here he found a great deal to do in making up arrears in the reduction of observations, but in 1899 he was able to publish the *New Madras General Catalogue* of 5303 stars containing the results of meridian observations made under

Pogson's directions in the years 1862 to 1887. Amongst other purposes, this catalogue was intended to supply the places of stars between North Polar Distances 130° and 150° as standard stars for the zones of the southern survey in extension of Argelander's great work.

Michie Smith was also much interested in Meteors, and he published the record of those observed at Madras between 1861 and 1890. In 1899 when the expected 33-year Leonid shower failed, through the main swarm having been diverted by the disturbing action of Jupiter, he nevertheless with the help of an assistant recorded a considerable number of fine meteors at the recently established hill station at Kodaikánal. Eighty-five Leonids were observed between November 13 and 16, of which one was rated equal in brightness to Venus, and six others were estimated as being of the 1st magnitude. A number of meteors was recorded at the Madras Observatory at the same time.

In 1894 Michie Smith observed the annular eclipse of the sun, and at the total eclipse of 1898 he secured at Sahdol, in Rewa State, some particularly fine large-scale photographs taken with a 40-foot lens.

But, undoubtedly, one of the outstanding events of his career was the establishment in the face of certain difficulties of the observatory on the hills at Kodaikánal in 1899. Here systematic work was at once begun in seismology and meteorology, and as soon as the necessary instruments could be erected—in the study of sun-spots, solar photography, prominences, and sun-spot spectra. The spectrograph was ready for use at the end of 1900, and in 1904 a spectroheliograph was installed, consisting of a 12-inch triple achromatic lens by Cooke & Sons of 20 feet focal length, fed by a Foucault siderostat with an 18-inch mirror, and the spectroheliograph proper. The design of the latter was that of Professor Hale, in which the image of the sun and the camera remain stationary, while the collimating lens with the prisms and slits are moved across at right angles to the optical axis. Despite difficulties arising from the general unsteadiness of the solar image except during a comparatively short time in the morning, work of the highest value in the study of solar physics has been achieved at this observatory. From 1899 to the time of his retirement in 1911, when he was succeeded by Mr. John Evershed, Michie Smith held the office of Director of both the Kodaikánal and Madras Observatories.

A man of wide scientific knowledge, he had interests outside astronomy, and he communicated to the Royal Society of Edinburgh a number of important papers dealing with the determination of surface tension by the measurement of ripples, atmospheric electricity, various absorption spectra, and other matters.

He died in India on 1922 Sept. 27.

He was elected a Fellow of the Society on 1894 May 9.

Edward John Stutter was born in 1842 at Bury St. Edmunds. At the age of 12 he went to Subiaco, in Italy, where he completed his education and eventually became a Benedictine monk there in 1860. From Subiaco he was sent to their dependent house at Ramsgate, and

in 1878 he became affiliated to the English Congregation of Benedictines at Downside Abbey, Bath.

He served for a short time at Bath, Stanbrook, and Coventry, until he was appointed to Acton Burnell in 1889. Here as chaplain to the Smythe family—who had harboured the Downside monks when driven from France by the Revolution—he remained for thirty-one years. Failing health caused him to return to Downside in 1920. In 1922 October he had left Downside for a short holiday, and was taken ill in Birmingham, where he died on November 4, aged 80. He was buried at Downside on November 6.

From his earliest days at Subiaco, Father Stutter evinced a keen interest in every form of scientific study, but it was astronomy which gave him the greatest pleasure. At Acton Burnell he built himself a small observatory, containing a 6-inch equatorial—chiefly made by his own hands—and an excellent 3-inch transit instrument. He had reconstructed the sidereal clock which used to be in the ill-fated Downside Observatory, destroyed by fire, and from his sitting-room it actuated the chronograph in the observatory. Once he was offered a post at the Vatican Observatory, but could not see his way to accept the position.

He spent the greater part of his last two years in his workshop at Downside; and although in his eighty-first year he continued to make scientific instruments and to instruct some of his younger brethren who shared his interests. At the time of his death he was interested in radio-activity, and also contemplated the construction of a quadrant electrometer.

He was elected a Fellow of the Society 1901 May 10.

George Lyon Tupman was born at Boulogne on 1838 September 7, being the eldest son of Commander George Tupman. He was educated at the Royal Naval School, New Cross, and entered the Royal Marine Artillery as Second-Lieutenant on 1855 October 26. He became in succession First-Lieutenant (1859), Second-Captain (1867), Captain (1873), Major (1879), Lieut.-Col. on retirement in 1880. He served on *H.M.S. Sidon* and *H.M.S. Forte* off the coasts of Africa and South America (1859–1864) and on *H.M.S. Prince Consort* in the Mediterranean 1868–71.

He took an active interest in astronomy from his youth, and one of his earliest papers (*M.N.*, 33) was a catalogue of meteor radiants based mainly on his own observations in the Mediterranean. He was one of the first to make estimates of the velocity of meteors, and was thus enabled to disprove the suggested explanation of stationary radiants, as being due to extremely high velocities, far in excess of the parabolic value.

Probably his most important contribution to the science was his work in connection with the Transits of Venus in 1874, 1882. He was appointed instructor to the observing parties in a course of training which they undertook at Greenwich in 1873–4, and was the chief of the party that observed the 1874 transit from Honolulu. As there was then no cable to the Sandwich Islands, he made numerous lunar observations for longitude, and subsequently made a full discussion of

the errors of the lunar ephemeris, and the resulting longitudes of the different observing stations. On his return he consented, at the request of Sir George Airy, to remain at Greenwich as a voluntary worker in carrying out the reduction of the observations made by the various British parties. He completed this work, which occupied four years, *con amore*, and his preliminary value of the solar parallax, 8″·813, is quite close to the value now accepted; but he later increased it to about 8″·85. He recognised the drawbacks to the Transit Method, and I have heard him lament that the conditions of contact in the actual transit were so much less satisfactory than in the clockwork model used for training the observers. In the latter case the accordance between observers was all that could be desired.

He observed the transit of 1882 from Christchurch, New Zealand. Not long after his return he set up the Hillfoot Observatory at Harrow, which was equipped with a transit circle and two equatorials, reflector and refractor. He observed a large number of faint stars on the meridian, and as he was a most careful observer, it is to be desired that these should be published in catalogue form, which was not done during his lifetime. The equatorials were used for the observation of occultations, especially during the lunar eclipses of 1884, 1888, 1895, comets and minor planets. The results were communicated to the Society in a number of papers. He used the cross-bar micrometer for comets, and contributed a paper (*M.N.*, **48**, 96) on the method of adjusting and the correction of errors. He frequently put the instruments at the disposal of visitors, and was delighted to give them hints on methods of observation.

Tupman was on the Council from 1873 to 1880, and again as Secretary from 1884 to 1889. His last paper to the Society was on the occultations observed during the lunar eclipse of 1895 March.

Tupman's interests extended not only to astronomy, but to various kindred sciences; geology, microscopy, natural history, archæology, and wireless telegraphy. He had a wireless station in his house up to the outbreak of war in 1914.

He married on 1876 October 26, Miss Rebecca G. Wetherill of Philadelphia, who survived him for two months only. They had no children.

He died on 1922 November 3, at the age of 84.

He was elected a Fellow of the Society on 1863 May 8, and held his Fellowship for the unusually long period of 59½ years.

A. C. D. C.

Peter Hately Waddell, son of the Rev. P. Hately Waddell, LL.D., was born at Girvan, Ayrshire, on 1854 June 15. He was educated at the High School and the University of Glasgow, where he had a distinguished career. After being ordained a minister of the Church of Scotland, he was appointed in 1879 to the charge of Whitekirk Church, East Lothian. In 1904 ill health compelled him to resign and he went to live at Fairknowe, North Berwick, where he died on 1922 November 22. As an author he wrote several important works—many of them of a theological or religious nature—and a number of articles for various

publications. In 1901 the degree of D.D. was conferred on him by Glasgow University in recognition of the value of his writings.

Amongst his interests astronomy held a high place, and he not only gave lectures on it, but he erected a small observatory equipped with a reflecting telescope.

He was elected a Fellow of the Society on 1895 January 11.

Major-General James Waterhouse, who died at Eltham on 1922 Sept. 28, was born 1842 July 24. At the age of 17 he joined the Royal Bengal Artillery and spent a large part of his life in India. He became an expert in photography, and when he was 24 he was appointed Assistant Surveyor-General in charge of the photographic section of the Surveyor-General's Office in Calcutta, an appointment which he held for 31 years. Amongst his other work he extended Vogel's researches in sensitising plates for green and red light by the use of dyes, and he became recognised as the chief authority on photography under tropical conditions. He was also specially interested in the methods of photographic reproduction and introduced several improvements. On retiring in 1897 he took up his residence at Eltham, where he continued his researches and also renewed his interest in the old Daguerreotype process, communicating to the Royal Photographic Society an account of certain matters which were almost forgotten. He was President of that Society in 1905–7, and amongst other honours received its Progress Medal, and the Voigtlander Medal of the Vienna Photographic Society. He had, too, interests outside photography. He had been Hon. Sec. of the Calcutta Zoological Gardens, 1894-97, President of the Asiatic Society of Bengal, 1880–90, Trustee and twice Chairman of the Indian Museum at Calcutta.

As an astronomer he observed the solar eclipses of 1871 and 1875, and the transit of Venus in 1874.

He was elected a Fellow of the Society on 1876 Dec. 8.

Fritz Cohn was born on 1866 May 12 at Königsberg, being the second of the three sons of a merchant of that city. From 1872 to 1883 he studied at the Königsberg Gymnasium, where he gained a reputation for talent and diligence; the death of his father in 1877 placed the family in reduced circumstances, which Cohn did his best to improve. He studied in succession at the Universities of Königsberg and Berlin, taking the subjects mathematics, physics, astronomy, geography, and history. In 1888 he wrote a thesis for his degree on Lamé's Function with complex parameter. In July 1891 he went to Königsberg Observatory as a computer. Save for a year's leave of absence granted for further study at Leipzig, he remained at the observatory till October 1909, being promoted in 1898 to an assistantship. During this period he wrote many astronomical articles, among them a discussion of the meteorological records of Königsberg for forty-five years, a new reduction of some observations made there by Bessel between 1813 and 1819, and a study of the declinations and proper motions of the stars selected for the international latitude observations. He observed double stars with the Königsberg heliometer, and while at

Leipzig determined the latitude of the observatory with the Wanschaff universal instrument. He formed two catalogues of the right ascensions of stars from his observations with the Repsold meridian circle, which had a travelling-wire micrometer, to which he added motor-driving. One of these catalogues was of the reference stars used for the opposition of Eros 1900–1901; this was a sort of introduction to the work on minor planets which he subsequently carried out.

At this time he wrote two articles for the *Enzyklopädie der Mathematischen Wissenschaften*, "On the Theory of Astronomical Instruments for Measuring Angles, Methods of Observation and their Errors" and "Spherical Astronomy and the Reduction of Observations."

In 1905 he was appointed Extraordinary Professor at Königsberg, and in 1909 he was promoted to the Chair of Astronomy at the Friedrich-Wilhelm University, Berlin, and to the Directorate of the Rechen Institut, into the work of which he brought the greatest energy and skill. He attended the Conference of Directors of National Ephemerides held at Paris in 1911, which arranged for the partition of work among the different nations, to avoid needless repetition. This left time for his staff to devote themselves more especially to the minor planets. Cohn organised a plan of distribution, both of observational and computational work, and arranged for the computation of definitive orbits of the older planets, and of provisional orbits of the new ones, sufficiently good to secure their recovery at subsequent oppositions. The annual supplement of the *Berliner Jahrbuch* gave finding-ephemerides for all planets passing opposition during the year. He also made some very practical improvements in the methods of orbit-computation, and published a useful paper on the subject in the *Vierteljahrsschrift* for 1918.

After the deaths of Wislicenus and Berberich, Cohn undertook the publication of the *Astronomische Jahresbericht*, well known as a valuable index to the astronomical work of each year; the MS. of the 1921 volume was completed a few days before his death.

Cohn married in 1898 a daughter of C. F. W. Peters, who was then Director of Königsberg Observatory, and leaves a son and two daughters. J. Peters, who contributes an obituary notice to *Astr. Nachr.*, No. 5208, writes in warm terms of the affection and esteem felt towards Cohn by all his associates.

He died on 1922 December 14 after an operation, at the age of fifty-six.

He was elected an Associate of the Society on 1913 June 13.

Jacobus Cornelius Kapteyn was born at Barneveld, Holland, on 1851 January 19. His father had a well-known boarding school at that village, where he brought up a very large family, several of whom later occupied important scientific positions. The future astronomer entered the University of Utrecht in 1869, and there he specialised in physical science. It would appear that during his student days Kapteyn was not particularly attracted to astronomy, but in 1875 he was appointed to the observing staff of the Leiden Observatory, and at once settled down to purely astronomical work. The first problem he took

up was the structure of the universe, and to various methods of attacking this problem he devoted the whole of his life.

In 1878 Kapteyn was elected to the newly instituted chair of Astronomy and Theoretical Mechanics at Groningen. His duties included the giving of lectures on mathematics and theoretical dynamics, but his inaugural address on "The Parallax of the Fixed Stars" indicated his true inclinations. There was no observatory at Groningen, but Kapteyn hoped to get one erected soon. When he had settled down to his new duties and found the erection of a telescope still far off, he looked in other directions for active participation in astronomical pursuits. He obtained permission to use the transit instrument at Leiden during his vacations for the determination of stellar parallaxes by differential observations in right ascension. Observations were made during 4 periods in the years 1885 to 1887 and discussed in Vol. VII. of the *Annals of the Leiden Observatory*. The parallaxes of 15 stars were determined with a probable error of $\pm 0''.03$.

The application of photography to astronomy indicated to Kapteyn a new outlet for his energies. He freely offered to undertake the measurement and reduction of plates taken elsewhere. In this way Kapteyn founded the Astronomical Laboratory at Groningen. One of the needs of astronomy at this time was a catalogue of all the stars in the southern sky down to a magnitude comparable with that of the Bonn Durchmusterung for the northern. Sir David Gill had undertaken to do this work at the Cape Observatory, and the work of securing the photographs was commenced early in 1885, and finally completed in 1889. Towards the end of 1885 Kapteyn wrote to Gill asking for a few specimen plates, and offered to devote six or seven years of his life to the measurement of the whole series. The offer was gratefully accepted, and actually Kapteyn spent not six or seven but twelve or thirteen years of his life in the measurement of the plates. The measures are published in three large volumes of the *Annals of the Cape Observatory*, and give the positions of 454,875 stars between declination $-18°$ and the South Pole. The star density is nearly 33 per square degree, or about twice that of the Bonn Durchmusterung. For the carrying out of his share of the work Kapteyn secured a few small grants, which enabled him to get some unskilled assistance. At one stage of the work the prison authorities of Holland allowed certain convicts to assist in it. Yet a great deal of the drudgery of the work was done by Kapteyn himself, who made all the duplicate measures and examined all discordances between the plates and previously published catalogues. The plates were measured in a machine devised by Kapteyn, which gave at once equatorial co-ordinates. The catalogue gives the positions to $0^s.1$ in R.A. and $0'.1$ in Dec., and the magnitudes from an empirical formula derived separately for each plate. In the introduction to the third volume of the measures Gill wrote: "It is my colleague and friend whose name appears in the title-page, to whom I am under the deepest obligation. At a time of stress and discouragement he lifted from my shoulders a load of responsibility by his noble and spontaneous offer to undertake the measurement of the plates, the compilation of the results, and the formation of the catalogue. . . . I now

realise that my many other duties and the difficulty of obtaining adequate assistance would have compelled me to defer a great part of the work of the catalogue to the years of my retirement from official life, and might even have prevented its completion."

The measurements of the Cape Photographic Durchmusterung was carried out completely in two small rooms of the physiological laboratory of Groningen University in the years 1886 to 1896. In the latter year the Astronomical Laboratory was officially recognised and provisionally housed in a private dwelling. After a few years it was moved to another temporary shelter, and finally in 1913, when a new physiological laboratory had been built, it returned to take up a permanent home in the very building in which it had been instituted. At the beginning of the measurement of the Cape photographs there was only one computer; it has gradually been increased, and since 1899 the Director has had an assistant.

The Cape Photographic Durchmusterung did not exhaust the activity of Kapteyn during the years 1885–1899. We have already referred to his observational work at Leiden, and the reduction and discussion of these observations must have occupied a good deal of time. He still seemed anxious to explore the sidereal system by the trigonometric determination of parallax, and at the conference in connection with the Astrographic Catalogue in 1889 Kapteyn proposed to make a wholesale determination of parallaxes. His plan was to expose the plates at three consecutive maxima of parallactic displacement, the plates being kept undeveloped till all the exposures had been secured. At the middle epoch, when the displacement would be in the opposite direction to that at the two extreme epochs, double the number of exposures were to be made. A simple measurement of the relative positions of the different star images would give the relative parallaxes of all the stars on the plates. This scheme was not adopted for general use, but Kapteyn arranged with Professor Donner of Helsingfors to test it for several selected regions. The first results were published in 1900 in the Publications of the Astronomical Laboratory at Groningen, No. 1. "The parallax of 248 stars of the region round BD+35°, 4013." In this paper the many sources of error are carefully discussed, and in particular those depending on the different brightness of the different stars. These results constitute a great advance on previous photographic determinations, and pointed out the way to new advances. Kapteyn indeed did not surmount the difficulties of the guiding error, which cannot apparently be done without reducing the intensity of the photographic images of all the stars considered to approximate equality. The probable error given by Kapteyn for his results was about $\pm 0''{\cdot}035$ for 1 plate (12 exposures) or $\pm 0''{\cdot}020$ for 3 plates.

Publications of the Groningen Laboratory succeeded one another rapidly, and within ten years twenty important papers had been issued. They all deal with questions which arose in Kapteyn's efforts to extend our knowledge of the sidereal system. The direct determination of parallaxes was continued, the plates being taken for the most part by Donner at Helsingfors. One paper gave the parallaxes of 3650 stars, while other papers dealt with the parallax of the h and χ clusters in

Perseus and the parallax of the Hyades. Several papers give compilations of parallax determinations, while others give useful formulæ. Kapteyn now turned his attention to proper motions, to supplement the knowledge derived from parallaxes. He started by discussing the theory of the apparent proper motions, assuming that the peculiar motions of the stars had a random distribution; this is published in No. 5 of the *Groningen Publications*. It was to be followed by an application to the Bradley stars for which accurate proper motions were known. Two papers indeed give the components of the motion towards the solar apex and at right angles to this direction, but Kapteyn found that the proper motions would not fit the theory. The hypothesis of random peculiar motion was therefore not closely followed by the stars, and so Kapteyn was led to the discovery of two star streams, *i.e.* that the distribution of the peculiar motions of the stars is not random, but is preferential in two opposite directions in space. This result was first announced at the International Scientific Congress in St. Louis in 1904, but it is best known to British astronomers through the paper read at the meeting of the British Association in South Africa in 1905. The discovery resulted from a consideration of the motions of the separate stars and not the mean motion of all the stars in different regions of the sky. This phenomenon was most unexpected by astronomers, but when once pointed out it was soon found to be a most conspicuous feature of proper motions, and it has since been confirmed by many researches. No scheme of the sidereal universe can now be acceptable which disregards the existence of this peculiarity.

Kapteyn commenced his researches by the determination of trigonometric parallaxes, from these he turned to proper motions, and gradually he turned to more and more statistical methods to unfold the scheme of the stellar system. No. 8 of the *Groningen Publications* is on the "Mean Parallax of Stars of Determined Proper Motion and Magnitude," and No. 18 on "The Number of Stars of Determined Magnitude and Determined Galactic Latitude." More recently he has returned to this subject with improved data, and in collaboration with Van Rhijn he published in 1918 and 1920 two most important papers, Nos. 29 and 30, "The Secular Parallax of the Stars of Different Magnitudes, Galactic Latitude and Spectrum," and "The Number of Stars between Definite Limits of Proper Motion, Visual Magnitude and Galactic Latitude for each Spectral Class." The titles of these papers explain well their scope, and the tables summarise in a most useful way the best information we have. The data for these tables have been drawn from many sources with varying degrees of accuracy, which had to be carefully scrutinised for the elimination of systematic errors which might easily have crept into the results and completely vitiated them. With his long experience Kapteyn well knew the reliability of the different sources of information, and throughout his life was unceasing in his efforts to get the gaps in the information filled up, himself assisting where necessary in the reduction of the plates. We may here mention that the *Groningen Publications* contain the proper motions of over 10,000 stars determined by Kapteyn and his assistants from photographic plates taken at Helsingfors, The Cape, Bonn, and Potsdam Observatories.

Having obtained the best data he could Kapteyn proceeded to their interpretation, and the final conclusions he reached are given in three papers in the *Astrophysical Journal* (reprinted as Mount Wilson Contributions Nos. 188, 229, 230). Assuming that the Sun is near the centre of the galactic system, and that the system is symmetrical about an axis perpendicular to the galactic plate, he obtained, on neglecting absorption of light in space, the luminosity law (*i.e.* the relative number of stars of different absolute magnitudes, assumed everywhere the same) and the star density at different distances from the centre. The extent to which these results could be relied on with confidence was carefully considered. We may expect soon to have complete data for star counts to apparent magnitude 17, and then it should be possible to find fairly reliable densities to a distance 4000 parsecs in the galactic plane and 1600 parsecs towards the poles, or roughly throughout the sidereal universe where the star density exceeds 1 per cent. of its value in the neighbourhood of the Sun. Kapteyn's final paper is modestly described as "First Attempt at a Theory of the Arrangement and Motion of the Sidereal System." In it he discussed the distribution of masses, forces and velocities throughout the stellar system, assuming that the distribution of stars may be represented by a series of similar spheroids as equi-density surfaces. The total number of stars is taken as $47{\cdot}4 \times 10^9$, and with a random velocity of 10·3 km. per sec., Kapteyn found the mean mass of the stars to decrease from 2·2 to about 1·6 times that of the Sun as we pass from the neighbourhood of the solar system to the regions where the star density is reduced to 1 per cent. Kapteyn sought an explanation of the two star streams in a rotation of the galactic system, some stars moving in one direction and some in the other. He explicitly stated the defects of his theory, where his assumptions did not quite fit in with the facts or with his conclusions. These he hoped to remove in the future, but death has now interrupted his work on this problem, which has been almost continuous for the past fifty years.

It is not possible here to enumerate all the fields of investigation in which Kapteyn worked. In many of his papers he did not separate the stars into groups according to their spectral types, but this does not mean that he ignored this classification. In his early discussion of the proper motions of the Bradley stars he separated the stars according to Secchi's types, and in some of his later papers he followed the Harvard classification. He found a progression in the average velocities of the stars in passing through the spectral types from B to M, and in several papers he paid special attention to clusters of the B type stars, particularly in the galactic regions of the southern hemisphere. Kapteyn realised how necessary it was to extend our observational data before new advances into the actual theory of the stellar universe could be made. When work was pressed to fainter and still fainter stars he realised the impossibility of observing all the stars, and in order to obtain as much information as possible on representative stars he proposed the "Plan of Selected Areas." In this scheme intensive methods were to be applied to 206 areas distributed nearly uniformly over the whole sky, and 46 special areas mostly near the galaxy. A great

deal of work has already been done by a number of observatories, but much more will have to be done yet before the purpose of the scheme will be accomplished.

It will be seen from the above sketch that Kapteyn was one of the really great men in Astronomy. His mind was able to take the widest and most comprehensive view of the great problem presented by the stellar universe, yet he never shrank from any labour however tedious, if required to advance the solution. The problem is one which can only be solved if attention is paid to a multitude of details, for it frequently depends on minute quantities which would be hopelessly vitiated by the slightest systematic error. Knowing exactly where extreme accuracy of the data was most required, and where the data were most liable to error, he made his conclusions with what accuracy was warranted, and never allowed himself to overstep that limit. No figures were ever regarded as final, and the recurrence of the same titles in his publications shows how he was striving to improve his results as more accurate data became available.

Kapteyn's enthusiasm and his willingness to help others in the reduction of their observations early gained him many friends. He was fired with the same love of truth as his friend Gill, and when the latter retired from the Cape and came to reside in London, Kapteyn frequently visited him there. His friends were to be found wherever the science of astronomy was cultivated. From 1908 to 1914 he made an annnal visit to America, and spent a considerable time at Mount Wilson Observatory, of which he was a Research Associate. The war interrupted these visits, but Kapteyn was able to come to this country in 1921, when he attended the meeting of the British Association at Edinburgh. That was the last time he spoke to a British audience, and it was most fitting that he should speak on his theory of the arrangement of the sidereal system, which forms his last published paper.

For his contributions to our science Kapteyn was honoured throughout the world, and it is pleasing to record that his own country was not slow to honour him. We might here mention honorary degrees from the Cape of Good Hope, Harvard, and Edinburgh; the Gold Medal of our Society in 1902 for his share of the Cape Durchmusterung, the Bruce Medal of the Astronomical Society of the Pacific in 1913, and several French and Dutch Orders.

He married Catharina Elisabeth Karlshoven in 1879 and had a son and two daughters, one of whom is the wife of our associate Professor Hertzsprung of Leiden.

He died in Amsterdam on 1922 June 18.

He was elected an associate of the Society on 1892 November 11.

J. J.

William Henry Wesley.—In a Society like ours, scattered over England, and indeed in these days over the face of the world, in which there is a new President every two years and a new Secretary every half-dozen, the more permanent elements are of vital importance. It is scarcely too much to say that an Assistant Secretary has it in

his power to make or mar the success of our Society as such. If so, Mr. Wesley made it. For forty-seven years he identified himself with our interests so closely that it was scarcely possible to think of the Society without including him in the picture. It was the rarest experience to enter the rooms without finding him busily at work, but ready to respond to another call at a moment's notice. Failing health in recent years made him fear that he might have to give up his life-work, and that fear was perhaps a greater cloud than the illness itself. Fortunately he was spared that tribulation, and he passed away peacefully in his sleep on 1922 October 17, after a day in which he had been more active than his illness had recently allowed, and had chatted with his son till nearly midnight. He was buried on October 20 at Tooting Cemetery, where the President, Officers, and other Fellows of the Society said a final good-bye to their old friend.

William Henry Wesley was born on 1841 August 23 at the village of Stapenhill in Derbyshire on the banks of the Trent, immediately opposite Burton-on-Trent where his father carried on the business of a printer and publisher: and educated at the Grammar School at Ockbrook, near Derby. He learnt French and German privately. In 1855 his father came to London and apprenticed his son to Mr. Williams, a well-known engraver. Those of us who saw the artistic skill and devotion of his later years cannot doubt but that this career was a matter of choice rather than chance, and are not surprised to learn that he spent nearly all his spare time in the British Museum making drawings.

In 1862 he was introduced to Professor Huxley, for whom he made many drawings: and in the same year he became an active member of the Anthropological Society, following up his great interest in this science by constant attendance at the meetings of the British Association. His connection with these scientific bodies brought him many introductions—among others, one to Herbert Spencer with whom he formed a long and intimate friendship. He engraved the symbols in Spencer's account of his father's system of shorthand. His work for Professor Huxley was followed by much work for St. George Mivart, Professor Flower, Sir John Lubbock, Dr. J. Barnard Davis, and many other biologists. From the first he was a man of Science as well as of Art: he made a minute study of anatomy, with special reference to the skull, which had previously been only roughly drawn: the result was the meticulous accuracy shown in the numerous admirable drawings for Dr. J. Barnard Davis's *Thesaurus Craniorum*. Photographs and drawings made since are crude in comparison with Wesley's wood-engravings.

In 1869 he married the daughter of Mr. Robert Henson, the well-known mineralogist in the Strand, and he found in Mrs. Wesley not only a devoted wife, but a colleague who also identified herself closely with the Society's interests, and shared his work almost till the end. During the War, residence at Burlington House, round which bombs were dropped on many occasions, and which was itself struck at least once, was not attractive: but the Wesleys, though Mrs. Wesley was by that time a confirmed invalid, would never hear of deserting their post. She lived to see the Armistice, but died on 1919 May 15, a few months

after the celebration of the fiftieth anniversary of her wedding-day. There were several children of the marriage, and one son for some years assisted his father. The war interrupted this arrangement, and afterwards his uncle offered him a post in his business; but the Society retains a grateful memory of his services to them.

Soon after his marriage Mr. Wesley began drawing on stone for illustration of papers in the *Philosophical Transactions*, chiefly for Professor Richard Owen. Mr. Walter White, at that time Assistant Secretary of the Royal Society, was a lifelong friend of Mr. Wesley's father and introduced the son to Mr. A. Cowper Ranyard, who was at his wits' end to find an artist to draw the Solar Corona of 1871 from the photographs. It is well known to his family that at first this suggested work repelled Mr. Wesley, and he only undertook it after much gentle persuasion from Mr. Ranyard to make the attempt: but once having started he threw himself heartily into it with the conspicuous success we all know of. The drawings of not only this but many subsequent Coronas will bear permanent witness to his skill and his adaptability to entirely new work.

He then seriously began the study of landscape painting and figure painting in water-colours, and having determined to make art his profession, he joined other rising artists of the time in renting a studio to work in. But on 1874 December 3 Mr. J. Williams, the R.A.S. Assistant Secretary, died in the 78th year of his age. (He had originally been Secretary to the old Spitalfields Mathematical Society, and took office with us shortly after the amalgamation of that Society with our own.) Mr. Ranyard made up his mind that Mr. Wesley was the proper man to succeed him, though again he encountered a strong difference of opinion on Mr. Wesley's part. Indeed, so little concerned was the latter to change his intended career, that although he allowed himself to be put forward as a candidate, he did not wait to hear the decision of the Council, which was conveyed to him by post. The decision is recorded in the Council Report (*M.N.*, **35**, 165):—

> Several gentlemen well fitted for the office made application for the post, when, after the most careful consideration of the different qualifications of the candidates, the choice of the Council fell on Mr. W. H. Wesley, who will enter upon his duties soon after the Anniversary Meeting.

But as soon as the die was cast, Mr. Wesley, as in the case of the Coronal drawings, took up the work with devotion, though his first task was quite uncongenial to him. It was to bring into order the correspondence files and account books of the Society, which were in confusion. Mr. Williams had been accustomed to enter the Fellows' subscriptions at one end of a small ledger, at the other end of which he kept his own private accounts (expenditure on luncheon, etc.). Mr. Wesley had no clerical help at that time and faced the labour single-handed. After attaining the required orderliness he next tackled the library, and compiled a catalogue of it which, when printed, formed an 8vo volume of over 300 pages, every title in which was written by himself. It is rare to find a man of artistic ambitions able to turn to such laborious work

and to accomplish it with a bibliographic accuracy which has never been questioned. The moments he could call his own were devoted to his beloved art, or to making drawings for Professor Owen, who could find none but Mr. Wesley to give him satisfaction. When these things are brought before us we cannot but feel that though some of us knew, or thought we knew, what Mr. Wesley had done for us, we scarcely knew what he sacrificed for us. It is satisfactory to remember that his great services were explicitly recognised by the Society, or by its Fellows in their unofficial capacity on more than one occasion during his later years. It is satisfactory also to reflect that his connection with Astronomy enabled him to use the Art he loved to do real service to Science. His beautiful pictures of the Corona have already been mentioned. The work of constructing them gave him an intimate knowledge of that wonderful structure, as shown on photographs, but opinion was current at the end of the nineteenth century that much more could be seen with the eye. Fortunately, an opportunity was provided for Mr. Wesley to test this point. The Total Eclipse of 1900 May 28 passed directly over the Algiers Observatory, and M. Trépied, the Director, with fine international courtesy, put at Mr. Wesley's disposal the great equatorial coudé of the observatory. The weather was beautifully clear, and a unique combination of skilled observer, big telescope, and good weather, gave a clear answer in the sense that the photographs had been libelled. From Mr. Wesley's full account in the *Observatory* for 1900 September, we may extract a couple of sentences:—

> [I] now maintain that good photographs, taken on a sufficiently large scale, are capable of showing all the details which can be distinguished by the eye at the telescope. I make this statement the more readily as it disagrees with my preconceived ideas.

The weight of this verdict is such that it is not likely to be upset in the near future.

Mr. Wesley also engraved Dr. Doeddicker's charts of the Milky Way, published in 1892, which Mr. E. B. Knobel (*Observatory*, 1922 November, p. 355) considers perhaps his most important work for astronomy. It is difficult to choose. "In these charts he had to plot every star by careful measurement from the positions given by Argelander. The result is perhaps the most beautifully executed star charts we possess." But the map of the inner portions of the Moon's surface, which occupied all his spare time, including holidays, from 1911 to 1915, has very great claims on our respect. Here the fundamental positions were plotted by Saunder (*M.N.*, 73, 216), but the drawing of detail from the photographs so as to represent mean libration, required all Wesley's skill and accuracy. The four portions which he undertook were completed, but only with something of a race against the threat of cataract to the eyes. The outer portions were to have been drawn by Franz, but he and Saunder died in 1912. They have since been drawn, with a success which surprised and delighted Wesley, by Miss Blagg, so that the whole map is now complete: but since its main purpose is to act as a vehicle

for names adopted by international agreement, the issue of it is hardly possible just at present.

But all Wesley's skill and accuracy and toil for the Society could not have endeared him to us without his very human qualities. He loved a joke. Miss Common, to whom the Society is deeply indebted for that help in recent years, without which Wesley might not have been able to remain in office, has kindly told me of the double pleasure it gave her to share *Punch* with him: and this same sympathetic appreciation gave others a twofold enjoyment of pictures and other works of art. One of the privileges which he greatly valued, and which he owed to his residence in Burlington House, was that of admission to the private view of the Royal Academy. His catalogue, marked with his notes, was always at the service of any Fellow paying a subsequent visit, and often started a good talk about artistic matters.

The question of retirement affects so many interests and occupations that a word or two more may be ventured on the lessons to be learnt from the instance before us. The relation of a man to his work may, with advancing years, take one or other of two turns, for which we are liable to use nearly the same word in two very different senses: it may become an increasing care to him, or it may be all he cares for. The former alternative points to retirement, the latter, away from it. It is no small cause for satisfaction to our Society, that our last two Assistant Secretaries have clung so devotedly to their office. Of Mr. John Williams it is recorded (*M.N.*, 35, 183) that "He could not be persuaded to resign his post of Assistant Secretary, even when he might well have done so": and in reading the account of his death we are reminded in several ways of our recent loss. If Wesley was reluctant to enter upon his life-work, he was more than reluctant to leave it, except in the inevitable way. H. H. T.

Proceedings of Observatories.

The following reports have been received from the Directors of the several Observatories, who are alone responsible for the same. Except where otherwise stated the reports refer to the two years 1921 and 1922.

Royal Observatory, Greenwich.
(*Director, Sir Frank Dyson, Astronomer Royal.*)

Transit Circle.—The number of transits observed in the years 1921 and 1922 were 8087 and 9633 respectively. The sun was observed 158 times in 1921 and 126 times in 1922. The moon was observed 115 times in 1921 and 105 times in 1922. The observation of Backlund-Hough stars was concluded at the end of 1921, and a catalogue is in the course of preparation. The stars at present under observation are those as bright as $7^{m}\cdot9$ in the Catalogues of the Astronomische Gesellschaft which lie between Dec. $+32^{\circ}$ and $+64^{\circ}$. These limits were chosen because the zone $+24^{\circ}$ to $+32^{\circ}$ and the region from $+64^{\circ}$ to the North Pole have been observed at Greenwich within the last 25 years. In all, the Working Catalogue contains 10,247 stars. This includes a considerable number of stars fainter than $7^{m}\cdot9$, which were inserted in sparse regions of the sky in order to give a uniform distribution of reference stars. From the beginning of 1922 azimuth error has been determined from observation with the travelling-wire micrometer of 70 stars whose polar distances vary from 13° to 45°, as well as from close circumpolars. As a large number of these stars are sufficiently bright to be observed in the daytime, the number of double transits has considerably increased.

Altazimuth.—In 1921 extra-meridian observations of the moon were obtained on 21 nights, and in 1922 on 4 nights.

Tabular places of the moon from 1902 to 1922 derived from Hansen's tables have been corrected for certain periodic terms in the same way as those from 1851 to 1901 had been previously corrected by Dr. Cowell. Corrections have also been applied to the observations to reduce them all to Newcomb's system. Following the lines of Cowell, Fotheringham, and Brown, in applying what appear to be well-established corrections to the differences between Hansen's tables and the observations, it is proposed to publish in a small volume the corrected differences between the tabular and observed places for the period 1851–1922, during which Hansen's Tables and the Transit Circle observations can be compared.

Cookson Floating Telescope.—In 1921 and 1922 the number of plates taken were 224 and 222 respectively. The plates are all measured; the variation of latitude at Greenwich for 1921 was published in *Monthly Notices* for 1922 March, and that for the year 1922 will be ready for publication next month.

Occultations.—Eight occultations of stars were observed in 1921. Details of the observation of the solar eclipse of 1921 April 7, and of

the occultation of Venus on 1921 July 1, have been communicated to the Society. In 1922 both phases of the occultations of Aldebaran on August 16 and November 6 were observed.

28-inch Refractor.—A catalogue of double stars from observations made at Greenwich during the years 1893–1919 has been published. The volume contains determinations of new orbits, notes on many double stars which show considerable motion but whose orbits cannot yet be determined, and the determination of hypothetical parallaxes.

A new working catalogue has been prepared of stars suitable for a large telescope of which observations are needed. 494 observations of 265 double stars were made in 1921, and 628 observations of 326 double stars in 1922. They are distributed as follows:—

Separation.	1921.	1922.
Under 0″·5	70	53
0″·5–1″·0	80	100
1″·0–2″·0	66	90
Over 2″·0	49	83

Thompson Equatorial.—The principal work has consisted of parallax determinations from photographs with the 26-inch refractor. Subsidiary work has been the determination of effective wave-lengths of stars in the Greenwich Astrographic Zone by means of a wire grating in front of the 30-inch reflector. A few experimental spectra have been taken with a prism crossed by a grating. The prism is mounted behind the 30-inch mirror used in the Cassegrain form while the grating covers the mouth of the telescope tube. By this arrangement a parallel beam passes to the prism and is condensed by a 6-inch lens of 28-inch focus on the photographic plate. This gives, at a single exposure, a principal spectrum bordered by diffraction spectra of known relative intensities.

The records of plates taken and measured for the two years are as follows:—

	1921.	1922.
Taken for parallax	951	1183
Measured for parallax	918	1060
Taken for effective wave-length	98	12
Measured for effective wave-length	84	0
Taken for spectra	2	17
Taken for positions of comets	11	3
Measured for position of comets	11	3

Communications have been made to the Society of the determinations of 107 parallaxes and of the effective wave-lengths of those stars contained in the Draper Catalogue within 3° of the North Pole.

Astrographic Equatorial.—The instrument was dismounted in July 1921 for alterations and adjustment previous to its being sent to Christmas Island on 1922 January 28. It was returned to Greenwich on 1922 December 4, and was remounted and ready for use on December 31. In the first half of 1921, 95 photographs of stars in the North Polar Sequence were obtained. These were measured and the

results communicated to the Society. The photographic magnitudes are in close accordance with the results obtained at Mount Wilson.

It is proposed to retake a number of the fields of the Astrographic Catalogue through the glass, and compare with plates taken from twenty-three to thirty years ago, for the determination of proper motions of stars.

Photoheliograph.—Photographs of the sun were obtained on 248 days in the year 1921 and on 225 days in 1922. At the Cape Observatory photographs were obtained on 331 days in 1921 and on 336 days in 1922. The combined record for 1921 leaves 11 days unrepresented, for which, however, plates are available at Kodaikánal. In the year 1922 there are 11 days not represented at Greenwich and the Cape, but photographs on 10 of these days were secured at Kodaikánal.

During the last two years effective progress has been made with the arrears of work. The combined series of photographs from 1919 June 28 to 1921 November 1 have been measured, and the computations have been completed from 1917 October 3 to 1921 June 26. The copy for press of the Photoheliographic Results for 1920 is nearing completion.

A preliminary reduction of 36 photographs distributed equally throughout 1921 and 1922 gives the approximate mean daily area (corrected for foreshortening) in millionths of the Sun's visible hemisphere as 350 and 250 respectively.

The progress of the present sun-spot cycle is represented by the following table:—

Year.	1913.	1914.	1915.	1916.	1917.	1918.	1919.	1920.	1921.	1922.
Mean Daily Area	7	152	697	724	1537	1118	1052	700	350	250

The values given for years after 1919 are subject to revision.

Wireless Time-Signals.—Signals from Paris, Bordeaux, Nauen, and Annapolis have been received regularly, and occasionally also signals from Lyons and Moscow. In particular, during the early summer of 1921 special series of rhythmic signals from Lyons and Bordeaux sent out for Australian longitudes were observed.

The European signals are generally received automatically on a syphon recorder operated by means of an oscillating valve relay system. After many trials considerable success has been achieved in recording the Annapolis signals in a similar manner, and they have been so recorded on 50 per cent. of the days since the middle of 1922.

Communications with reference to the signals were made to the Society in 1922 January and 1922 December.

Solar Eclipse.—Owing to cloudy weather no results were obtained at Christmas Island at the Eclipse of 1922 September 21. A report on the expedition will be furnished by the observers to the *Joint Permanent Eclipse Committee of the Royal and Royal Astronomical Societies.* It is yet uncertain whether an expedition will be made from Greenwich to observe the Eclipse of 1923 September 10.

Printing.—The Greenwich Observations for 1915, 1916, and 1917 have been distributed. The volume for 1918 is in the binder's hands, and the printing of the volume for 1919 is approaching completion. In

addition to the regular observations the following publications have been distributed:—

Catalogue of Stars for 1910.
Vol. IV. of Astrographic Catalogue.
Variation of Latitude 1911–1918.
Double-Star Observations 1893–1919.

Staff.—Mr. L. S. T. Symms was appointed Junior Assistant 1921 October 24.

Royal Observatory, Edinburgh.

Work has continued upon the same lines in 1922 as in previous years. Close attention has been given to the determination of time. A paper was published (*M.N.*, 1922 Jan.), showing the large semi-systematic discrepancies between the determinations at different observatories. By perfecting the clocks and chronographic connections these are definitely located in the Transit Circle. Two new clocks have been set up during the year: the first, a clock by Mr. W. H. Shortt, which is very nearly a free pendulum, receiving an impulse every 30 seconds only, and oscillating in a high vacuum under constant temperature; and the second, a clock, Riefler, No. 450, purchased for the Survey of India, and entrusted to the Royal Observatory for test. The tests show an unquestionable superiority in Mr. Shortt's clock over any of the others, including our standard clock, Riefler, No. 258. The microchronograph has been in daily use to give photographic records of the relative errors of these clocks reliable to $^{s}\cdot001$, with very definite results.

The printing of the sections of the Astrographic Chart taken at the Observatory of Perth, W.A., which were measured at Edinburgh, having been undertaken by that Observatory, the MSS. of the first 6^{h} of Zone $-40°$ have been handed over. Other sections are ready and will be despatched to Australia at an early date.

Observations of the solar rotation have been made as heretofore.

The Koch photo-electric spectro-photometer which has been under construction has been completed and reduced to regular working order. A series of stellar spectra obtained with the photo-visual telescope and objective prism have been examined with it, and the results discussed for their bearing upon the estimation of the continuous spectrum as recorded photographically, with special reference to stellar temperatures (*M.N.*, 1923 Jan.). The instrument is highly satisfactory, and can be applied to numerous problems.

Cambridge Observatory.
(*Director, Professor A. S. Eddington, F.R.S.*)

The preparation of the Zodiacal Catalogue from observations made in the years 1900–18 has been continued. The magnitude equations of the observers have been determined by a comparison with the Cape Zodiacal Catalogue.

Further determinations of proper motion have been made by the

method of superposition of two plates placed film to film in the measuring machine, one of the pair of plates having been photographed through the glass. The old plates available are mainly those taken by A. R. Hinks and H. N. Russell in the years 1903–5 for the determination of stellar parallax. Plates for comparison with these have now been photographed through the glass, usually two or three exposures of twenty minutes being given. Fifteen regions have already been measured and reduced, and the investigation of other regions is proceeding. Attention has been given to obtaining plates in readiness for future comparisons. The Kapteyn selected areas (Declination 0° to +60°) and other miscellaneous regions are being photographed. 250 plates have been secured as follows:—Kapteyn areas, 45 regions (126 plates); miscellaneous fields, 21 regions (56 plates); "through glass," 22 regions (68 plates).

The attempt to determine the change of effective wave-length of δ Cephei during its light-variations (mentioned in the last report) was not very satisfactory. Difficulties arose through the rapid change of focal length of the triple objective of the Sheepshanks telescope and other causes, which have not been successfully overcome.

The work done by research students in connection with the observatory includes Mr. L. J. Comrie's predictions of occultations and satellite phenomena. Miss Payne has determined the proper motions of stars in the neighbourhood of N.G.C. 1960 and of the cluster itself. Miss Douglas, in conjunction with the Director, has made a statistical investigation of the progression of velocity with absolute brightness of stars, which has been communicated to the Society.

Solar Physics Observatory, Cambridge University.
(*Director, Professor H. F. Newall, F.R.S.*)

Spectroheliograph.—Photographs of the sun's disc in K_{232} light were obtained on 112 days in 1921, and on 94 days in 1922; and photographs of the prominences at the sun's limb were obtained on 100 days in 1921, and on 81 days in 1922.

The solar activity in 1922 was very considerably less than that in 1921.

For the year 1920 October 1–1921 September 30, the Director of the Kodaikánal Observatory has sent 330 spectroheliograms showing the sun's disc in K_{232} light; and for the year 1921 October 1–1922 September 30, 326 spectroheliograms.

The examination of the Kodaikánal photographs has been continued, especially in the study of systematic distribution of the masses of bright calcium flocculi. Of 2325 flocculi examined by Mr. Butler, 1252 were found to be elongated in form rather than circular or irregular, and 679 exhibited this elongation with an inclination directed equatorward to the west in both hemispheres. The range of inclination is greater than that for the axes of elongated spot-groups, in many cases being as much as 40°. The phenomenon is sustained throughout periods of small and of large solar activity, during the years 1915–1920, with the maximum activity in 1917 (*M.N.*, 82, 334).

Study of Sun-spot Records.—The study of the proper motions of sun-spots has been continued by the Director and Mrs. Beech.

Solar Prominences.—Copies of the manuscript records of the daily observations of prominences made at Kodaikánal during the years 1912–1922 have been received from Mr. Royds and are being systematically examined by Mr. Moss.

Solar Rotation.—A modification of the usual procedure in determining the solar rotation by spectrographic observations has been devised by the Director, who with the assistance of Mr. R. Sekiguchi, of the Imperial Marine Observatory of Japan, has secured a large number of observations during the summer months of 1922 (*M.N.*, 81, 115).

Spectroscopic observations, made by various observers for many years past, have shown that the absorbing vapours round the sun are rotating round the sun's axis with angular velocities which depend upon the latitude. The implications of the law which is generally accepted for the variation of angular velocity with latitude have been examined (*M.N.*, 81, 101).

Stellar Spectra.—The spectrum of the variable star β Lyræ has been studied by Mr. Baxandall. Three series of plates have been measured by him: (1) a series obtained at S. Kensington, (2) a series obtained at Cambridge, and (3) a long series kindly lent by the Director of the Allegheny Observatory.

The manuscript of a memoir on the spectrum of Nova Aquilæ III. has been revised and enlarged, and the reproductions of photographs of typical phases of spectrum have been made from the photographs sent from the Astrophysical Observatory, Victoria, B.C. The work has been in the charge of Mr. Baxandall, who has also compiled a revised list of spectral lines of unidentified origin in various celestial spectra.

Since the last report (1921 February) Mr. Milne has completed several papers dealing with certain problems connected with the theory of radiative equilibrium of stars. Dealing first with the distribution of temperature near the boundary of a star, he has found an integral equation, in virtue of which the law of darkening of a stellar disc from centre to limb may be deduced (*M.N.*, 81, 361). Secondly, from the temperature distribution, the distribution of energy in the continuous spectrum can be calculated if the coefficient of absorption is the same for all wave-lengths (the case of "gray material"), and in this case it appears that the position of maximum energy in the spectrum is displaced slightly to the violet side of the maximum for a black body of the same effective temperature (*M.N.*, 81, 375). Thirdly, the effect of a strong absorption line is considered, and it is shown that discontinuities in the coefficient of absorption are accompanied by a decrease in the boundary temperature and by changes in the energy curve, which depend on the relative positions (in respect of wave-length) of the selective absorption and of the maximum energy in the spectrum (*M.N.*, 81, 510). In a fourth paper Mr. Milne deals with the theoretical distribution of temperature in the parts of a stellar atmosphere outside the photosphere (*M.N.*, 82, 368). A memoir published in *R.S. Ph. Trs.*, A, 223, 201, with the title "Radiative Equilibrium: the relation between the spectral energy curve of a star and the law of darkening of the disc towards the

limb, with special reference to the effects of scattering and the solar spectrum," completes the investigation with regard to absorption and extends it to include scattering of radiation.

Atmospheric Electricity.—Observations and experiments relating to the electrical phenomena of thunderstorms and showers have been continued by Mr. C. T. R. Wilson, who has obtained during the summer months many records of the changes produced in the electric field by lightning flashes. Information gained in these and previous investigations at the Observatory has been utilised in an article on "Atmospheric Electricity" contributed by Mr. Wilson to Sir Richard Glazebrook's *Dictionary of Applied Physics.*

Work on Ionising Radiations.—During 1921 many improvements were made in the details of the method of securing stereoscopic photographs of the tracks of ionising particles. From 1921 November to 1922 August the apparatus was in constant use, and several hundred pairs of pictures of the tracks of α and β particles were obtained. The action of X-rays has been specially studied. A paper containing some of the results obtained with α-rays has been communicated to the Cambridge Philosophical Society. The main work on X-rays and β-rays is nearly completed for publication.

Dunsink Observatory.
(*Acting Director, Mr. Charles Martin.*)

The Roberts reflector, during the year 1921, was used in connection with variable star work. Light curves of RR Leonis, Y Aurigæ, and UU Cassiop. have been determined, and the results communicated to the Society.

The time-service to Dublin has been continued as in previous years, the meridian circle, in conjunction with the Paris signals, being used to determine mean time.

The Observatory has suffered a great loss in the resignation of Professor H. C. Plummer, F.R.S., who was appointed Director of the Observatory in 1912, and who left in 1921 June to take up a mathematical post in England. The Professorship of Astronomy in the University of Dublin has been suspended.

The ordinary routine work of the Observatory has been carried on as usual during 1922. Students are given a day each week for instrumental instruction, and are welcome for night work.

The observatory instruments are not in good repair.

A large sum has been expended in repairing the observatory buildings and walls; this work is not yet completed.

Durham University Observatory.
(*Hon. Director, Mr. Harold Thomson.*)

Since the last report the observatory has sustained a heavy loss by the death of the Honorary Director, Col. E. H. Grove-Hills, F.R.S.

At the present time the principal work of the observatory is in

connection with the variable stars for so many years observed by the late Mr. C. Grover at Rousdon Observatory, Dorset.

By the kind permission of Sir Wilfred Peek, the telescope employed by Mr. C. Grover was transferred to the Durham Observatory during 1921, and observations of the variable stars contained in the Rousdon programme were commenced during September of that year by Mr. F. Sargent. Since that time upwards of 700 observations have been made on these stars.

Experimental photographs have also been taken with the Zenith telescope. The results obtained are not yet satisfactory, and an investigation is proceeding into the causes of the errors found.

Mr. Sargent has maintained his series of observations on the planetary surfaces, and has recorded interesting abnormal surface movements on Jupiter which have been reported to the Society.

Systematic observations have been made of Venus and the position angle of the cusps.

During the recent inferior conjunction the planet was seen as a complete circle of light.

The usual meteorological observations have been maintained and transmitted to the Meteorological Office.

The Cathedral clock has been regulated from the observatory as hitherto.

Liverpool Observatory. (*Mr. W. E. Plummer.*)

Avoiding all matters of routine, the most interesting feature in the conduct of the Observatory for the past year has been the negotiations, inaugurated at the instance of the Tidal Committee, with the object of transferring the interests and conduct of the Institution from the control of the Mersey Docks and Harbour Board, who have carried on the Observatory since 1857 to the present time, to a joint Board, created *ad hoc.*

As stated in the preamble of the agreement, not yet fully ratified, the object is "to obtain by co-operation between the Board and the University, the fullest commercial, scientific, and educational value from the Liverpool Observatory, and from the property and funds placed at the disposal of the Committee."

In carrying out this intention it seems probable that the astronomical observatory will disappear, to be replaced by one of a more geophysical character, in which enlarged opportunities for the pursuit of meteorology, seismometry, geodesy, magnetism, and tidal investigation will be provided.

Norman Lockyer Observatory, Salcombe Hill, Sidmouth.
(*Director, Major W. J. S. Lockyer, M.A., Ph.D.*)

The usual routine of photographing the spectra of stars with the 12-inch McClean prismatic camera has been continued, and numerous spectra of special interest have also been secured with the larger dispersion of the 9-inch Kensington instrument. Successful night

work has been carried out on 120 nights, and 906 negatives have been obtained during the year.

The determination of parallaxes by the spectroscopic method has now become an important part of the routine work of the observatory, since it was found that the large number of stellar spectra already photographed was suitable for this research. The smallness of the staff necessitated some outside assistance for this undertaking, and an application to the advisory council of the Scientific and Industrial Research Department for a temporary research assistant met with their approval. Mr. W. B. Rimmer, who had worked at the observatory during the summer of 1920, was appointed, and he commenced this research in 1921 July. The investigation was restricted to the determination of the spectroscopic parallaxes of stars of types F_0 to M. At the present time the parallaxes of 500 stars have been completely reduced, and a paper summing up the results is well in hand.

An attempt has been made to determine also the spectroscopic parallaxes of stars of type B, and a preliminary paper on the subject has already been communicated to the Society.

All the buildings, instruments, and apparatus are in a satisfactory condition, and several valuable additions have been made to the observatory's equipment and library. Among these may be mentioned a sidereal clock by Frodsham, presented by Lieut.-Col. F. K. McClean, and two large handsome bookcases presented by Lady Lockyer.

On 1922 July 22, a portrait medallion of the late director, Sir Norman Lockyer, erected in the observatory by his relatives and friends, was unveiled by the Astronomer Royal, Sir Frank Dyson, F.R.S., in the presence of a distinguished company of guests. A full account of the proceedings has been published in *Nature* for 1922 August 5.

The Director attended the Rome meeting of the International Astronomical Union in May, as a member of the commission on stellar parallaxes. At the conclusion of the meeting he was also elected a member of the commission on the spectral classification of the stars.

The observatory is thrown open to the public every Wednesday afternoon from 2.30 to 4.0 P.M., and the number of visitors who attended during the past year was 366.

The annual general meeting of the Norman Lockyer Observatory Corporation took place on June 12.

Radcliffe Observatory, Oxford.
(*Director, Dr. A. A. Rambaut, F.R.S., Radcliffe Observer.*)

The programme of work with the 24-inch photographic refractor on Kapteyn's "Selected Areas" has been continued; twenty-four plates were taken on Kapteyn's original plan and twenty control plates; the latter have been taken with the film at the back for comparison with those obtained and developed ten years previously. Other plates include seventeen taken at the suggestion of the late Professor Kapteyn for determination of proper motions of all stars contained in the Radcliffe determinations of stellar parallaxes (the measurement of these plates has been undertaken by Professor P. J. van Rhijn), fifteen plates during

the annular eclipse of the Sun on 1921 April 7, and a photograph of the region of Nova Cassiopeiæ (1572) taken on 1922 September 15 for Dr. Steavenson.

Visual observations were made with the 18-inch refractor on 1921 February 21 for altitude and azimuth adjustment of the polar axis on which this and the 24-inch are mounted. The adjustments were found to be still very satisfactory. The 18-inch was also used in observations made during the passage of the earth through the plane of the ring-system of Saturn on 1921 February 21 and 22.

The Barclay 10-inch was employed on observations of Denning's Nova Cygni until 1921 November 24. This instrument has also been requisitioned for visitors on open nights.

Telescopic and naked-eye observations of the annular eclipse of the Sun of 1921 April 7 (partial at Oxford) were secured under favourable conditions.

Preparations have been made for the measurement of the plates of the "Selected Areas." All forms required for the work have been printed ready for use.

The quinquennial volume, lii. (Meteorological), has been completed and distributed. Part 1 of vol. liv. has also been printed, and a limited number of copies distributed in the usual manner. Vol. liii., an astronomical volume, is in the press. It is entitled *Radcliffe Determinations of Stellar Parallax.*

University Observatory, Oxford.
(Director, Professor H. H. Turner, F.R.S.)

Volume VI, zone $+59°$ of the Vatican Catalogue is completely printed, and will be distributed shortly; the reductions for the remaining zones $+58°$ to $55°$ are all completed. Mr. Hough has provided plates for the whole of zone $-22°$, which are being measured as opportunity offers, meanwhile Hyderabad Observatory has completely measured this zone, so that possible systematic differences between plates taken in the two hemispheres can be checked.

During the past year considerable attention has been paid to seismological reductions; the results for 1917 have been printed and distributed, and incidentally a method was suggested for estimating the focal depth of a shock from the time of arrival of waves on the opposite side of the earth; an apparent periodicity of 21 minutes in the frequency of earthquakes has been investigated, and another of 4 years is under scrutiny.

The Coats Observatory, Paisley.
(Principal McLean, B.Sc., C.E., Convener.)

The number of visitors to the observatory in 1922 was 304.

The Milne seismograph recorded 21 earthquakes.

The Committee record with profound regret the death of Sir Thomas Glen-Coats, Bart., an enlightened patron and generous benefactor of the Observatory, which was founded by his father and extended by his brother, Mr. James Coats, jun.

Imperial College, South Kensington.
(*Professor A. Fowler, F.R.S.*)

Apart from occasional spectroscopic observations of spots and prominences, the astronomical instruments have been used exclusively for the instruction of students.

Work in the spectroscopic laboratory has been continued in various directions. Dr. M. A. Catalán, of Madrid, succeeded in extending our knowledge of the series spectra of manganese, and thence deduced that the ionisation potential of this element cannot be far from 7·4 volts (*Phil. Trans.*, A, 223, 127). Among other investigations, Dr. S. Datta, of Calcutta, obtained results of considerable importance relating to the absorption of potassium vapour. Mr. H. Barrall has been occupied with precise determinations of the wave-lengths of silicon lines, which are used in the determination of the radial velocities of early-type stars. Work with the vacuum spectrograph has progressed satisfactorily, and details of the series spectra of silicon at successive stages of ionisation are almost ready for publication.

A "Report on Series in Line Spectra," prepared by Professor Fowler, has been published by the Physical Society. A considerable amount of time was occupied with work connected with the Rome meeting of the International Astronomical Union, and in preparing for publication the first volume of the *Transactions* of that body.

Stonyhurst College Observatory.
(*Director, Rev. A. L. Cortie, S.J.*)

The magnetic, meteorological, and seismological observations have been carried on as usual, and regular returns have been communicated to De Bilt, to the Meteorological Office, and to the Seismological Committee.

Observations of the Sun's surface in 1922 were made on 256 days, and include 248 drawings. Of these, 220 are complete and show all spots and faculæ, the remaining 28 are complete for the spots, but not for the faculæ. The mean daily disc-area of the spots (in units 1/5000 of the visible disc), stands at 1·73, a considerable decline from 3·14 for 1921, and 4·05 for 1920. The mean daily ranges in terrestrial magnetic horizontal direction and force, accompanying this decline in spot-area, were greater than in 1921.

There were two large spot-groups: that of February 27 to March 8, with a disc-area of 21·5 units, the centres of the chief spots forming the groups being situated between +6°·8 and +11°·1 in latitude, and 105°·5 and 136°·5 in longitude; and that of December 22 to 1923 January 3, with a maximum disc-area of 13·0 units, in position, +6°·2 in latitude, and from 85°·6 to 93°·5 in longitude. Other moderately large spots were those of February 8 to 17, with disc-area 9·2, and of March 1 to 12, with disc-area 9·6 units. In the whole year there were 33 groups in N. latitude with total area 72·6, and 26 groups in S. latitude with total area 38·6. On the whole, the greater activity is in the N. hemisphere, but is tending to pass to the S. hemisphere. There were 93 spotless

days in 1922, as compared with 29 in 1921, the respective and relative percentages of all days of observation being 36·3 and 12·5.

The spectra of the March and December sun-spots were examined, and the observations confirm the permanent nature of the sun-spot spectrum.

The large grating spectrograph has been employed mainly in experimental work. With the stellar spectrographs a fair number of photographs of the brighter stars has been secured, especially of γ Cassiopeiæ. It is intended, if possible, to use our accumulated store of stellar spectrograms for the determination of stellar parallaxes by the spectroscopic method. Dr. Hemsalech has kindly presented to the observatory excellent photographic copies of suitable wedges. Other researches in hand deal with the relations between solar and terrestrial magnetic phenomena.

Lee Observatory, Chorlton-cum-Hardy, Manchester.
(Mr. Albert Alfred Buss.)

The equipment of this observatory has during the past year undergone an almost complete change, interrupting observations from May 14 till August 13. The main instrument now installed in permanent position is a 5-inch Cooke clock-driven equatorial of $72\frac{7}{8}$-inch focal length. Thorp original Rowland grating solar spectroscope, Hilger double compound-prism spectroscope, two star spectroscopes, Thorp's solar polarizer, and full equipment of eyepieces and special appliances. There is further the old $3\frac{1}{8}$-inch Bardou refractor equatorially mounted on portable stand, and a separate original Rowland grating solar spectroscope for same, at disposal as a stand-by.

During the first portion of the year 35 observational dates were secured for the old outfit, while since the resumption on August 13, some 44 additional dates were recorded. The transit of the fine spot group in March afforded excellent opportunities for observing displacements of various spectrum lines, such as the D_1 and the D_2 lines of sodium, as well as exceptionally pronounced absorption effects in D_3 helium. Further, the "claw-effect" was well observed on that occasion, as also during several later spot transits. This effect invariably assumes the shape of a rosetree-thorn or "claw," emerging with its broad base from the normal position of dark Hα usually some considerable distance away in latitude from the spot (mostly on the northern side of a north-latitude spot, and on the southern side of a south-latitude spot), curving over and tapering to a point, which latter in nearly all cases is on the red side of Hα and merges with the dusky spectral band of a spot-umbra, the phenomenon undoubtedly depicting spectroscopically the outline and movement of the solar tornadic cone. It thus strikingly indicates the indraught from the higher chromospheric regions towards and downwards to the spot-level, but not actually entering the spot-cavity, rather spreading out laterally near the general level of the upper photosphere, and thus partaking in the formation of the penumbral striæ and filaments; in exceptional cases this leads to the vortex appearance seen through the telescope.

Solar activity persisted though diminishing in strength from March into May, but was rather subdued during the summer months, reviving again in October, when on the 22nd of that month the passage over the west-limb of a very active spot-group permitted the spectroscopic observation of many special limb phenomena. Again during the last week of December the transit of the large group in low northern latitude afforded the spectroscopic observation of remarkable and intensely eruptive disc-phenomena.

Wolsingham Observatory Report, 1921–1922.
(*Rev. T. E. Espin.*)

The search for new double stars in the selected zones has been continued by me with the 24-inch reflector, and by Mr. Milburn with the 17¼-inch reflector, and the results communicated to the Society. Towards the end of 1922 a powerful driving clock made by Mr. A. J. Pullon was added to the 17¼-inch. The clock is similar to that which drives the 24-inch. Some plates have been taken with the stellar camera, and observations of certain suspected variable stars have been continued.

Mr. A. M. Newbegin's Observatory, Sutton, Surrey.

The observations in 1922 were confined entirely to solar spectrum phenomena, as in former years. The results for prominences are given below:—

Number of observations = 107
,, prominences = 959
Mean daily number = 8·96
,, area = 3·30 sq. minutes
,, height = 34″·62

Distribution.	Number.	Area (sq. minutes).
North	499	1·97
South	460	1·33
East	493	1·51
West	466	1·79

Metallic prominences were observed on very few days; only one of them was of Type III. Observations of Hα displacements at the limb were also made on a number of days.

The results as a whole seem to indicate a slight increase of activity.

The Rev. T. E. R. Phillips's Observatory, Headley, near Epsom, Surrey.

Observations of the Planets and the measurement of double stars have been continued, but absences from home and other causes have much interfered with regular observational work. Since the last report

a memoir dealing with observations of Jupiter by members of the Jupiter section of the British Astronomical Association in 1916–17 has been published, and that for 1917–18 has been prepared and is now in the press.

In 1922 October the object-glass of the Coleman Equatorial belonging to the Society was sent to Messrs. Cooke & Sons to be cleaned, and a slight zonal aberration was corrected at the same time. Subsequent tests have shown its performance to be exceedingly fine. A discussion of observations of the third satellite of Jupiter made with this instrument was published in *Monthly Notices*, 82, 96.

Mr. Reynolds' Observatory, Harborne.

During the last two years the 30-inch reflector has been devoted to nebular work. An examination of a number of the Franklin-Adams plates has also been made, principally of the zone $-40°$ to $-90°$, with the object of getting information as to the distribution and dimensions of the spiral nebulæ in these regions.

Three papers dealing with the galactic distribution and angles of inclination of the spirals have been contributed to the *Monthly Notices.*

The 30-inch Common mirror, originally in the Reynolds reflector at the Helwân Observatory, has now replaced the 28-inch at Harborne.

Dr. W. H. Steavenson's Observatory, West Norwood.

On moving into a new residence at the end of 1922 April, it was possible to re-establish this observatory after an unavoidable interval of seven months.

The 6-inch Wray equatorial was erected on May 1 and has been in regular use since that date. The instrument has been employed chiefly for the observation of Novæ in their later stages. In all, 296 observations of these bodies have been made during the year. Most of them have already been communicated to the Society.

Special attention has been given to Nova Persei (1901) in view of its marked and irregular variability, and it has been observed on every possible occasion.

Some 200 of Struve's nearly equal pairs have been examined with a view to determining the present difference of magnitude of the components.

Comets Skjellerup (1922 *b*) and Baade (1922*c*) were observed for position on several nights, and the measures were communicated to the director of the Comet Section of the B.A.A.

The usual physical observations of the major planets have been continued.

The new position of the Observatory, as read off from the Ordnance Survey, is—

Long. $0^h\ 0^m\ 23^{s}\cdot1$ W.
Lat. $51°\ 26'\ 9''\cdot9$ N.

The Dominion Observatory, Ottawa, Canada.
(*Director, Dr. Otto Klotz.*)

Meridian Work and Time Service.—Observations with the meridian circle were obtained on 148 nights. The observations were, as heretofore, divided between the programme of latitude stars under observation and transit observations for determination of clock error and measurements of longitude; the present programme will be completed within the next month or two. A redetermination of pivot errors is in progress and nearly completed; these were last measured in 1910; preliminary computations show a small but measurable change in the form of the pivots.

Field observations for latitude and longitude, in continuation of the previous season's work, were carried out at four stations in the Mackenzie river basin, the most northerly station being within the Arctic circle. The longitudes were determined by wireless telegraphy, time signals from several stations being received simultaneously in the field and at Ottawa.

The time service has been maintained as in previous years. There are now thirteen secondary master clocks synchronised continuously from the observatory; these in turn control 575 clocks and dials of various kinds, including two tower clocks. Time signals are sent out by telegraph and telephone, and the time recorded on the various seismographs at the observatory.

The wireless time signals from Annapolis and Lafayette are received regularly; their comparison with Ottawa time continues to show the peculiar anomalies found previously here and elsewhere; a method has been developed for recording wireless time signals on the chronograph without appreciable systematic error.

A new Riefler clock was received and installed during the year, and has been performing satisfactorily; comparisons between the two Riefler clocks are made twice a day.

Astrophysics.—The 15-inch equatorial has been used as formerly in the determination of the radial velocities of stars; detailed studies of stars of the β Canis Majoris type have been continued. Stars of this type appear to exhibit rapid oscillations of radial velocity with periods of the order of $\frac{1}{4}$ of a day, superimposed on oscillations with a period of the order of a month. The short period oscillation of δ Aquilæ was found to have a period of about $3^h\ 48^m$ with an amplitude of about 40 km., which is suspected to be variable. Many of these systems are apparently involved in nebulosity or darkening interstellar material, as is strongly suggested in the case of θ Ophiuchi, σ Scorpii, SS Cygni, and others.

Over 1200 spectrograms have been secured, and most of them measured; about 100 direct photographs have been secured with a wide-angle camera, and are intended to be used for determining photographic light curves of variable stars.

Solar Physics.—During the year there was an abnormal amount of smokiness, haziness, and cloudiness. The spectrographic work was continued in the usual way of recording nine strips of spectrum at each observation, using various combinations of solar spectra from centre, limbs, and midway positions with electric arc and iodine spectra. Over

1000 such observations were obtained. The new double spectrocomparator was used in some of the spectrum measurements with satisfactory results. Errors of measurement traced to mechanical effects of oil between nut and thread have been finally eliminated or controlled, and revision and remeasurement of some of the early observations made. A pair of arcs has been constructed for use at pressures below atmospheric pressure, and these will be used with the standard arcs in making simultaneous observations of solar spectra.

Terrestrial Magnetism.—Two parties were engaged in the field work of the magnetic survey in the northern parts of Alberta, Saskatchewan, and Manitoba. 73 stations were occupied; of these 8 are relocations or approximate relocations of former stations, from which it is hoped valuable data as to secular change may be obtained.

Seismology.—There were recorded 116 earthquakes during the year; the records were reported to a mailing list of 120 stations; reports were received from 44 stations, covering in all 493 station-months.

A pair of Milne-Shaw seismographs was received and installed in a specially constructed vault 15 feet underground. One of these is fixed, while the other, for purposes of experimental work, will be arranged to allow rotation into any desired azimuth without change of the constants.

Gravity.—As in the previous year, a gravity observer accompanied the longitude party in the Mackenzie basin, a total of 9 stations having been now occupied in that region. The most northerly station was at Arctic Red River, within the Arctic circle. The rates of the chronometers used were obtained from the same wireless signals used for the longitude work.

Publications.—During the year the observatory issued and distributed nine publications.

The Dominion Astrophysical Observatory, Victoria, B.C.
(*J. S. Plaskett, Director.*)

The Staff.—The scientific staff of the Observatory remains as previously: J. S. Plaskett, Director, Mr. W. E. Harper, Dr. R. K. Young, Mr. H. H. Plaskett. During the summer vacation of 1921 Professor J. W. Campbell of the University of Alberta, and during the summer vacation of 1922 Professor S. L. Boothroyd, Cornell University, Professor D. Buchanan, University of British Columbia, and Professor J. W. Campbell, University of Alberta, were employed as Research Assistants.

Observing.—The observing weather during the two years has been rather disappointing. 1921—The dome was opened on 226 nights and 1395 star spectra obtained. On 131 nights 5 or more spectra, and on 81 nights less than 5 spectra were obtained, while on 14 nights no usable observations were secured.

1922—The dome was opened on 210 nights and 1529 spectra obtained. On 138 nights 5 or more spectra, and on 63 nights less than 5 spectra were obtained, while on 9 nights no usable observations were secured.

Practically all the observing time has been devoted to photographing stellar spectra, with the exception of two hours on each Saturday evening, when visitors are allowed to observe with the telescope.

Observing Programmes.—After the completion of the observing for the first radial-velocity programme of 800 Boss Stars, in 1921 August, a second general programme, consisting of about 1500 unobserved stars in Harvard, 50 between 5·5 and 6·5 visual magnitude and north of the equator, was prepared. Little work has yet been done on this programme, as several restricted programmes were given preference and this was only used to fill in gaps. The Director is observing all the known and several other suspected absorption-line O-type stars within reach, and is continuing his work on eclipsing variables. Mr. Harper and Dr. Young have jointly undertaken the determination of the spectroscopic parallaxes of all the suitable stars observed here for radial velocity, and in consequence most of their observing time has been devoted to securing spectra of stars of known parallax to define the curves, but a number of spectroscopic binaries have also been followed. Mr. H. H. Plaskett has obtained wedge spectra of stars and sun for study of intensity distribution, and has used one-, two-, and three-prism dispersion to secure spectra of three O-type stars for discovery and measurement of the wave-lengths of the enhanced helium components of the Balmer lines. The research assistants have mostly observed spectroscopic binaries and assisted in the other programmes.

Researches.—Observation was completed in 1921 August, measurement of spectra shortly afterwards, and the publication Vol. II., No. 1—The Radial Velocities of 594 Stars—appeared and was distributed early in 1922, thus completing the first radial-velocity programme. The first list of 100 new spectroscopic binaries previously published was followed by Vol. I., No. 26—Eighty-eight Spectroscopic Binaries—and several have since been discovered. The Director has done the greater part of the observing and measuring on his programme of O-type stars, but a considerable amount yet remains before completion. Several interesting side lines in this investigation have developed, and as part of this programme may be mentioned the discovery and investigation of the very massive (about 160 times the sun) O-type star B.D. 6° 1309. Another O-type binary, B.D. 44° 3639, has had its orbit completed. In addition the spectroscopic orbit and dimensions of the eclipsing variable TV Cassiopeiæ have been completed and published. Mr. Harper and Dr. Young have developed some new methods for determining line intensities in spectroscopic parallax work, and have obtained satisfactory curves for the various spectral types, so that the parallaxes of about 400 stars should soon be obtained. In addition to this work, Mr. Harper has obtained the orbits of Boss 4622, of a very interesting visual double star Boss 497, each component of which is a spectroscopic binary with double lines and in which consequently the mass ratios are known, of Boss 3793 foll., and of ϕ Aquilæ. He has also measured the radial velocity of 81 stars between types F and M which have been observed for defining the curves for absolute magnitude, and whose radial velocity has not hitherto been published. Dr. Young, in addition to the work on spectroscopic parallax, has obtained orbits of Boss 5442 and H.R.

8427, and has carried the observations of several long-period binaries to completion. Six months of his time in 1922 has been spent in connection with preparation for and observation of the Australian Eclipse, in collaboration with Professor Chant. Successful plates were obtained for the relativity displacement and for the measurement of the distribution of light and the polarisation of the Corona. Mr. H. H. Plaskett has succeeded in discovering the enhanced helium components to the Balmer lines in the spectra of the three O-type stars. The accurate measurement of their wave-lengths has enabled independent determinations of the fundamental atomic constants to be made from these stars and has shown their close agreement with laboratory values. He has also investigated the physical conditions in the atmospheres of these stars on the basis of Saha's ionisation theory, with the additional consideration of the relative abundance of the elements, and has obtained their temperatures and proposed a scheme of classification. Considerable work has also been done on the application of Merton and Nicholson's wedge method to the determination of the spectral intensity distribution in the stars and sun, and interesting results are expected.

Research Assistant Professor S. L. Boothroyd has obtained the spectroscopic orbit of H.R. 8803 with curiously behaving Calcium H and K, and of Boss 4870, while Professor J. W. Campbell has obtained the orbits of H.R. 5992 and H.R. 6532, and measured the radial velocity of ten stars.

Equipment.—A new ultra-violet stellar spectrograph for use at the focal plane has been obtained from Hilger, and after some slight adaptations gives beautifully defined spectra to the limit of atmospheric transmission. It will be used chiefly in the investigation of the spectra of early type stars and for faint objects. Two new types of camera lenses for the Cassegrain spectrograph have been obtained, one of medium focus for use with high dispersion, and the other of very short focus for low dispersion, which give fine definition over the whole field. Additional bearing wheels have been placed under the main ribs of the dome to relieve pressure on the former overloaded ones.

Publications.—The publications of the *Dominion Astrophysical Observatory*, Vol. I., Nos. 13–30 inclusive, completing the volume, and Vol. II., Nos. 1–4 inclusive, have appeared and been distributed; four other numbers are in press and several in preparation.

Helwân Observatory, Ministry of Public Works, Egypt.
(*Director, Mr. H. Knox-Shaw.*)

Time Service.—The time service was maintained as usual, transits being taken on 134 nights during the two years. For a month in the autumn of 1922, when the transit circle was being overhauled and a travelling-wire micrometer was being fixed to it, the error of the standard clock was determined from the Lyons W/T signals. In 1921 April and May the special Lyons time signals were observed, and in the autumn of 1922 the observations of the Lyons and Bordeaux signals was commenced as a regular piece of work.

Reynolds Reflector.—The telescope was used on 131 nights during sixteen months of the two years. Owing to there being no assistant available for this work, and the director having duties outside the observatory, no work was done with the telescope after the middle of 1922. 159 photographs were taken with the reflector, of which 94 were on nebulæ and 47 on Jupiter VIII.

The search for Jupiter VIII. was unsuccessful in 1921, but the satellite was found in 1922 February by the help of an ephemeris supplied by Mr. Jackson. Fifteen good plates were obtained of the satellite, and the positions derived from them are in the press.

Solar Radiation.— Bulletin No. 23 was published in 1922, containing the results of pyrheliometric observations of the solar constant made at Helwân during the years 1915–1921, and a study of the question whether the assumption of constancy of atmospheric absorption involved in the usual method of determining the solar constant is justified. The conclusion arrived at was that observations made at Helwân and elsewhere provided evidence of a progressive change in the atmospheric absorption during the course of an observation, which vitiated the results derived by this method. Observations for determining the solar constant have consequently been discontinued at Helwân.

General.—The usual meteorological, magnetic, and seismological observations were made. A Milne-Shaw seismograph was erected in the autumn of 1921.

Mr. C. C. L. Gregory resigned the post of Chief Assistant in July 1921, and his place has not been filled.

Bulletins Nos. 20–24 were published during the two years, and No. 25 is in the press.

Nizamiah Observatory, Hyderabad.
(*Director, Mr. T. P. Bhaskaran, M.A.*)

The work of the observatory has been continued almost on the same lines as in previous years.

Astrographic Equatorial.—Vol. iv., zone $-20°$ of the Hyderabad Astrographic Catalogue was printed and distributed during the past two years. This completes the portion of the catalogue originally assigned to Hyderabad in 1909.

An extension of the programme to the three southern zones, $-21°$ to $-23°$, has been sanctioned by the Government, and work is now being pushed on with a view to completing this section as early as possible.

Zone $-22°$, all the plates in this zone have been secured and measured. The reductions have also been completed and the measures have been copied and sent to Professor Turner.

Zone $-21°$, 177 plates with centres in Decl. $-21°$ have been taken, and all of them have been measured and reduced.

In zone $-23°$, 145 plates have been obtained and passed for measurement; and of these 74 have been measured to the end of the year. The computation of "Standard Co-ordinates" of Reference Stars from the Cordoba Catalogue is also proceeding. The reductions will be

taken on hand after these computations have been finished. It is hoped to make a commencement with the printing of zone −21° very shortly.

The Grubb 15-inch Telescope.—This equatorial was erected about the end of the period under report, and a start has been made with the adjustments. Regular work is expected to be commenced during the ensuing spring.

There has been no change in the staff of the Observatory.

Kodaikánal and Madras Observatories.
(Director, Mr. John Evershed, F.R.S.)

The routine work with the spectroheliographs and the visual observations of solar phenomena was continued in 1922 on the usual lines.

The grating spectrograph was employed early in the year in securing a good series of plates of the D lines in Sun and arc. The results indicate a zero pressure in the Sun, and if the pressure shifts of D_1 and D_2, as determined by Humphreys and reduced to the atmospheric pressure at Kodaikánal, are added to the Sun-arc shifts, the results approximate rather closely to the Einstein shift of 0·0124 A at the centre of the disc. There is no increase of shift at the limb such as occurs with the iron lines.

The prism spectrograph has been found to be very efficient in the ultra-violet as far as λ 3800, and a fine series of Sun and Fe arc spectra has been obtained with it in the H and K region for comparison with grating plates taken in former years. The results seem to indicate an appreciable instability of wave-length in some of the stronger solar lines, both at the centre of the disc and at the limb.

Photographs of Venus spectra were taken with the prism spectrograph in April and in June. Plates taken with this equipment in 1921 December, when the angle V-S-E exceeded 140°, gave values of the shifts Sun-arc practically identical with those in direct sunlight; and the same results are now obtained from the 1922 series with the planet on the opposite side of the Sun. As these prism spectra are of perfect definition, and superior to the grating plates of 1918–1920, which gave smaller wave-lengths in the light reflected by the planet compared with direct sunlight, it is concluded that the latter are affected by a systematic error, the origin of which is not easy to determine. It now appears certain that the lines are shifted towards red in light from all parts of the Sun, there being no difference between the visible and invisible hemispheres. The general shift of the iron lines increasing towards the limb is therefore impossible to explain by motion alone, and our researches seem to support the Einstein gravitational effect, combined with a radial outflow of the iron vapour, varying according to level, and possibly varying with the sun-spot period.

The prism spectrograph has also been employed in photographing the spectrum of Sirius with very high dispersion. The remarkable character of the lines in Sirius has been described in the *Monthly Notices* for 1922 May. In October spectrum photographs were obtained

in Australia of Canopus and of Achernar, but neither of these stars resembles Sirius in the character of their lines.

Dr. Royds has studied the effect on the wave-length of certain unsymmetrical lines of copper when other substances are introduced into the electric arc. A comparatively large displacement occurs when sodium metal is added, amounting to 0·048 A towards red: smaller shifts occur with other substances, such as iron, nickel, aluminium, etc., or alkaline carbonates. In the flame arc with carbon poles the calcium triplet λλ 4578-81-86 undergoes similar displacements, but smaller in amount, and mostly towards violet. The results strongly support the view that the widening and displacement of unsymmetrical lines is caused by the electrical field due to neighbouring ions; and the introduction of different substances into the arc causes differences in the field near an atom by altering the number of ions in the arc.

It has been previously found that in the first subordinate series of barium the second member is unsymmetrically widened in the opposite direction to that of succeeding members. Other examples of series lines anomalous in this respect have been found, and the anomaly appears to follow the behaviour in an electric field.

Eclipse Expedition.—Preparations for the solar eclipse of September 20 involved a large amount of construction work during the first seven months of the year. The main purpose of the expedition was to photograph the star-field surrounding the Sun on a very large scale, using a 12-inch photo-visual lens of 21 feet focus, and to determine the deflection of light near the Sun by comparison with photographs taken later in India with the star-field at the same altitude in the morning sky. In addition, two prism spectrographs were constructed for photographing the displacements of the green corona line at the east and west limbs simultaneously by means of a simple optical device.

The work was beset by serious difficulties from the beginning. The large lens appeared to be defective, but this was finally traced to the action of the mirror used, and of three plane mirrors that were available, made by three British firms of repute, not one was found to be free from serious defects. A mirror was therefore ordered from Hilger, but was not received in time; and this one alone appears to be good. The 16-inch cœlostat lent by the Joint Eclipse Committee was found to give a varying rate of drive, notwithstanding the new driving clock that had been provided for it: this involved the construction and careful grinding of a new driving-screw, and the re-figuring of the sector, entailing strenuous work up to the day of despatch of the instruments to Australia.

All this work was of no avail, however, and the large Einstein camera failed at the critical time, mainly on account of slight outstanding irregularities in the driving of the cœlostat combined with astigmatism of the mirror. The spectrographs also failed to record the corona line, which appears to have been abnormally weak at the time of the eclipse.

The weather conditions at Wallal, W. Australia, were about as perfect as can ever be realised at an eclipse, and so a unique opportunity for solving the deflection problem was lost, owing to bad workmanship in the apparatus.

Funds for the expedition were provided by a special grant from the Government of India of a sum equivalent to about £300, and the members of the expedition were the guests of the Australian Government while at Wallal.

Union Observatory, Johannesburg.
(*Union Astronomer, Dr. R. T. A. Innes.*)

There has been no change in the staff, and work has been continued on the usual lines.

During the two years 1921–1922 Circulars 51 to 55 were issued, and with these were included 71 maps of the Southern sky on the scale of $1^\circ = 36$ mm. So far 137 of these maps have been printed and distributed, leaving 419 still to be done to cover the whole sky south of -19°. Circular No. 56, dealing with a possible short period irregularity in the motion of the moon and the choice of certain small objects on the moon, in addition to Mösting A, for the determination of the moon's position, is in the press and almost ready for issue.

Orbits were computed for the Southern comets 1922 *a* (Reid) and 1922 *d* (Skjellerup).

Various minor planets were observed at the request of Northern observatories.

The Blink-microscope has been mainly employed on plates for proper motions. The examination of the Cape Zone was duly finished, and the results are now being prepared for press. As explained in Circular 53, about one-third of the Melbourne Zone -66° was lost in transit (probably stolen), so the work in this zone was finished and has been duly published.

In 1921, and in connection with the desire of the Imperial Government to divest itself of financial obligations incurred in the Union of South Africa, a proposal to amalgamate the Cape and Union Observatories was mooted. The Union Astronomer's report was asked for, but it did not for obvious reasons favour the proposal. There are, as it is, too few active observatories in the Southern Hemisphere. The proposal was strongly opposed by English astronomers and ultimately fell through.

A wireless receiving-apparatus (with a seven-valve amplifier) has been in course of erection. So far, with a temporary aerial, time and weather messages from Lourenzo Marques are the only ones received regularly.

Towards the end of 1922 the Union Astronomer went to England to superintend the completion of the mechanical portion of the 26-inch refractor under construction by Sir Howard Grubb & Sons. It is understood that the objective will be completed early in 1924.

Lovedale Observatory. (*Dr. A. W. Roberts.*)

During the year 1922 work has been interrupted at this observatory.

The Director was called upon by the Union Government to take over the post of Senior Native Commissioner for South Africa, and this has

necessitated considerable travel during the year, not only through South-West Africa, but also over the northern and eastern parts of the Union.

The delicate conditions of many of the native questions and difficulties made it impossible for the Director to give any part of his time to astronomical work.

During 1923 it is hoped, however, that Variable-Star work will be taken up as formerly.

Adelaide Observatory.
(*Director, Mr. G. F. Dodwell, Government Astronomer.*)

1921.—The routine work was continued as in previous years (see earlier reports). In addition a large amount of attention was given to the reception of long distance wireless signals and to the determination of longitude at Deakin and Kimberley, two field stations in the south and north respectively, near the 129th meridian, the boundary between South and West Australia, and between the Commonwealth Northern Territory and West Australia. An expeditionary party, including the Directors of Perth and Adelaide Observatories, and Messrs. C. A. Maddern and C. M. Hambidge of Adelaide, carried out the field observations at Deakin and Kimberley, and were successful in receiving wireless signals from Lyons, Bordeaux, Annapolis, Adelaide, and Perth, the overseas signals circling the earth in both directions. A longitude connection was obtained on this occasion with Greenwich, Paris, Washington, Ottawa, Helwan, Sydney, Perth, and Adelaide. At Kimberley, mountains in the vicinity were named after the Astronomer Royal, Sir Frank Dyson, General Ferrié, and Admiral Hoogewerff, who had specially facilitated the work.

Later in the year preparations were commenced for observations of the total solar eclipse of 1922 September 21, in Central Australia. Mr. A. L. Kennedy, B.E., who had been with Sir Douglas Mawson's Antarctic Expedition, was appointed Chief Assistant in November, in place of Mr. J. B. O. Hosking, who had received an appointment in Melbourne Observatory.

1922.—Routine work was carried on as usual. In March, the Director left to attend the International Astronomical and Geodetic Conference at Rome as a representative of the Commonwealth Government and of the Australian National Research Council, and went on from there to Greenwich to complete the boundary determination by observations for elimination of personal equation, etc., the portable transit instrument used at the boundary being set up, through the kindness of the Astronomer Royal, in the Greenwich transit pavilion.

On the journey back to Adelaide through America, advantage was taken of the opportunity of visiting several of the American Observatories. A South Australian expedition, including members of the Adelaide University and of the Observatory, went to Cordillo Downs to observe the total eclipse on 1922 September 21. The Beltana Pastoral Company provided transport for equipment over 400 miles from the railway to Cordillo Downs Sheep Station and return, and hospitality at the home-

stead. Mr. Kennedy took up the instruments, including a 40 ft. coronagraph lent by the Lick Observatory, and a quadruplet "Einstein" camera of 65 in. focal length lent by Allegheny Observatory, some months beforehand.

Excellent weather conditions were experienced and some valuable photographs were obtained. The comparison plates for the Einstein photographs are expected to be taken early in 1923 March.

Melbourne Observatory.
(*Director, Dr. J. M. Baldwin, Government Astronomer.*)

During 1921 and 1922 the only meridian observations have been for local time, as much time as possible being devoted to the arrears of reduction. The investigation of the proper motions for the stars included in the catalogue of astrographic standard stars (1900) has been completed, but, as was to be expected, earlier observations for stars south of 75° S. dec. were lacking in many cases. The formation of the supplementary catalogue of astrographic standard stars has been proceeded with. Further minor alterations have been made to the 8-inch transit circle, and the travelling wire micrometer has been received from Troughton and Simms. As a result of inquiries in Europe and America, some additional changes will be made to bring the method of observation into line with best modern practice. A new programme consisting of observations of fundamental stars will be commenced when the instrument is ready.

Through lack of assistance no progress has been made with the astrographic work, but as one assistant is now available a start will be made to take the remaining chart plates.

The 8-inch equatorial has been used for observations of variable stars of long period, and 1626 observations of stars mostly south of −20° were made in 1921 and 650 in 1922. The results have been forwarded to Harvard College Observatory.

The magnetic variation instruments at Toolangi have been in operation, but the routine measurement of the curves cannot yet be undertaken. At the end of 1922 the vertical force instrument, which has never been entirely satisfactory, was brought to Melbourne for thorough inspection. The classification has been forwarded to De Bilt as in previous years. In February 1922 an intercomparison of the absolute instruments with a set belonging to the Department of Terrestrial Magnetism of the Carnegie Institution was carried out, and at the same time a determination of the station difference between Melbourne Observatory and Toolangi was made. Since this time the absolute observations have been made at Toolangi once a month, and the absolute observations at Melbourne have been discontinued.

The seismograph records have been read off and the results forwarded to Oxford, and the verification of weights and measures, the time service, miscellaneous testing, registration of tides, etc., have been continued as in previous years.

An opportunity was given to the Director by the Victorian Govern-

ment to visit Europe, and he attended the Rome Conference in May 1922 as one of the Australian delegates. After the Conference he visited some of the leading observatories of Europe and America, and also made inquiries into weights and measures legislation and administration on behalf of the Victorian Government. In his absence Mr. Merfield was Acting Government Astronomer.

Through the munificence of a private donor, a party was sent from Melbourne to observe the total solar eclipse of September 1922 at Goondiwindi, the instruments being for the most part furnished by the Melbourne Observatory, and the Government arranged for the Director to act as leader of the expedition. A cœlostat was lent by the Greenwich Observatory. A good series of photographs of the corona was obtained, and it is hoped that the spectrographic results will also be satisfactory. Observations of magnetic declination, temperature, and pressure were also made. In addition, Mr. Hosking was granted permission to go to Wallal as a member of the Lick Observatory Eclipse Party.

During 1921 two vacancies on the staff were filled, but the staff is still inadequate to carry on the work of the observatory. A sum of £200 has been placed on the estimates for the current year for temporary assistance for the preparation of the astrographic catalogue, and £480 towards the printing of the first volume. This, although insufficient for completing the first volume, will enable the printing to be commenced.

Sydney Observatory.
(*Director, Mr. W. E. Cooke, Government Astronomer.*)

At the commencement of 1922 the astrograph was adapted for observing the solar eclipse, and later forwarded to Goondiwindi. An effort was made to measure the displacement of rays of light from stars passing close to the sun (Einstein effect), but the definition was so bad that no use can be made of the photographs taken for this purpose.

Incidentally, an excellent negative of the eclipsed sun was obtained, with an aperture reduced to 7 inches and an exposure of 10 seconds.

A complete series of photographs of the progress of the moon across the sun's disc, from first to last contact, was taken with a photo-heliograph, and the times of the contacts deduced from measures of the "horns."

Special observations of the shadow-bands were taken, and an effort made to delineate the exact position of the edge of the moon's shadow on the ground as it passed the towns of Grafton and Beaudesert, near the 153rd meridian.

The photographing of the plates for the International Catalogue was necessarily suspended, but is now being resumed.

At the Observatory the routine work was continued. During the year our first Star Catalogue, giving the positions of 1068 "intermediate" stars situated between 51° and 65° south declination, was published and distributed. The first two volumes of the Astrographic Catalogue were sent to the printer. We completed and prepared for printing the first volume (51°–3°) of the meridian zone observations for the Reference Stars

of the Astrographic Catalogue, but authority for printing was withheld; and we have nearly completed the observations and computations for the second volume (53°–5°). The measurement of the plates with one star micrometer has proceeded continuously, and we have now completed a second micrometer and shall in future keep the two systematically employed. Owing to the necessary preparations for the eclipse the Government Astronomer was unable to accompany his confrères to the Rome meeting of the Astronomical Union, and the usual course of lectures at the University was not given.

Notes on some Points connected with the Recent Progress of Astronomy.

Discovery of Minor Planets in 1922.

The following announcements of discovery were made in 1922:—

Letter and Number.	Date. 1922.	Discoverer.	Magnitude.	Letter and Number.	Date. 1922.	Discoverer.	Magnitude.	Letter and Number.	Date. 1922.	Discoverer.	Magnitude.
LJ	Jan. 4	Ba	15	LY 978	May 18	Be	14	MM	Oct. 13	R	14
LK 972	18	Wf	13	LZ	19	Be	13	MO	14	R	13
LM	24	Wf	15	MA	19	Be	14	MP	Sept. 24	Be	13
LN	Feb. 20	Wf	14	MB	19	Be	14	MQ	Oct. 19	So	13
LO	25	R	14	MC 979	June 29	R	14	MR	23	R	12
LP	25	Wf	15	MD	May 21	Wd	12	MS	July 19	B	15
LQ	Mar. 15	Wf	14	ME	July 30	R	13	MT	Nov. 10	J	14
LR 973	18	R	13	MF	15	Be	13	MU	15	J	13
LS 974	18	R	13	MG	Aug. 17	R	13	MW	18	Sn	13
LT 975	27	P	13	MH	27	R	11	MX	Dec. 14	Wf	13
LU 976	27	J	13	MJ	27	R	14	MY	14	Wf	14
LV 977	Apr. 6	J	13	MK	Sept. 21	R	13	MZ	14	Wf	14
LW	28	Be	13	ML	21	Wf	14	NA	12	J	11
LX	May 28	Wf	14								

The initials stand for: Ba = W. Baade, Bergedorf, Hamburg; B = S. I. Bailey, Arequipa; Be = S. Beljavsky, Simeis, Crimea; J = E. Jekhowsky, Algiers; P = J. Palisa, Vienna; R = K. Reinmuth, Königstuhl, Heidelberg; Sn = A. Schwassmann, Bergedorf, Hamburg; So = J. Comas Sola, Barcelona; Wd = H. E. Wood, Johannesburg; Wf = M. Wolf, Königstuhl, Heidelberg.

Planets LM, MG were both noted "doubtful." LY, NA were found independently by Wf May 30, and by Be Dec. 19, respectively. MW was subsequently found on a Heidelberg plate of Nov. 12. LH proved to be identical with 267 Tirza, LL with 414 Liriope, MN with 574 Reginhild, MV with 629 Bernardina. NA at discovery was moving daily $-1^{m}\cdot2$, S. 21′; its orbit is likely to be interesting.* A planet designated Algiers J is probably identical with 257 Silesia.

A number of planets photographed before 1922 are also announced;

* NA proves to be identical with the long-lost planet 132 Æthra, discovered by Professor Watson in 1873, but never seen since that year.

some of these were previously identified erroneously with known planets, others have been picked up on re-examining plates. Details in each case will be found in *Astr. Nach.* for 1922.

1895 CD[a], 1902 HW[a], 1908 DK[a], 1912 *a*, 1913 *p*, 1914 *e*, 1915 *g*, 1916 *o*. Also planets Σ 78 to Σ 118 (both inclusive) were found between 1914 and 1919, but several of them are identical with planets found elsewhere (Σ denotes Simeis); thus Σ 104 is probably 461 Saskia.

The following planets, unnumbered at the date of the last Report, have received permanent numbers:—Σ 45 = Σ *by* 951, Σ 61 = 1921 JY, 952; JT, 953; JU, 954; JV, 955; JW, 956; JX, 957; KC, 958; KF, 959; KH, 960; KM, 961; KP, 962; KR, 963; KS, 964; KT, 965; KU, 966; KV, 967, KW, 968; KZ, 969; LB, 970; LF, 971.

The redetection of 574 is useful, as it had not been seen since 1905, when it was discovered. The orbits of 592 Bathseba and 677 Aaltje have recently been improved. 592 is identical with 1909 HX, 1914 WC, 1916 Σ *au*, 1919 FS, but not with 1908 DK[a], 1912 PB, PC (these two symbols refer to a single planet) 1913 *p* or 1914 *e*. 677 is not identical with 1912 *a* as was at first thought, but an image of it has been found on a 1912 plate. 805 is identical with 1915 WW, but not with Σ 105; Σ *ef* is identical with 885.

1920 HS = 310 Margarita, HU = 602 Marianna.

Σ *ea* = Σ 92 was observed again at Simeis 1922 July 24.

The following planets have been named:—752 *Sulamitis*, 761 *Brendelia*, 814 *Tauris*, 819 *Barnardiana*, 847 *Agnia*, 848 *Inna*, 907 *Rhoda*, 932 *Hooveria*, 941 *Murray*, 965 *Angelica*, 971 *Alsatia*. By Professor Barnard's request the name of 907 was changed to *Rhodu* (his late wife's Christian name); the name *Barnardiana* was transferred to 819. A planet discovered at Washington 1921 October has been named *Anacostia*.

W. Baade followed the interesting planet 944 until 1922 Mar. 31, when its magnitude was 16·5; as its orbit extends to that of Saturn it will be invisible for about 10 years.

Professor M. Wolf suspects that the light of 641 *Agnes* is variable by 1 or 2 magnitudes.

A. Klose gives a full discussion of the orbit of 189 *Phthia* in *Astr. Nach.*, 5150–1.

The computing section of the British Astronomical Association is preparing ephemerides of Pallas and Juno. The *B.A.A. Handbook for 1923* contains ephemerides of Vesta and Eros; the latter will be well placed for northern observers at the end of the year, it will then go rapidly south. A. C. D. C.

Discovery of Comets in 1922.

Comet 1922 *a* = 1921 V. was discovered by Mr. W. Reid at the Cape on January 24: it was too far south for most northern observers. Prof. Barnard obtained one observation, the others were confined to the Cape, Johannesburg, and Cordoba, where it was followed till the end of April. Perihelion was passed in October 1921, and the comet must

have been an easy object in the northern hemisphere at that time. There is some evidence of deviation from a parabola, but, if elliptical, the period probably exceeds 1400 years.

Comet 1922 b = 1922 I. was found by Mr. J. F. Skjellerup at the Cape on May 6; Mr. Reid found it independently a few hours later. It was moving rapidly northwards and was well observed. It proved to be a short-period comet; indeed it has the shortest period of any comet whose orbit is well determined, with the single exception of Encke's. It was followed till the end of July.

Messrs. Crawford and Mayer were the first to announce the ellipticity of the orbit (*Lick Obs. Bull.*, 341). They suggested that the comet might be identical with Grigg's Comet 1902 II.; only six very rough observations extending over eleven days were made by him, and no one else saw it, owing to a delay in the announcement. Further investigation renders this identity very probable. If so, there have been four revolutions in the interim, giving a mean period of 4·97 years. This appears to be very close to the present period. New elements were found for Grigg's Comet, assuming a period of 4·90 years; these are given below, and do not differ too much from the present elements, in view of the fact that two fairly close approaches to Jupiter must have taken place in the interim. The comet should be observed again in 1927, and the question of identity can then be placed beyond a doubt.

Comet 1922 c was discovered photographically by Dr. W. Baade at Bergedorf on October 19. It was an easy telescopic object, and the number of observations is very great. It is likely to be followed for some months more. It has a large perihelion distance, 2·255 units, and passed perihelion a week after discovery.

The following interesting notes on its physical details were made by Prof. Frost at Harvard (*Bull.*, No. 778), and by Dr. Baade at Bergedorf (*B.Z.*, 34). On October 24 the nucleus was surrounded by an elliptical nebulosity, which had an extension 120″ long in P.A. 80°. The nucleus was 5″ south of the axis of symmetry. There was a fainter tail 20′ long in P.A. 240°, only 20° from the direction of the Sun. A Bergedorf photograph on December 9 showed the longer tail 17′ long in P.A. 283°, the shorter one 100″ long in P.A. 20°. Sun's P.A. 241°.

Comet 1922 d was discovered by Mr. J. F. Skjellerup on November 25. It was moving rapidly southward, and observations have been practically confined to the southern hemisphere. It comes north again in March and April, when it may still be within telescopic reach. Its orbit shows a close resemblance to that of Comet 1892 VI., but as that comet was observed for ten and a half months without deviating sensibly from a parabola identity appears to be excluded, and it presumably affords another case of a family of comets with a common origin. As the date of perihelion is given as January 3, it is not quite certain whether this comet will be catalogued under 1922 or 1923.

Comet 1922 e was found by Mr. Nakamura in Japan on November 29, being of mag. 13. It was supposed to be Perrine's periodic comet of 1896 and 1909. The period of that comet being about six and a half years, it is only observable at alternate perihelia. In case of identity perihelion passage must have taken place about October 17.

The following are the approximate orbits of the above comets. The year of perihelion passage is 1922 unless otherwise noted :—

Comet.	1921 V.	1922 I.		1922 *c*.	1922 *d*.
Computer.	H. E. Wood.	G. Merton.	M. Ebell.	J. M. Vinter-Hansen. B. Strömgren.	Prof. Chacón.
T in G.M.T.	1921 Oct. 17·46	May 15·1928	May 15·1699	Oct. 26·2007	1923 Jan. 3·4938
ω	184° 3′ 53″	355° 0′ 42″·4	354° 58′ 46″·8	118° 22′ 20″	264° 12′ 14″
Ω	278° 26′ 56″	215° 28′ 41″·3	215° 31′ 9″·1	220° 30′ 32″	261° 55′ 31″
i	35° 22′ 18″	17° 30′ 39″·4	17° 29′ 7″·1	51° 26′ 49″	23° 21′ 29″
ϕ	...	44° 1′ 38″·2	43° 53′ 29″·1		
log q	0·25798	9·948877	9·948907	0·35366	9·96600
log a	...	0·464578	0·462184		
Period	...	4^{y}·97584	4^{y}·935		

All the above orbits are referred to the equinox of 1922·0.

Several improved orbits of comets observed in previous years have recently been published, and are collected here. T is in G.M.T.

Comet. Discover or Name. Computer.	1909 I. Borrelly-Daniel. L. Gabba.	1847 V.—1919 III. Brorsen—Metcalf. P. Duckert.	P. Duckert.	1916 II. Neujmin. Neujmin.
T	1909 June 5·2576	1847 Sept. 9·5656	1919 Oct. 16·8816	1916 Mar. 11·3239
ω	5° 0′ 24″·5	129° 25′ 20″·0	129° 30′ 57″·6	193° 47′ 33″·0
Ω	305° 36′ 4″·3	310° 51′ 25″·0	310° 49′ 16″·0	327° 33′ 0″·5
i	52° 4′ 20″·8	19° 7′ 38″·0	19° 11′ 34″·9	10° 37′ 11″·6
ϕ	84° 7′ 59″·0	76° 6′ 28″·1	76° 12′ 49″·7	34° 26′ 33″·2
log q	9·925747	9·688086	9·685666	0·126926
log a	2·2066	1·221948	1·226150	0·489016
Period	203^{y}	68^{y}·0648	69^{y}·0597	5^{y}·414
Equinox	1909·0	1925·0	1925·0	1916·0
Reference	*Brera Obs. Pub.*, 57	*Ast. Nach.*, 5144	*Ast. Nach.*, 5144	*Ast. Nach.*, 5197

Comet. Discoverer or Name. Computer.	1919 IV. Schaumasse. Schaumasse.	1918 IV. Wolf. M. Kamensky.	—— Wolf. M. Kamensky.	1902 II. Grigg. Crommelin.
T	1919 Oct. 20·5079	1918 Dec. 13·48435	1925 Oct. 28·4	1902 June 23·295
ω	105° 59′ 41″·31	172° 56′ 24″·9	160° 1′ 54″	320° 45′ 54″
Ω	33° 39′ 17″·95	206° 42′ 42″·6	205° 5′ 48″	220° 43′ 30″
i	27° 19′ 39″·30	25° 17′ 30″·9	29° 3′ 54″	12° 29′ 30″
ϕ	45° 10′ 50″·38	33° 57′ 58″·2	25° 0′ 54″	49° 50′ 0″
log q	0·0681107	0·1992295	0·370545	9·832739
log a	0·6047147	0·5544989	0·609260	0·460131
Period	8^{y}·07369	6^{y}·788103	8^{y}·20145	4^{y}·90 (assumed)
Equinox	1919·0	1920·0	1920·0	1902·57
Reference	*J. des Obs.*, No. 5, 11	*Ast. Jour.*, 807	*Ast. Jour.*, 807	*Obs.*, 1922 Sept.

It was formerly thought that 1909 I. might be identical with the comet of 1802 (Pons). Signor Gabba considers that this is untenable, but suggests that 1909 I. may be identical with the comet of B.C. 137.

The orbits of Brorsen-Metcalf are a first attempt, not yet definitive, to connect the apparitions of 1847 and 1919.

Mr. Neujmin exposed two plates on 1920 November, $16^d\ 8^h{\cdot}7$ G.M.T. He found a faint object on each plate in R.A. $2^h\ 0^m{\cdot}4$, N. Decl. 26° 29′; mag. 15 to 16. The motion was approximately correct. Assuming T 1921 August 18·875, the place is represented within 2° 20′, which is a possible discordance, as perturbations were not calculated. His orbit, given above, is from three normal places extending over 96 days. Schaumasse's Comet passed within 0·35 of Jupiter in 1913 August, and suffered large perturbations. Wolf's Comet also suffered huge perturbations by Jupiter; the distance (minimum) on 1922 September 26 was 0·1220. The perihelion distance was greatly increased, the orbit reverting almost exactly to its size and shape before the close approach to Jupiter in 1875, which brought the comet into the region of visibility from the Earth. The next perihelion passage, 1925 October 28, occurs when the comet is in opposition, at distance 1·40 from the Earth. There is therefore a possibility of its being detected. If not seen then it is probably permanently lost.

Owing to the rough nature of the observations of 1902 II. a considerable range of variation is possible in the elements without producing residuals of improbable size.

Mention should also be made of some periodic comets due at perihelion about 1922, which have been the subject of computation. Mr. G. Merton (*M.N.*, 82, 381) investigated the perturbations by Jupiter of Metcalf's Comet 1906 VI., showing that its detection last year was practically hopeless, but that there are good prospects at the next return in 1929. Mr. F. R. Cripps (*B.A.A.J.*, 32, 282) investigated Daniel's Comet 1909 IV., showing that the perihelion passage would probably be delayed from 1922 October to the late Spring of 1923, when it would be unfavourably placed. H. Mahnkopf (*B.Z.*, No. 2) assigned 1922 June 13 as the computed perihelion date for Taylor's Comet 1916 I. Unsuccessful search for it was made by Dr. Baade from 1921 September to 1922 January 23. A. C. D. C.

Solar Activity in 1922.

Sun-spots.—The progressive decline of the solar cycle towards minimum is clearly marked by the mean daily spotted area of 250 in 1922 as against 350 for the year 1921 and 700 for 1920 (unit, millionths of the visible hemisphere, corrected for foreshortening). The activity of the year was diversified—three large groups and six of secondary importance contrasted with several small groups and a large number of spotless days. The second half of the year showed a decided decrease in general activity.

The three spot groups which call for special attention are those which were seen on the Sun's central meridian on March 1, March 7, and December 29 respectively. The first of these was composed of two

streams, at first distinct but merging later with each other. The group was situated in latitude 9° N. and had a total area of nearly 2000 millionths. The second group of note, in latitude 12° N., was composed of one regular and two composite spots in a cluster, with a long train of very small followers indicating a large region of disturbance. The third group was a stream of normal character. It appeared in latitude 6° N. with an area of about 1400 millionths, and was thus a considerable revival after nine months of diminishing activity. A small ephemeral spot in latitude 31° N. on June 24 was a possible first indication of the new cycle, but the general activity of the year and the presence of the recent large group of spots would point to a minimum not earlier than 1924. H. W. N.

The Prominences.—The areas and numbers of prominences for the year 1922 have both decreased considerably as compared with the figures for 1921.

The mean daily areas from the Kodaikánal photographic records are as follows:—

	North.	South.	Total.
1922—January to June . . .	1·90	1·27	3·17
„ July to December. . .	1·58	1·70	3·28

The mean daily numbers have decreased from 11·1 during the first half of the year to 10·4 during the second half.

The distribution in latitude was nearly the same during the two periods of six months, but there has been a steady increase of latitude in the northern zone of greatest activity amounting to 5° per annum, the zone advancing from +40° during the earlier months of 1921 to +50° during the later months of 1922.

A similar zone of activity in the south has, during the same period, moved at the same rate in the opposite direction from −53° to −43°. Such fluctuations are not without precedent during this period of the solar cycle, and the regular increase of latitude in both hemispheres culminating at the poles is not likely to begin until after the year of minimum activity of sun-spots.

Metallic prominences were scarce during the year, only 41 being recorded, of which 33 were north of the equator. Displacements of the hydrogen lines at the limb and on the disc also show a considerable excess in the northern hemisphere and a great falling off in frequency compared with the previous year.

The excess of displacements towards red observed in every year hitherto is again in evidence, and the proportion is precisely the same as in 1921, both at the limb and on the disc. J. E.

Solar Research in 1922.

Magnetic Fields.—The outstanding discovery of the year is G. E. Hale's detection of local magnetic fields on the sun's surface at places at which a sun-spot subsequently appears, or from which one has disappeared. Accounts of this, under the title of "Invisible Sun-spots,"

are contained in *M.N.*, 82, 168; *Proc. Nat. Acad. Sci.*, 8, 168; *P.A.S.P.*, 34, 59.

Rotation.—The Report of the *Commission de la rotation solaire* (President, H. F. Newall), presented at the Rome conference (*Trans. Internat. Astron. Union*, 1, 48), reviews all the existing spectroscopic determinations of rotation. The differences between the results are found to be larger than the probable errors of measurement, but the question of differences between the rotations of the two hemispheres, being independent of systematic error, is recommended as a problem for any one observer and instrument.

Strong evidence of variability through the solar cycle is found by J. Halm (*M.N.*, 82, 479), from a discussion of the existing material, in continuation of H. F. Newall's investigation (*M.N.*, 82, 101).

An account of the instruments and methods of observation used by R. F. De Lury in his spectroscopic investigation of the rotation is given by him in *Pub. Ottawa Obs.*, 6, part 1.

A memoir on his determination of the rotation period for 1903–8 from the calcium flocculi is published by P. Fox (*Pub. Yerkes Obs.*, 3, part 3). He finds the rotation to be the same as that found by other observers for the faculæ; it is intermediate between that of the reversing layer and that given by the high level lines. He is unable to detect any definite variation in the rate of rotation during the cycle. The determination is in general agreement with the similar one by P. Kempf for 1906, which has not previously been referred to in these summaries (*Pub. Potsdam Astrophys. Obs.*, 23, Nr. 71, 1916).

Sun-spots, Prominences, and Flocculi.—Further evidence is brought forward by A. L. Cortie (*M.N.*, 82, 170), on the connection between sun-spots and terrestrial magnetic disturbances; he finds the magnetic period to be a function of the solar latitude of the associated sun-spot period.

The frequency distribution of "revival" spot-groups over the solar surface is considered by H. W. Newton (*M.N.*, 82, 171).

W. J. S. Lockyer discusses further the relationship between solar prominences and the corona (*M.N.*, 82, 323).

The first part of a detailed study of the systematic distribution of the calcium flocculi is published by C. P. Butler (*M.N.*, 82, 334). The large majority of the groups of flocculi have axes whose inclinations point equatorwards to the west, but the range of variation of the inclinations is greater than that of the axes of the spot-groups. Comment is made on the greater longevity of flocculi as compared with spots.

The motion of the K_3 filaments from Meudon spectroheliograms during four complete rotations in 1919–21 are exhibited in graphic form by L. d'Azambuja (*Ann. de l'Obs. de Paris*, 6, 1).

In *C.R.*, 174, 1536, E. Merlin gives formulæ for computing heliographic co-ordinates, taking account of parallax corrections.

Spectroscopic Work.—C. E. St. John and H. D. Babcock (*Ap. J.*, 55, 36), in an extensive investigation fail to confirm A. Perot's announcement of the variability of certain telluric lines with the sun's zenith distance. They conclude that high velocity radial convection currents in the earth's atmosphere are absent, and that atmospheric lines form a

reliable standard of reference even in work requiring the highest precision. They find also that solar wave-lengths from the centre of the disc are remarkably constant, and radial convection currents in the sun, whilst not absent, are small, being downward at high levels and upward at low levels. However, A. Perot (*C.R.*, **174**, 215) gives further evidence in support of his previous announcement; he finds the wave-lengths of telluric lines to increase with increasing solar elevation.

An interferometer investigation of the magnesium *b* triplet in the solar spectrum, from the point of view of detecting pressure shifts and the Einstein displacement, is described by A. Perot in *Journ. de Phys.*, ser. 6, 3, 101. Utilising the fact that the *ratio* of the wave-lengths of similar lines possessing different pressure coefficients is unaltered by the Doppler or Einstein effects but is altered by pressure, he finds that the pressure in the solar layer contributing the magnesium lines is negligibly small. Correcting then for the pressure difference of one atmosphere between the solar and terrestrial sources, and correcting also for the velocity of descent (1·6 kms. sec.) found in a previous investigation from observations along a solar meridian, he finds a residual difference of ·013A, the computed Einstein amount being ·011A. In a note of another investigation (*C.R.*, **174**, 933) by the same method he finds the pressure in the layer contributing certain iron lines to be of the order of 30 cms. of mercury. P. Salet (*C.R.*, **174**, 151) finds very small pressures from certain iron lines.

In accordance with a prediction of the Saha theory, C. E. St. John finds increased ionisation over solar faculæ, the spectrum showing a change in the direction from Go to F (*Pop. Astron.*, **30**, 228). Results are given for titanium.

The application of the quantum theory of band spectra to the measurement of temperature is considered by R. T. Birge (*Ap. J.*, **55**, 273), using the position of maximum intensity as an indicator. From the solar cyanogen band λ 3883 the temperature comes out as 4000° ± 500.

The spectrum of the corona in 1918, as observed by the eclipse expedition from the Lowell Observatory, is described by V. M. Slipher (*Ap. J.*, **55**, 73); λ5303 radiation extended about a solar diameter above the sun's surface.

Solar Radiation.—The year has seen the publication of Vol. 4 of the *Annals of the Astrophysical Observatory of the Smithsonian Institution*, by C. G. Abbot, F. E. Fowle, and L. B. Aldrich. It covers the solar constant determinations of the years 1912–20, and includes the establishment of a station at Calama, Chile, and the subsequent transfer of the Mt. Wilson work to Mt. Harqua Hala, Arizona, and the Calama work to Mt. Montezuma, Chile. The mean value of the solar constant of radiation over a complete period comes out as 1·94 cal. min.$^{-1}$ cm.$^{-2}$. The volume contains investigations of the variability of the sun, and of the empirical method of solar-constant determination by observations at a single air mass, by the use of a pyranometer. See also *Proc. Nat. Acad. Sci.*, **8**, 177.

A study of the relation between solar-constant measures and sun-spot numbers is given by A. Ångstrom (*Ap. J.*, **55**, 24).

Following a method originally suggested by W. H. Julius, W. J. H. Moll and J. van der Bilt have attempted to deduce the distribution of brightness over the sun's disc from actinometer observations of solar radiation during the eclipse of 1914, Aug. 21 (*B.A.N.*, No. 30, 170). Differences between the ascending and descending curves prevented the method from giving trustworthy values. W. H. Julius urges that the method is still capable of giving good values of the limb intensities (*B.A.N.*, **33**, 189).

Theoretical.—From an analysis of Evershed's and St. John's observations of limb-centre displacements, W. H. Julius concludes that the displacements depend to a large extent on the proximity of other lines, and considers this as confirmation of the importance of anomalous dispersion (*Arch. Néerland*, **6**, 92).

It is concluded on different grounds by A. Pannekoek (*B.A.N.*, No. 19, 113) and E. A. Milne (*Phil. Trans.*, 223A, 225) that scattering in the outer layers of the sun is negligible compared with general absorption.

Further speculations as to the external magnetic field of the sun in relation to the structure of the corona are given by C. Stormer (*C.R.*, **174**, 1447).

W. Anderson draws a distinction between the solar temperatures corresponding to the "pores" and "rice-grains" (*A.N.*, **215**, Nr. 5148), and offers an explanation of detached prominences (*A.N.*, **216**, Nrs. 5166 and 5179).

A pulsation theory of sun-spots and solar periodicity is considered further by E. Bélot (*C.R.*, **174**, 283). Suggestions on the same problem are given by T. Terada (*Proc. Phys. Math. Soc. Japan*, ser. 3, 4, 163).

A paper by R. Dietzius on "The Distribution of Intensity over the Solar Disc" (*Wien. Sitz.*, **131**, Abt. 2A, H. 2) is not yet accessible.

E. A. M.

Stellar Spectroscopy in 1922.

Nebulæ.—An important contribution to the study of diffuse galactic nebulæ and of the spectra of associated stars is made by E. Hubble (*Ap. J.*, **56**, 162). Nebulæ belonging to the Milky Way have mostly emission spectra, and are associated with stars of type Bo or earlier without (in general) bright lines; nebulæ belonging to the Local Cluster have mostly continuous spectra and are associated with stars of type B1 or later. There is a continuous transition from Wolf-Rayet stars in small planetary nebulæ through those in large planetary nebulæ to the stars of types earlier and later than Bo already mentioned. It is suggested that the stars are the source of the luminosity of the nebulæ, though the phenomenon is not simply one of reflected starlight. The same author also gives a description of 588 faint nebulæ occurring in seven well-defined clusters (*Pub. Yerkes Obs.*, **4**, part 2).

Novæ.—General. A study of the absolute magnitudes of the Novæ is given by K. Lundmark (*P.A.S.P.*, **34**, 207). It appears that Novæ originate from both giant and dwarf stars, and the possibility is inferred that all stars may have passed through the Nova stage.

A. Belopolsky (*A.N.*, **216**, Nr. 5161) gives a comparison of the

spectra of the Novæ of 1892, 1901, 1912, 1918, and 1920, as observed at Pulkovo; he suggests that in general four stages may be discerned in the evolution of the spectrum after the outburst.

An interesting summary of some of the features of Novæ is contained in a short paper by E. E. Barnard (*Proc. Amer. Phil. Soc.*, **61**, 99).

Nova Cygni, 1920.—Observations of the disc of this Nova are given by K. Lundmark (*P.A.S.P.*, **33**, 316). He is unable to say whether it shows the characteristic structure of that for Nova Aquilæ, 1918. The probable diameter in nebulium radiation is 2–3″. Confirmation of this estimate is contained in a note by E. E. Barnard (*P.A.S.P.*, **34**, 164). Photometric observations are given by P. M. Ryves (*M.N.*, **82**, 299), F. Kaiser (*A.N.*, **216**, Nr. 5162), and N. Barabascheff (*A.N.*, **217**, Nr. 5201) The latter also gives spectrophotometric observations of Hα, Hβ and λ5010, and of the regions near Hα and Hβ; and effective wave-lengths are given by I. Balanowsky (*A.N.*, **215**, Nr. 5146).

Other Novæ.—In *Harvard Coll. Bull.*, No. 773, announcement is made of a tenth-magnitude Nova discovered by S. I. Bailey; it is named Nova Scorpii, 1922. Notes on various Novæ are given by E. E. Barnard (*P.A.S.P.*, **34**, 164, and *M.N.*, **82**, 457). Magnitude observations of Nova Persei, 1901, are given by W. H. Steavenson (*M.N.*, **82**, 403), and of Nova Aquilæ, 1918, by P. Dawidowitsch (*A.N.*, **215**, Nr. 5159). K. Lundmark finds the spectrum of Nova Aquilæ IV., 1920 (Wolf), to be of type Ro, and gives other particulars (*P.A.S.P.*, **33**, 314).

Photometry and Spectro-photometry.—The Report presented by the *Commission de photométrie stellaire* (President F. H. Seares) of the International Astronomical Union at the Rome meeting contains a table of standard magnitudes of the stars of the North Polar Sequence (*Trans. Internat. Astron. Union*, **1**, 69), based on the observations of all the observatories contributing to the work. The table gives the adopted mean photographic magnitudes (to m20) and photo-visual magnitudes (to m17), with the residuals for the different observatories. It gives also the colour indices derived directly and those derived by the method of exposure ratios. The report contains information as to the colour equations of the different scales, and a table of less well-observed supplementary stars is given. Comparison is made with the colour indices derived from the determinations of effective wave-length by C. Davidson and E. Martin (*M.N.*, **82**, 65), and by E. Hertzsprung (*Ap. J.*, **55**, 370). Further evidence on the same subject is contained in papers by F. H. Seares and M. L. Humason (*Ap. J.*, **56**, 84) and by F. H. Seares (*Ap. J.*, **56**, 97). Certain difficulties in accepting the colour determinations of Seares for faint stars in the region of the North Pole are discussed by E. Hertzsprung in *B.A.N.*, No. 17, 89, and in *B.A.N.*, No. 17, 91, he gives a fresh "giant and dwarf" diagram deduced from colours, apparent magnitudes, and proper motions.

Hertzsprung has also published a list of mean colour equivalents (in the form c_2/T) and hypothetical angular semi-diameters of 734 stars (*Leiden Obs. Ann.*, **14**, part 1), of which a summary is given in *Observatory*, **45**, 356.

Sundry spectro-photometric methods for determining stellar luminosities are described by B. Lindblad (*Ap. J.*, **55**, 85, and correction, **55**,

413). He obtains criteria of absolute magnitude for stars of types A and B from the relative intensities of certain regions of their spectra. O. Bergstrand and H. Rosenberg (*A.N.*, **215**, No. 5159) give a list of stars which they suggest to be suitable for the determination of standard effective wave-lengths, and K. Lundmark and W. J. Luyten (*M.N.*, **82**, 495) contribute a valuable discussion, critical and historical, of the whole subject of the meaning and determination of the colour equivalent of a star.

In *M.N.*, **82**, 472, J. Halm applies a formula he had previously found to give a method for the determination of photographic magnitudes when the magnitude of one comparison star is known; and in *B.A.N.*, No. 10, 51, J. Schilt describes a thermo-electric method of measuring photographic magnitudes.

In the course of their work with their three-colour photometer, Ch. Nordmann and Le Morvan have found that the star 13 Cephei, of type A, has an intensity distribution corresponding to an effective temperature of 5100°, and that the star θ Urs. Maj. (F8p) has a distribution in certain regions of its spectrum the same as a normal star of type A (*C.R.*, **174**,, 101; **174**, 669; **175**, 573). They have also determined the temperatures of three stars of Type N (*C.R.*, **174**, 1690).

Spectroscopic Parallaxes.—A method of determining the absolute magnitudes of A-type stars from their spectra is described by W. S. Adams and A. H. Joy (*Proc. Nat. Acad. Sci.*, **8**, 173, and *Ap. J.*, **56**, 242), and the parallaxes of 544 stars thus determined are given. The method depends on an observed correlation between detailed spectral type and mean absolute magnitude, and the parallaxes of the individual stars show as good an agreement with those determined otherwise as do the late-type stars of the earlier lists. G. Stromberg (*Ap. J.*, **55**, 11) makes use of the differing dependence of probable error upon the value of the parallax for trigonometrically and spectroscopically determined parallaxes to deduce the systematic errors of the two scales, and concludes that the Mount Wilson spectroscopic parallaxes are appreciably free from systematic error. The parallaxes of 50 stars of type K deduced from objective prism spectra are given by H. Shapley and B. Lindblad (*Harvard Coll. Circ.*, 228) and of 87 stars by H. and Martha B. Shapley (*Harvard Coll. Circ.*, 232), the latter containing a comparison with Mount Wilson results. In *M.N.*, **82**, 226, W. J. S. Lockyer describes the use and calibration of graduated wedges for measuring the relative intensities of lines in stellar spectra, and in *M.N.*, **83**, 47, D. L. Edwards gives the results of the application of the method in the determination of spectroscopic parallaxes of 49 stars of type B.

Measurement of the Total Radiation from the Stars.—The growing importance of this line of advance demands a paragraph to itself. A full account of the technique of the manufacture and use of vacuum thermo-couples for application to problems of astrophysics is given by S. B. Nicholson and E. Pettit in *Ap. J.*, **56**, 293; and in various notes (*P.A.S.P.*, **34**, 132; **34**, 181; **34**, 290) they give the results of the measurement of the total radiation of various stars of types M and N (including interesting long-period variables like *o* Ceti and R Aquarii), with and without water transmission. Some discussion of the measure-

ments on *o* Ceti is given by E. A. Milne (*Observatory*, **45**, 223). In *Ap. J.*, **55**, 21, and in *Proc. Nat. Acad. Sci.*, **8**, 49, W. W. Coblentz gives the results of vacuum thermo-couple observations made with the 40-inch reflector at the Lowell Observatory. The former paper gives the water transmission for 27 stars, and the latter the effective temperatures of 16 stars as estimated from the relative distribution of energy in 5 spectral regions.

Classification.—The Report presented by the *Commission de classification spectrale des étoiles* (President W. S. Adams) at the Rome meeting (*Trans. Internat. Astron. Union*, **1**, 95) contains comprehensive suggestions for the revision of the notations associated with the Harvard system of classification, in the direction of making them more detailed and more informative. The principles underlying the proposed changes are clearly set out. A provisional classification of the stages of the spectrum of a Nova is given, and a new class (S) of red stars is formed (see *Harvard Coll. Bull.*, 778); the class Md is abolished.

The Henry Draper Catalogue has now been published up to 18 hours R.A.

H. H. Plaskett, as the result of a detailed study of the spectra of three O-type stars (*Pub. Dominion Astrophys. Obs.*, **1**, 323), suggests a revision of the Harvard classification of O-type stars, abolishing the presence of bright lines as a criterion and using the continuous change in the intensities of characteristic absorption lines as in the other spectral classes.

Ionisation in Stellar Atmospheres.—Numerous extensions, and in some parts criticisms, of Saha's quantitative theory of ionisation at high temperatures and its applications to solar and stellar spectroscopy have been made, and it is difficult to summarise them. An extension of the theory to gaseous mixtures has been made by H. N. Russell (*Ap. J.*, **55**, 119), who also applies the theory to the comparison of solar and sun-spot spectra, finding general agreement with the predictions of the theory. He has identified lines of Rb in the sun. Difficulties concerning the comparative behaviour of Na and Ba are brought to light by him in *Ap. J.*, **55**, 238, and are examined further by H. N. Russell and K. T. Compton (*Pop. Astron.*, **39**, 227) and by M. N. Saha (*Phil. Mag.*, **44**, 1128); the latter modifies the theory by the introduction of "steric" factors. H. H. Plaskett makes extensive use of the ionisation theory in his memoir on three O-type stars (*loc. cit.*), employing it to deduce their temperatures but modifying it by taking account of the relative abundance of the different elements. He discusses the difficulty concerning Ba and Na in the sun, and puts forward an alternative method of calculation of ionisation based on electron collisions.

A valuable study of the application of the Saha theory to the reversing layers of stars is made by A. Pannekoek (*B.A.N.*, No. 19, 107); this paper discusses many other points in the theory of stellar atmospheres. In *B.A.N.*, No. 21, 127, the same author suggests doubly ionised calcium as the possible source of unknown lines in the coronal spectrum.

In one of his earlier papers (*Phil. Mag.*, **40**, 824, 1920) Saha suggested that the method was capable of application to the dissocia-

tion of compounds in stellar atmospheres; in *M.N.*, **82**, 396 (see also *Observatory*, **45**, 190) R. d'E. Atkinson considers the dissociation equilibrium of titanium oxide, and applies it to deduce the pressure in the atmospheres of stars showing this compound in their spectrum; the pressures come out remarkably low.

In this connection may be mentioned W. H. Wright's illuminating discussion of the continuous radiation found in certain celestial spectra beyond the Balmer limit, in the light of the quantum theory and the Bohr atom (*Nature*, **110**, 810).

Experimental work illustrating the Saha theory has been done by A. S. King (*Ap. J.*, **55**, 380; *P.A.S.P.*, **34**, 220) and by A. A. Noyes and H. A. Wilson (*Proc. Nat. Acad. Sci.*, **8**, 303).

Special Studies.—High dispersion spectra of Sirius are described by J. Evershed (*M.N.*, **82**, 392) and compared with the solar spectrum. The stellar lines are considerably widened, relatively, and are more intense. Various explanations are considered, but no satisfactory one is found.

In *A.N.*, **217**, Nr. 5200, J. Weber accounts for a difference of 1·1 km. sec. between the stream motion of Sirius and that deduced from its radial velocity as due to the Einstein shift.

By using revised wave-lengths for oxygen and nitrogen lines, S. Albrecht finds a small correction in the direction of reducing the amount of Campbell's K-term in the radial velocity of B-type stars (*Ap. J.*, **55**, 361).

Bright H and K lines are found by F. C. Leonard (*P.A.S.P.*, **33**, 323) in the spectrum of the fainter component (C) of the double companion of o^2 Eridani, the spectrum itself being of M-type with bright hydrogen lines. Various notes on stars showing bright lines in their spectra are given by M. L. Humason (*P.A.S.P.*, **34**, 133), P. W. Merrill (*P.A.S.P.*, **34**, 134), and the two authors jointly (*P.A.S.P.*, **34**, 180; **34**, 351). Some comments on the spectra of M-type stars with bright lines are made by C. P. Butler (*Observatory*, **45**, 15). Wave-lengths of bright lines in M-type stars are given by W. S. Adams and A. H. Joy (*P.A.S.P.*, **34**, 175). The same authors in *P.A.S.P.*, **34**, 174, give a useful list of 35 dwarf stars of type M, whilst further new and suspected M-dwarfs are given by W. J. Luyten (*P.A.S.P.*, **34**, 342). Additional stars with peculiar spectra are given by Miss Cannon (*Harvard Coll. Circ.*, 231). The irregular variable R Scuti is found by A. H. Joy (*P.A.S.P.*, **34**, 349) to have a periodic recurrence of titanium-oxide bands in a spectrum otherwise classed as G or K; the star resembles o Ceti in the displacement of certain bright lines to the violet. An account of changes in the spectrum of o Ceti in 1919 is given by C. F. Shane (*L.O.B.*, 335).

A study of the Pleiades group, with details of magnitudes, colour-indices, spectra where known, etc., is published by R. Trumpler (*L.O.B.*, 333). Criteria for membership of the group are stated. The luminosity law, and the frequency law for spectral types are discussed, and estimates for the parallax of the cluster are obtained. The group contains no red or yellow giants. F. H. Seares (*P.A.S.P.*, **34**, 56) suggests that the abnormally low colour-indices for F and G stars in the Pleiades

found by Trumpler is due only in part to their being dwarfs, but is due to errors in classification which are a function of magnitude, the stars in question being faint. Further evidence is contained in *Harvard Coll. Bull.*, 764.

In *L.O.B.*, 337, M. L. Heger gives a further study of the sodium D lines in stars showing stationary or anomalous H and K lines. The sodium lines completely share these anomalies. In the same *Bulletin* she gives notes on the spectra of the B-type stars used in the preceding investigation, including γ Cassiopeiæ.

Fresh studies of early B-type stars of the β Can. Maj. type are given by F. Henroteau (*Pub. Ottawa Obs.*, **5**, Nos. 2 and 8; **8**, No. 1; *Journ. R.A.S. Canada*, **16**, 58).

H. H. Plaskett's study of three O-type stars may be again referred to under this heading (*Dominion Astroph. Obs.*, **1**, 323).

Two fresh stars having bright reversals of the H and K lines are announced by H. Deslandres and V. Burson (*C.R.*, **175**, 121), bringing the total number known up to 12, out of 25 examined. All are giants, and all save one are of types G and K. Theoretical ideas arising out of these observations are developed by the first-mentioned author in *Nature*, **110**, 847.

Harvard plates of the spectroscopic binary β Aurigæ show that the following side of each component shows the K line sharper than the preceding side (*Harvard Coll. Bull.*, 767). In the same *Bulletin* it is announced that hydrogen emission lines have appeared in the spectrum of a certain B-type star between 1908 and 1922.

Astronomical Interferometer Measures.—F. G. Pease (*P.A.S.P.*, **34**, 342) finds some evidence of variation in the angular diameter of α Orionis, and of correlation between this and the magnitude changes. The necessity for an accurate knowledge of the effective wave-length in interferometer measures is emphasised by J. A. Anderson (*Ap. J.*, **55**, 48) who shows how it may be determined experimentally. Contributions to the theory of interferometer measurements are made by J. A. Anderson (*loc. cit.*), H. S. Jones (*M.N.*, **82**, 513), and M. Hamy (*C.R.*, **174**, 342; **175**, 1123). A suggested new form of interferometer with increased resolving power is described by A. Danjon (*C.R.*, **174**, 1408).

Spectroscopic Binaries.—Under the title "A Very Massive Star" (*M.N.*, **82**, 447), J. S. Plaskett gives details concerning the spectroscopic binary B.D. 6° 1309 (H.D. 47129), which is of type Oe5 with stationary H and K lines. It appears that minimum values of the masses of the components are $86\odot$ and $72\odot$, and the absolute magnitude of the system must be brighter than $-6{\cdot}3$. The system is thus the most massive known.

Orbits of spectroscopic binaries are given by R. F. Sanford (*Ap. J.*, **55**, 30), Miss Maury (*Harvard Coll. Circ.*, 232), in the papers of F. Henroteau already cited, and in numerous publications of the Dominion Astrophysical Observatory, Victoria, by J. S. Plaskett, W. E. Harper, J. W. Campbell, R. K. Young (I., 26, 27, 28, 29; II., 2, 3).

Radial Velocities.—A list of stars with variable radial velocity is given by W. W. Campbell (*P.A.S.P.*, **34**, 167). Studies of the radial velocities of stars of type N and deduction of the solar motion are given

by J. H. Moore (*P.A.S.P.*, **34**, 169; *L.O.B.*, 342). In *B.A.N.*, No. 16, 85, E. Hertzsprung mentions certain desiderata in the determination of radial velocities.

Wave-lengths.—Reference must be made to the appearance of the valuable "Report on Series in Line Spectra" (Physical Society of London) by A. Fowler, and also to a similar work, "The Origin of Spectra," by Foote and Mohler.

The Report of the *Commission des étalons de longueur d'onde et tables de spectres solaires* (President C. E. St. John) (*Trans. Internat. Astron. Union*, **1**, 35), contains amongst other tables a full table of proposed iron tertiary standards.

Instruments.—The temperature regulation of the spectrograph attached to the 72-inch reflector of the Dominion Astrophysical Observatory is described by J. S. Plaskett, *Journ. R.A.S. Canada*, **16**, 91.

Miscellaneous.—Statistical studies of the relation of spectral class to magnitude, etc., are given by Miss Leavitt (*Harvard Coll. Circ.*, 230), H. D. Curtis (*P.A.S.P.*, **34**, 33; *Journ. R.A.S. Canada*, **16**, 210), and of the relation of spectral class to distribution in space by H. Shapley and Miss Cannon (*Harvard Coll. Circs.*, 229, 239), and by H. Shapley (*Harvard Coll. Circ.*, 240). The important paper by F. H. Seares on the masses and densities of the stars should be mentioned here (*Ap. J.*, **55**, 165).

From the coincidence of the time of photographic and photovisual maxima of variables in Messier 5, H. Shapley has been able to deduce that the difference in velocity of blue and yellow light in interstellar space is less than one part in ten thousand million (*Harvard Coll. Bull.*, 763; *Nature*, **109**, 318).

P. Doig (*M.N.*, **82**, 372) has examined the spectral relations between the components of double stars, and finds them in general agreement with the requirements of the giant and dwarf theory, on the assumption that the less massive star evolves the more quickly.

The correction to the Schwarzschild boundary temperature due to non-grey absorption is given by E. A. Milne (*M.N.*, **82**, 372), and the Schwarzschild formula is extended to the outer parts of a stellar atmosphere. In *Phil. Trans. Roy. Soc.*, 223A, 201, the same writer examines the theoretical relation between the law of darkening of a stellar disc from centre to limb and the continuous spectrum on various hypotheses as to the structure of the stellar atmosphere.

The refraction of a light ray by a stellar atmosphere is considered by M. Ferrier (*C.R.*, **174**, 1404). In *C.R.*, **175**, 746 and 1395, Maggini brings forward evidence in favour of attributing the anomalous behaviour of the H and K lines in certain stellar spectra to anomalous dispersion.

E. A. M.

Double Stars.

Measures and New Pairs.—During the years 1921–1922 particulars of 172 new double stars found at northern observatories have been published, besides 111 southern pairs discovered by Dawson at La Plata.

K. Bohlin (*A.N.*, **214**, 247) gives observations of 9 double stars.

W. Doberck (*A.N.*, 212, 499) publishes observations of 36 well-known double stars in continuation of previous series. Further observations for about 200 stars are given in *A.N.*, 217, 97.

At Edinburgh Observatory the work of reobserving stars in high declination has been continued, and the measures of about 250 stars, mostly under 5″, have been published (*M.N.*, 81, 567).

Rev. T. E. Espin has continued his work with the assistance of Mr. Milburn, and gives the 15th and 16th series of observations in *M.N.*, 81, 232, and 82, 187. Besides measures of about 100 previously known double stars, observations of the new double stars Es 1806–Es 1933 and Mil 150–Mil 241 are given. These lists contain respectively 68 and 60 with separation under 5″.

W. S. Franks (*M.N.*, 81, 150) has published the measures of 71 wide pairs showing relative motion. The names are given of other wide pairs for which the observations show no motion.

C. Luplau Janssen, G. Haarth, and others have made further measures of double stars at the Urania Observatory, Copenhagen. The lists (*A.N.*, 215, 1 and 367) contain measures of 115, and 100 double stars including many Σ pairs and some wide pairs.

F. P. Leavenworth has continued his observations of binaries and other well-known stars with the 10½-inch refractor of Minnesota, and gives measures of 83 in *A.J.*, 33, 158. Another series of observations is given in *A.J.*, 34, 151.

C. P. Olivier has published the measures of 100 double stars, mostly recent discoveries of fairly close pairs in *A.J.*, 34, 111. This includes the new double stars Ol. 86–93, and in addition single measures of 13 other new double stars, Ol. 94–Ol. 106, are also given.

E. Pettit in *A.J.*, 33, 150, has given measures of 95 double stars, including 5 new double stars with separation under 5″.

T. E. R. Phillips (*M.N.*, 81, 566) has given further measures of a selected list of 32 double stars.

R. Trumpler has given particulars of a new double star (*Publ. A.S.P.*, 33, 270).

R. G. Aitken has now completed the reobservation of the first 3000 double stars discovered by himself, with the exception of only 102. In his fifth and sixth notes (*Publ. A.S.P.*, 33, 60, and 34, 52) he has drawn special attention to 43 stars showing decided motion.

G. van Biesbroeck is making good progress with a similar reobservation of the Hussey stars north of Dec.–20°. In *Popular Astronomy*, 29, 278, he announced that he had by 1920 December completed measures of 421 pairs, and that he had made first measures of 331 others, so that more than half the stars had been looked up. For about 60 pairs, *i.e.* 1 in 12, he has found a change in position angle of at least 10°, or in separation of 0″·5 in an interval of about eighteen years on the average. Details of the change are given for 37 stars. These include 4 with a period of the order of fifty years, and the variable star X Ophiuchi (period of variation 335 days) of spectral type Md6 or Md8, for which the position angle has changed 24° and the separation has remained unaltered at 0″·22. This star is further considered by C. H. Gingrich (*Pop. Astr.*, 30, 555).

P. W. Merrill (*A.P.J.*, 56, 40) has used the 20 feet interferometer attached to the 100-inch reflector at Mount Wilson for double star work. Further measures are given of Capella, for which a new orbit is deduced, and measures are also given for κ Ursæ Majoris and ν^2 Boötis. 85 stars to magnitude 5·7 have been examined, but with the exception of 6 doubtful cases no change in the visibility of the fringes with the position angle was observed.

We must here mention some observatory publications dealing with Double Stars.

"A Catalogue of Double Stars" has been issued by the Royal Observatory, Greenwich. It has been compiled from all the Greenwich observations made during the years 1892–1919. Notes are given for all pairs showing relative motion since the first observations. At the end of the catalogue detailed information is given of the orbits of 25 stars.

Publications of the "Washburn Observatory, vol. x, part 4," contains the observations of double stars made by the late Director, Professor G. C. Comstock, during the years 1907–1919. The separate observations and yearly means are given as also the means of observations of the same stars made by Professor Comstock during the years 1892–1906, for which details were previously published. The observations relate to a limited series of stars which were observed many times. For example, 70 Ophiuchi was observed 124 times and ζ Herculis 88 times.

B. H. Dawson has been engaged on Double Star work with the 17-inch refractor of La Plata Observatory since October 1912, with a break in the years 1914–1916. The observations for the years 1912–1917 were published in vol. iv, part 1 of the Observatory publications, and during the past year part 2 has been published giving the observations to 1921·5. Over 7500 measures in all have been made. This includes all Herschel stars south of -30° and other southern stars requiring observation. Many of the stars observed are wide pairs. The catalogue includes 111 new pairs, about half of which were picked up accidentally, while the others are the first results of a systematic search begun in 1920.

Reference here might also be made to the measures of double stars on parallax plates taken at the Leander McCormick Observatory. The results are published in the Observatory publications, vol. iii, part 2, by C. P. Olivier. The measures for 95 pairs and estimates for four others are given, including Olivier 66–80, S. A. Mitchell 1–9, Alden 1–8. The numbering of the Olivier double stars does not agree with the preliminary announcement in *A. J.*, No. 772; 12 were then announced.

Orbit Determination.—A considerable number of papers dealing with the methods of determining orbits have been published, and we give the following references:—Comstock (*A.J.*, 33, 139 and 163) with applications to μ^2 Boötis and γ Coronæ Borealis; Doberck (*A.N.*, 214, 89) with tables for the computation of orbits and application to ξ Boötis (see also Jubilee number of the *A.N.* with application to Castor); Howard (*Pop. Ast.*, 30, 386) with application to Sirius; Meyermann (*A.N.*, 215, 13 and 179) with application to Sirius; and Schnauder (*A.N.*, 215, 441; 216, 129) with application to ζ Cancri. Aitken (*P.A.S.P.*, 33, 270)

gives a note on β1212. Van den Bos determined the orbit of Σ554 = 80 Tauri in *M.N.*, **81**, 474, and in *B.A.N.*, 22, determined the masses of the two rather unequal components (mags. 5·8, 9·0) as 1·85 and 0·72 times the mass of the sun. He determined the orbits of β648 and OΣ77 in *B.A.N.*, 12 and 15 respectively, and in the earlier number gave a note on Σ1834. Dawson gave the orbits of the two southern binaries β744 and Brisbane 3574 in *A.J.*, **34**, 17. Pavel has determined the orbit of 70 Ophiuchi (*A.N.*, **212**, 347), Rabe that of Castor (*A.N.*, **216**, 49), and Voûte that of ξ Boötis (*A.N.*, **212**, 483).

Miscellaneous Papers.—A large number of papers dealing with general conclusions about double stars have been published. In *M.N.*, **81**, 2, Jackson and Furner have given the hypothetical parallaxes of 556 double stars, and by a determination of the speed of the star motion have checked the assumed average mass of twice that of the sun. This paper has been discussed by Seares (*Ap. J.*, **55**, 165) in dealing with the masses and densities of stars. In a discussion of the Σ stars (*M.N.*, **83**, 4) Jackson has concluded that amongst 649 systems in relative motion, 449 are certainly binary. Hypothetical parallaxes are given. Luplau-Janssen and S. Fjeltofte (*A.N.*, **214**, 423) have given 15 hypothetical parallaxes. The densities of double stars are also considered by E. Bernewitz (*A.N.*, **213**, 1) who gives data for 63 binaries. Meyermann (*A.N.*, **216**, 301 and 355) has determined the masses of 59 double stars and discussed the density of o^2 Eridani. In *A.N.* **216**, 385, he has given the masses and velocities of 48 double stars.

J. A. Miller and J. H. Pitman (*A.J.*, **34**, 127) have determined the masses of visual binaries from trigonometric and spectroscopic parallaxes, and have computed the relative radial velocities of the components in a number of systems. A determination of the radial velocity by spectroscopic observations should give a good check on the parallaxes.

Questions concerning the eccentricity of double-star orbits have been taken up by several investigators. Wilson (*A.J.*, **33**, 147) found that a mean maximum value of about 0·4 for periods of 25–100 days was largely attributable to giant stars, while with dwarf stars the eccentricity increased regularly with the period. Perrine (*A.J.*, **33**, 180) considered the relation between orbital eccentricity and the relative masses of the components. He found that a large eccentricity was associated with great difference in the magnitudes of the components of visual double stars. Russell (*Pop. Ast.*, **30**, 167) has given the preliminary results on the average eccentricity of double-star orbits from a statistical study of the angle between the radius vector and the direction of motion.

F. C. Leonard has given some particulars about exceptional spectral types in double stars. For the double faint companion of o^2 Eridani he has found (*P.A.S.P.*, **33**, 272 and 323) spectral type B9 for star B and Mdp (bright H and K lines) for star C. For Σ2245 with equal components he gives the spectral types as G5 and B9 and for the nearly equal components of γ Circini G and B5 (Harvard determination). P. Doig (*B.A.A. Journal*, **33**, 66) has dealt with the colours and spectra of the components of unequal double stars and their bearing on the theory of stellar evolution. He finds the faint star of "earlier" type in giant and of "later" type in dwarf systems. J. J.

The International Astronomical Union.

As explained in a previous note,* the International Astronomical Union, in common with certain other Unions, was founded at the Constitutive Assembly of the International Research Council which was held at Brussels in 1919. It became formally established in July 1920, when three countries had signified their adherence. Since then the Union has made rapid strides, both in regard to membership and to the organisation of international co-operative work. Its general purpose is to unite into a single group the various international astronomical organisations which existed before the war.

The importance of the Union as an aid to progress became especially manifest during last year, when the first General Assembly was held at Rome by invitation of the Reale Accademia dei Lincei. A detailed account of this meeting, including the many valuable reports presented for discussion, has been issued as vol. i. of the *Transactions of the International Astronomical Union*, under the editorship of the general secretary, Professor Fowler.

From the report of the Executive Committee, it appears that at the date of the Rome meeting the countries which had adhered to the Union were Australia, Belgium, Brazil, Canada, Czecho-Slovakia, Denmark, France, Great Britain, Greece, Holland, Italy, Japan, Mexico, Norway, Poland, Rumania, South Africa, Spain, and the United States. With the exception of Greece and South Africa, all were represented by delegates at Rome, the number of those participating being eighty-three. Many of the meetings were also attended by members of the International Geodetic and Geophysical Union, which met in Rome during the same period.

The inaugural ceremony took place in the great hall of the Campidoglio on May 2, when the two Unions were honoured by the presence of H.M. the King of Italy, H.R.H. the Crown Prince, His Eminence Cardinal Maffi, the Minister of Public Instruction, and the Mayor of Rome. Subsequent meetings were held in the beautiful rooms of the Reale Accademia dei Lincei at the Palazzo Corsini, and were presided over by M. Baillaud, President of the Union. Sectional meetings, which were open to all delegates and guests who were interested, were held separately for discussion of the reports and recommendations of the thirty-two Standing Committees, these reports being subsequently presented in their final form to the General Assembly. For details of the multitude of decisions which were reached during the Conference, reference should be made to the volume which has already been mentioned. It will suffice here to give a few examples, which may serve to indicate what are at present considered to be some of the functions of the Union.

Many of the recommendations are intended to promote international agreement on points which may be expected to facilitate the study of work emanating from different sources. Such, for example, are the recommendations referring to standard astronomical notations; the

* Council Report, 1920, *M.N.*, 80, 433.

adoption of a uniform system of standard star places by the directors of national ephemerides; the reduction of meridian observations to the equinox 1925, and the continued use of this equinox up to 1940; the adoption of the centre of gravity of the solar system as origin in investigations of the perturbations of comets; the adoption of a more completely descriptive notation for the spectral classification of stars; and the general adoption by radial velocity observers of the international system of wave-lengths. An attempt to secure uniformity in the forms of time-signals issued by radio-telegraphy was also among the recommendations.

One of the principal aims of the Union is to promote international co-operation among astronomers of different countries, so as to avoid unnecessary duplication and to enable more complete series of observations to be secured; and also to arrange for the co-ordination of the work of observers in certain fields. It was decided, for example, that central stations for the collection and discussion of various classes of solar statistics should be continued or established; at Stonyhurst for drawings of sun-spots, at Greenwich for heliographic positions and areas, at Mount Wilson for polarities and magnetic classification of spots, at Arcetri for visual observations of prominences, at Kodaikánal for photographic records of prominences, at Cambridge for calcium flocculi, and at Meudon for motions and forms combined with radial velocities in successive sections of the sun.

Adequate arrangements were also made for the continuation of the investigation of the variation of latitude, in close co-operation with the geodetic section of the Geophysical Union, a Central Bureau having been established at Mizusawa under the direction of Professor Kimura.

In connection with comets, it has been arranged that the computing section of the British Astronomical Association shall prepare a continuation of Galle's *Cometenbahnen* from 1893 to 1922, a contribution being made by the Union to meet the cost of printing. The same body will also co-operate with the Students' Observatory at Berkeley in systematic work on the prediction of comet returns and the computation of search ephemerides.

It was further decided that a central office for all matters relating to double stars should be established under the direction of Dr. Aitken, with a grant in aid for the distribution of circulars of information; important recommendations were also made in connection with the preparation of working lists and other details of observation.

The observation of variable stars calls especially for extensive co-operation among observers in different parts of the world, and the adoption of the proposals of the committee should do much to promote this branch of astronomy, while not interfering in any way with existing organisations. The committee is prepared to provide data to meet the needs of all classes of observers, and to give information as to stars requiring special attention. The Lyons Observatory has offered to undertake the determination of positions, and a grant has been made to the Cracow Observatory for the computation of yearly ephemerides of eclipsing binaries.

The reports of many of the committees will be of great value, as indicating the present state of our knowledge of the respective subjects,

and the most pressing needs for further developments. The subjects of standard wave-lengths, stellar photometry, variable stars, the spectral classification of stars, the determination of radial velocities, and the reformation of the calendar, are treated with especial fulness, and the reports will be of immediate value to many investigators.

The Central Bureau for astronomical telegrams is now under the direction of Professor Strömgren, and telegrams intended for the bureau should be addressed: "Observatory, Copenhagen."

It was considered that some of the committees had already fulfilled the purposes for which they were appointed, and that others could be combined with advantage; the number of committees has accordingly been reduced to twenty-six. The Executive Committee has taken the view that it is only necessary for the Standing Committees to include the principal workers in the respective subjects, and that co-operation with other workers should be organised in each country by the National Committees, if adequate organisations do not already exist.

The funds of the Union, it may be observed, are provided by subscriptions, on a scale depending upon population, from Governments or national academies, and are sufficient to cover expenses of administration and to allow of a limited number of grants for special purposes. The Bureau de l'Heure, established at the Paris Observatory, has a primary claim on the funds, as the scale of subscriptions was initially fixed to meet its needs.

The following were elected to form the Executive Committee of the Union for the years 1922–25:

President: Professor W. W. Campbell (U.S.A.).

Vice-Presidents: Professor Cerulli (Italy), M. Deslandres (France), Mr. Hough (Gt. Britain), Professor Shin Hirayama (Japan), Professor De Sitter (Holland).

General Secretary: Professor Fowler (Gt. Britain).

The next meeting of the Union will be held at Cambridge in 1925.

A. F.

List of Public Institutions and of Persons who have contributed to the Library, etc., since the last Anniversary.

(*Institutions exchanging publications with the R.A.S. Library are omitted from this list, in order to economise printing.*)

Bologna Observatory.
British Association.
British Astronomical Association.
Calcutta, Institution of Engineers.
Christchurch (N.Z.), Magnetic Observatory.
Christiania, Norske Meteorologiske Institut.
Cracow Observatory.
Danzig, Naturforschende Gesellschaft.
Gotha Observatory.
Hamburg Observatory.
Institute of Physics.
Kalocsa, Haynald Observatory.
Kobe, Imperial Marine Observatory.
Kyoto, University Observatory.
Liverpool, Literary and Philosophical Society.
Lourenço Marques, Observatorio Campos Rodrigues.
Madrid, Real Academia de Ciencias.
Masaryk University.
Meudon Observatory.
Poulkovo, Central Astronomical Observatory.
Prague, National Observatory.
Prague, University Observatory.
Rome, Specola Vaticana.
Royal Institution of Great Britain.
Schweizerische Naturforschende Gesellschaft.
Stepney Public Library.
Sydney Observatory.
Turin, Royal Observatory.
Washington, Smithsonian Institution.

Dr. C. E. Adams.
Dr. E. Anding.
Prof. H. G. van de S. Bakhuyzen.
Prof. E. E. Barnard.
M. Ch. Buisson.
J. C. Clancy, Esq.
L. J. Comrie, Esq.
M. E. Delporte.
M. H. Deslandres.
Herr Robert Dietzius.
G. F. Dodwell, Esq.
M. E. Esclangon.
Rev. T. E. Espin.
Prof. J. Evershed.
M. Ch. Fabry.
M. H. Fabry.
M. C. Flammarion.
Rev. Josep Jan Frič.
Mrs. Grove-Hills.
Rev. J. G. Hagen.
Prof. G. E. Hale.
Adam Hilger, Ltd.

M. R. Jarry-Desloges.
H. Spencer Jones, Esq.
Dr. Otto Klotz.
W. D. Lambert, Esq.
Sten Lothigus, Esq.
McGraw Hill Publishing Co.
Capt. R. A. Marriott.
K. Nakamurasan.
M. Fr. Nušl.
M. E. Öpik.
Prof. C. D. Perrine.
Prof. William H. Pickering.
H. H. Plaskett, Esq.
T. J. J. See, Esq.
Prof. Harlow Shapley.
M. Paul Stroobant.
Herr Ernst J. Thost.
V. Vaisala, Esq.
Issei Yamamotosan.
Signor G. Zappa.

ADDRESS

Delivered by the President, Professor A. S. Eddington, F.R.S., on presenting the Gold Medal of the Society to Professor Albert Abraham Michelson.

The Council have awarded the Gold Medal to Professor Albert Abraham Michelson for his Applications of the Interferometer to Astronomical Measurements.

Professor Michelson has had a long and distinguished scientific career. He has already received the highest honour which this country can bestow—the Copley Medal of the Royal Society. But you will not be in doubt as to why the Royal Astronomical Society chooses this moment to pay tribute to his achievements; it is because he has recently been responsible for one of the most remarkable triumphs of practical astronomy—the measurement of the angular diameter of a star.

To the naked eye the stars and planets equally appear as points of light. A telescope magnifies the planets into discs, but no telescope is large enough to render visible the disc of a star. We can calculate that a lens or mirror of 20 feet aperture would be needed to show us even the largest star disc; the construction of such an instrument, if not hopeless, is far distant. We have considerable knowledge as to the size of stars, but until this recent advance it was all found by indirect calculation; no test had made out the image to be other than that of a geometrical point. At the risk of going over familiar ground I must consider briefly the mode by which a telescope forms an image—in particular how it reproduces that detail and contrast of light and darkness which betrays that we are looking at a disc or a double star and not a blur emanating from a single point. This optical performance is called resolving power. Resolving power is not primarily a matter of magnification but of aperture; provided we use an eyepiece of reasonably high power the limit of resolution is determined by the size of aperture of the object-glass.

To create a sharply defined image the telescope must not only bring light where there ought to be light, but it must also bring darkness where there ought to be darkness. The latter task is the more difficult. Light waves in the æther tend to spread in all directions, and the telescope cannot prevent individual wavelets from straying on to parts of the picture where they have no business. But it has this one remedy —for every trespassing wavelet it must send a second wavelet by a slightly longer or shorter route to interfere with the first, and so produce darkness. This is where the utility of a wide aperture arises—by affording a wider difference in route of the individual wavelets, so that those from one part of the object-glass may be retarded relatively to and interfere with those from another part. A small object-glass can furnish light; it takes a big object-glass to furnish darkness.

Recognising that the success of an object-glass in separating double

stars and other feats of resolution depends on the production of darkness in the proper place by interference between the waves from different parts of the aperture, Michelson asked himself whether the ordinary circular aperture was necessarily the most efficient for giving the required interference. Any deviation from the circular shape is likely to spoil the definition of the image—to produce wings and fringes. The image will not so closely resemble the object viewed. But, on the other hand, we may be able to sharpen up the tell-tale features. It does not matter how different the image-pattern may be from the object, provided that we are able to read the significance of the pattern. If we cannot reproduce a disc, let us try to produce something which is *distinctive* of a disc.

A little reflection suggests that we ought to increase the resolving power by blocking out the middle of the object-glass and using only two extreme regions on one side and the other. For these regions the difference of light-path is greatest, and the corresponding wavelets are the first to interfere; they are the most efficient in furnishing the dark contrast needed to outline the image properly.

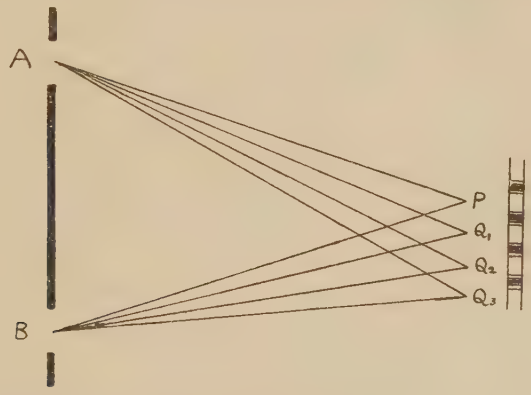
But if the middle of the object-glass is not going to be used, why trouble to construct it? We are led to the idea of using two widely separated apertures, each involving a comparatively small lens or mirror —after the pattern of a range-finder. That is much easier to construct than a huge lens circumscribing both apertures.

It is one thing to detect a small planetary or nebular disc; it is another thing to make a close measure of its diameter. It is one thing to detect the duplicity of a star; it is another thing to measure the separation of the components. Michelson's first experiments were directed not towards performing feats of resolution beyond what had previously been attained, but towards improving the accuracy of measurement. He applied his method first to measuring the diameters of Jupiter's satellites—discs which are easy to detect, but very difficult to measure reliably with an ordinary micrometer. But it is easier to understand the application of the method to measurement of double stars than to the diameters of discs; and I shall therefore speak more particularly of the double star problem first, although that is not the historical order.

Consider light coming from a distant point and passing through two small apertures, A and B, the rest of the object-glass being blocked out. From each aperture the light disturbance diverges in all directions, and our problem is to find the nature of the luminous pattern formed in the focal plane, this pattern constituting the image which is viewed and magnified by the eyepiece. At the point P (where, according to geometrical optics, the single point-image ought to appear) we have full illumination because the waves from the two apertures have equal paths, AP and BP, and reinforce one another. A little to one side we have another point of full illumination, Q_1; the paths, AQ_1 and BQ_1, are unequal, but differ by exactly a wave-length, so that the waves again arrive in the right phase to reinforce one another. Similarly we shall have a series of points of full illumination, Q_2, Q_3, etc., where the path-difference amounts to 2, 3, etc. wave-lengths. Intermediately there will

be points of darkness where the path difference is $\frac{1}{2}$, $1\frac{1}{2}$, $2\frac{1}{2}$ wave-lengths, and the waves arrive in opposite phase and cancel one another. The light-pattern viewed by the eyepiece is thus a line of alternate bright and dark fringes. At first sight it might seem that the fringes would continue of equal brightness to a great distance; but actually they soon fade away, because in the more oblique directions there is interference, not merely between the two apertures, but between different parts of the same aperture. In fact the fringes appear as fine detail in the midst of the small diffraction disc which would be formed by either of the apertures singly.

Arrange that the two apertures are movable, and widen their distance apart. The points Q_1, Q_2, Q_3 now come closer together, that is to say, the fringes contract. Decrease the distance and the fringes spread out.



It is a simple matter to find a formula giving the width of the fringes for any given separation of the apertures.

When the object viewed is a double star—two points of light—each point will produce its own line of fringes, and these will be superposed if the double star is a close one. It may happen that the two systems of fringes are in step, in which case the alternate bright and dark spaces will be conspicuous; but if they are at all out of step the pattern will be blurred. Remembering that we can alter at will the width of the fringes by varying the separation of the apertures, we can adjust them so that the bright fringes for one component coincide with the dark spaces for the other component. If the two systems are of equal brightness one system will fill the gaps in the other, leaving merely a line of uniform brightness; if the two components are of unequal brightness the same adjustment will give minimum visibility of the alternation of light and shade. Varying the distance between the apertures this critical position can be fixed with considerable precision: and for close double stars of not too unequal magnitude the separation is measured in this way much more accurately than with a micrometer.

The disc of a single star or planet is likewise measured by finding a position of the apertures for which the fringes disappear. It is not now

a problem of two points producing overlapping fringes, but every point of the circular disc produces a fringe-system, and the effects must be summed. When the diameter of the disc is 1·22 times the width of the fringes (*i.e.* the distance from one bright fringe to the next) the integrated effect is uniform illumination and the fringes disappear altogether. It is not much use trying to see in one's head a result which is more fittingly the subject of algebraic calculation; but we may notice in a general way that with this ratio the two outer quarters of the disc fall at places where the central half is producing dark spaces. That indicates roughly how the different portions of the disc compensate one another. The observation consists in varying the separation of the aperture until the fringes disappear; the diameter of the disc is then 1·2 times the fringe-width calculated for that separation.

The possibilities of a method of this kind had been explored to some extent before Michelson took up the problem; indeed Stéphan in 1874 had attempted unsuccessfully to detect the discs of stars by this means. We owe to Michelson the practical demonstration of its success. His first paper appeared more than thirty years ago in the *Philosophical Magazine* for July 1890. The next year he followed up the theory by measuring the diameters of the four satellites of Jupiter at the Lick Observatory. The method proved entirely successful, and his measures were afterwards closely confirmed by Hamy at the Paris Observatory in 1899, using the same device. The great value of the method seemed to be proved; it was thoroughly tested; and it forthwith lapsed into oblivion.

In 1919 Michelson again took up the matter with energy. He made observations in August with the 40-inch refractor at Yerkes which were found to be encouraging; and he went on to use first the 60-inch and then the 100-inch at Mount Wilson with the more ambitious design of surpassing the highest resolving power yet reached. At Mount Wilson he had the co-operation first of J. A. Anderson and afterwards of F. G. Pease. A great success was quickly obtained with the double star Capella. Capella is a spectroscopic binary with a period of 104 days. It was known that the distance of its components must nearly approach the limit of visual detection, but attempts to observe it visually had failed. (I may remark that that is a rather controversial statement to make—particularly at the R.A.S.—but the controversy is now ancient history.) With two narrow apertures in the beam from the 100-inch mirror the fringes were observed and then brought to minimum visibility by varying the position angle and separation of the apertures. The changing position angle and distance were traced through the revolution. Anderson's measures, afterwards continued by Merrill, have given a very accurate orbit. The separation of the two components varies from ″·04 to ″·05. From a comparison of this visual orbit with the spectroscopic orbit we find the parallax of Capella and also the mass. The parallax is ″·063, and the components are respectively 4·2 and 3·3 times as massive as the sun. The parallax does not differ much from that given by trigonometrical and spectroscopic determinations; but these were very rough values, whereas the interferometer parallax is presumably of the highest order of refinement. I suppose that the mass

determinations are about the best we have for any star.* But what is especially important is that Capella is the *only giant star* for which we know both the mass and the absolute luminosity.

I may perhaps be pardoned for referring to a personal interest in this first big result of Michelson's method. Capella now supplies the chief lacking constant in the radiative theory of stellar equilibrium, for which I had waited five years. It is, I think, generally conceded that the absolute magnitude of a giant star mainly depends on its mass, and theoretical formulæ can be found expressing the law of dependence. But we need to know one pair of corresponding values in order, as it were, to anchor the formulæ. Hitherto that correspondence could only be guessed roughly from statistical knowledge of the average luminosity of giant stars and an estimate of the corresponding average mass based on our general knowledge of the masses of stars (which, unfortunately, relates chiefly to dwarf stars). Having now the exact figures for Capella, we can substitute a precise determination instead of the provisional estimate; the change is by no means unimportant, the first estimate having been considerably in error.

Capella would have been slightly beyond the theoretical resolving power of the 100-inch if used in the ordinary way, though an elongation might have been detected. Resolving power was actually gained by blocking out the mischievous central portion of the aperture. But now came the final step—to produce fringes with a greater path-difference than any telescope yet constructed could provide. In 1920 a 20-foot interferometer designed by Michelson and Pease was constructed, in which the two interfering apertures could be separated to a distance of 20 feet. This was used in conjunction with the 100-inch mirror, which helped to bring the two beams together to produce their fringes. One might say that Michelson was now employing a 20-foot mirror; only, since he was only intending to use two small areas at its edges, he economically constructed those particular areas and left the rest of the mirror to imagination.

On 1920 July 10, the great 20-foot beam was placed across the telescope. On December 13 success was attained, and the diameter of Betelgeuse was measured. We can guess something of the hopes and disappointments, the difficulties and exasperations of that interval. I think we should all like to know more of the patient struggle which preceded the great achievement. So I have induced Dr. Pease, who was in charge of the instrument, to lift the veil a little. He has kindly taken great trouble to reply to my request, and what follows is extracted from his letter nearly in his own words.

"On July 10 the beam was placed on the telescope and I attempted to line it up, but no adjustment of mirrors would seem to bring success. The following morning I found that one of the images was formed by a slot between the mirror and the frame of the beam, so the next day we lined up the mirrors by sighting downward through the system to a lamp at the focus. That night we set on Vega, but could not get any fringes. The mirrors gave distorted images, and it was found difficult

* The masses of α Centauri, Sirius, and Procyon may have about the same accuracy, but I do not think that any others reach this standard.

to adjust the paths to equality by moving the inner west mirror. Michelson was present this night, and we decided to have the mirrors refigured and to introduce a wedge system and compensator near the eyepiece to adjust for path difference. The next trip was in August. On the 6th after much patient labour, using the new wedge system, I found fringes with the outer mirrors placed 18 feet apart. I left the instrument set for the assistant to observe the fringes. He saw them, shifted the wedges, and endeavoured to reset; but neither of us could find them again though we hunted for hours. These were the first fringes ever seen on any star using the Michelson periscopic system. On the 8th Michelson came up, and advised a step by step process which we find he uses with great success. So, with mirrors set 7 feet apart I soon picked up the fringes, and Michelson then saw them for the first time. The next night we placed the mirrors 12 feet apart, and this night Mr Hale saw them for the first time. On October 16, I spent half a night trying to pick up the fringes, only to find that one of the mirrors had slipped three inches. On the 17th Anderson was present, and whilst we obtained zero fringes from α Andromedæ and α Tauri we could not get those from the outer mirrors set 6 feet apart. Fringes did show on Betelgeuse using two openings 7 feet apart. We had hitherto used a very fine motion for moving the wedges, and decided to substitute a quicker motion.

"On December 12 Anderson went up the mountain with me, and his previous experience (on Capella) was of greatest assistance. Using the revised wedge motion we very easily picked up the zero fringes on Pollux, but did not get fringes from the outer pencils. On the memorable 13th we covered the end of the tube with canvas, except for the necessary openings, and placed a plank across the end of the tube for the assistant to sit upon and adjust the mirrors. We turned on Algol and endeavoured to pick up the fringes, moving the west mirror $\frac{1}{8}$ mm. at a time and running through with the wedge, but could find nothing. I was working with a 0·34 inch eyepiece and it fell out on the floor; I put in a 1-inch eyepiece—and the fringes were there. (Simple rule for finding fringes—just drop an eyepiece on the floor and substitute another.) The mirrors were then 81 inches apart. With the telescope still on Algol we shifted the mirrors to 121 inches and obtained the zero fringes immediately and the outer fringes shortly. Observations on γ Orionis showed both sets of fringes with but slight adjustment of the wedge and compensator. Turning to Betelgeuse, the beautiful orange and black zero fringes were present; but no trace of the outer fringes showed though we looked again and again."

Midnight had now arrived, and the observers adjourned to the "galley" for cocoa and toast with their heads full of sums—for the lack of fringes to Betelgeuse was presumably the real thing. Its fringes had totally disappeared at 10 feet separation, although the other stars showed them. "Presently Anderson speaks up and says, 'It's a whale of a thing—let's see, as big as the orbit of Mars.' Returning to the telescope, we set on Procyon and find the fringes immediately; the slightest motion of the wedge bringing them in. It was quite certain therefore that the instrument was in adjustment, and that the fringes on Betelgeuse were

not present at 10 feet separation." The deduced diameter was 0″·045 —about the same size as a halfpenny fifty miles away.

Michelson's visits to Mount Wilson were limited to the summer months, and he was not present when this result was obtained. When he returned in 1921 he found his collaborators much occupied in trying to find some plan of obtaining definite measures of the visibility of the fringes instead of vague judgments. He suggested the plan of using two apertures, one fixed and the other variable; a difference in the size of the apertures reduces the visibility to a definite extent, depending on the ratio of the two apertures. Finally, a comparison apparatus was constructed with one square aperture of 4 inches and the other a square variable from 4 inches to zero, in order to afford a definite scale of visibility.

In the early trials it took days to find the fringes, but as gradual improvements were made a few hours' work on the first night of a series of observations sufficed; the subsequent settings being made in a few moments.

I need only touch very briefly on later developments. Diameters of Antares, Aldebaran, Arcturus, and β Pegasi have since been measured. But, of course, the discs of most stars are far below the limit for a 20-foot instrument. Professor Hale is now constructing a 50-foot interferometer, and it is estimated that thirty or forty stars will be within its grasp. There is no need for a large mirror, and the use of the 100-inch in conjunction with the first interferometer was rather a luxury. The 50-foot is of different design, and will not depend on any other telescope. All the diameters of stars measured up to the present confirm very closely the theoretical values that had been predicted for them. The enormous actual size of these stars—the earth's orbit could be placed entirely inside Betelgeuse—is a picturesque feature of the results; but that was a confirmation of facts already established almost beyond doubt.

It is not unlikely that interesting bypaths may be opened up. Considerable fluctuations in the diameter of Betelgeuse have been found, which may or may not be due to varying definition. The star is an irregular variable showing also changes of line-of-sight velocity, and the correlation of varying diameter with these other fluctuations would be of great physical interest if it turns out to be genuine. Michelson has pointed out that it is theoretically possible to determine by the interferometer the distribution of light over the disc—the law of darkening at the limb. That is a conceivable development for the future.

Why was this method successfully tested in 1891 not applied to stellar diameters until nearly thirty years later? The trial on a large scale involved difficult and expensive instrumental construction, but I think it would scarcely have been delayed so long if there had been more positive assurance of success. But, like most great advances, it was an adventure; the adversary whose strength was perhaps over estimated was atmospheric disturbance. We know how seldom the atmospheric conditions permit us to take full advantage of a modest 24- or 28-inch aperture. No doubt it was feared that an attempt to use much greater path-differences would be doomed to failure from this cause. It may be of interest to explain why the unsteadiness of the air did not prevent success. Taking first a single star, we deal with two

beams pursuing parallel courses 20 feet apart. The more local variations of the density of the air will be different for the two beams; and when a rarefied region crosses one beam but not the other, altering the equivalent length of path, the fringes will jump right or left. Now consider the two components of Capella, and trace the beams backwards from one of the apertures towards the two stars. They are inclined at $\frac{1}{20}$ of a second of arc, so that by the time they reach Capella they have diverged by 126 million kilometres. But in the atmosphere they are still almost coincident; even at a height of fifty miles their centres are only 2 cm. apart. Whatever happens in the path of one beam, happens also to the other; and when the fringes of one component jump, the fringes of the other take the same jump. Thus the mutual obliteration or reinforcement of the fringes will not be disturbed, provided only that the image is steady enough to be held for an instant. We do not mind if the fringes dance, provided that they both dance to the same tune.

The use of a wire grating over the object-glass of a telescope is a feature in many modern investigations on the photometry and colorimetry of stars. Let us trace back the history of this valuable adjunct to the telescope. The coarse grating, consisting of alternate strips and slits, was first constructed and used by Schwarzschild for the measurement of double stars in 1895. He states that the idea came to him from reading Michelsón's paper in the *Philosophical Magazine* already referred to. Schwarzschild's original grating contained only eight slits, and was a natural development of Michelson's apparatus with two slits. If Michelson is not exactly the father of the modern coarse grating, I think we may fairly regard him as its grandfather.

Our Medal has been awarded for a particular part of Professor Michelson's work, and I do not propose to survey his varied contributions to science which lie outside the terms of the award. But I cannot refrain from mentioning the Michelson-Morley experiment. Indeed it would not require much sophistry to bring this experiment under "the applications of the interferometer to astronomical measurements." It was certainly an application of the interferometer; and its purpose was the measurement of the true velocity of the earth in space, which—one might innocently suppose—was an astronomical measurement. Unfortunately modern ideas condemn this "true velocity of the earth" to the domain of metaphysics; but in the eighties it could scarcely have been regarded otherwise than as an astronomical motion. Had the result of the determination not been null, it would have been regarded as an astronomical measurement of the highest importance. We should have been able to settle the question whether the stellar system is the stationary central system, the spiral nebulæ being subordinate to it; or whether the mean of the spiral nebulæ is at rest, the stellar system being an undistinguished unit in a larger system. However, it is no use speculating on what might have been. But the experiment has other astronomical associations; it is one of a series of researches on the problem of aberration in which Bradley's discovery of aberration and Airy's water-telescope experiment are conspicuous landmarks. And, finally, it is the gateway which leads to the great developments associated with the name of Einstein.

Michelson first tried the experiment in 1881. The results were not satisfactory. The arrangements were scarcely sensitive enough and there was a difficulty in avoiding straining the apparatus when it was rotated for the purpose of comparing the velocity of light in different directions. An error in theory was pointed out by Lorentz, but this was easily corrected. In 1887 Michelson, working in conjunction with E. W. Morley, greatly improved the apparatus. The difficulty in rotating it without strain was surmounted by erecting it on a stone pillar floating in mercury; this was kept rotating slowly but continuously, and the fringes could be observed without stopping the rotation.

The experiment was intended to determine whether of two rays whose tracks were at right angles one was delayed more than the other by the true velocity of the earth (*i.e.* its velocity relative to the æther) being in that direction. The delay would show itself by a displacement of interference fringes. The expected delay was very small, but that did not trouble Professor Michelson, who had provided for detecting a delay of less than a ten-thousand-billionth of a second of time.

Can we not imagine his feelings as he took the crucial observations? For a century astronomers and physicists had talked glibly of this true velocity of the earth without ever being able to run it down. It had shown a curious elusiveness. For instance, it must produce quite a large aberration-displacement of a fixed star; but we cannot measure that displacement because, very provokingly, we do not happen to know where the star ought to be if it was not displaced. The water-telescope should have greatly altered the displacement; no doubt it did; but it just wiped out the change by introducing an alteration of refraction as the star's ray passed into the moving water. Now at last it was to be fairly trapped. The dodges by which it had eluded first order experiments could scarcely succeed against Michelson's ten-thousandfold increased delicacy designed to capture the squared terms. How bitter must have been the disappointment when it was found that the result of the experiment was again null and the true velocity of the earth had hidden itself once more. It was not until some years later that it was realised how it had hidden itself; it had gone to the length of tampering with the dimensions of the stone pillar—falsifying the measurements so as to conceal the discrepancy in the balance sheet. No wonder the innocent scientist was deceived. Lest you should be shocked at such depravity on the part of an astronomical entity, I hasten to add that summary punishment has been inflicted; it has been expelled from astronomy, and handed over to metaphysicists. But to return to Michelson's disappointment—for I have heard from his own lips that it was a very great disappointment—the legitimate hope of fixing one more important astronomical constant was destroyed. "The great tragedy of science—the slaying of a beautiful hypothesis by an ugly fact" had occurred. But would that all experimental disappointments might have so happy an ending! Michelson and Morley did not attain the humble success they were seeking; to-day their experiment is known throughout the world as the gateway to one of the greatest developments of physical science.

In 1880 Michelson made a standard determination of the velocity

of light, which was a very great advance on the previous determination by Foucault. Forty years later he is attacking the problem again, hoping for still greater accuracy—more particularly with regard to the influence of the earth's rotation on the velocity. In 1881 he began his experiments on the relative velocity of earth and æther; forty years later, dissatisfied because those experiments were made in a cellar, he is repeating the work on the mountain-top to give the æther a fairer chance of slipping by. I have already mentioned the thirty years interval from his first success in astronomical measurement with the interferometer to his new achievements. I think that if we were to subject Professor Michelson to harmonic analysis, his periodogram would show a strong peak between thirty and forty years. The phase of maximum intensity is now in that department of his varied activities which most nearly concerns astronomy. It is with the deepest interest that we watch this wave rising to its crest; and we wish him health and good fortune in the enterprises he has so skilfully planned.

In the absence of the American Ambassador, who is unwell, Mr. Post Wheeler, Counsellor of the Embassy, has kindly attended to receive the Medal on behalf of Professor Michelson.

Election of Officers and Council.

The Meeting then proceeded to the election of Officers and Council for the ensuing year, when the following Fellows were elected:—

President.

J. L. E. Dreyer, Esq., M.A., Ph.D., D.Sc.

Vice-Presidents.

Sir F. W. Dyson, M.A., LL.D., F.R.S., Astronomer Royal.
A. S. Eddington, Esq., M.A., M.Sc., F.R.S., Plumian Professor of Astronomy, Cambridge.
E. B. Knobel, Esq.
H. F. Newall, Esq., M.A., D.Sc., F.R.S., Professor of Astrophysics, Cambridge.

Treasurer.

Lieut.-Col. F. J. M. Stratton, M.A., D.S.O.

Secretaries.

H. Spencer Jones, Esq., M.A., B.Sc.
Rev. T. E. R. Phillips, M.A.

Foreign Secretary.

H. H. Turner, Esq., D.Sc., D.C.L., F.R.S., Savilian Professor of Astronomy, Oxford.

Council.

A. E. Conrady, Esq., Professor of Optical Design, Imperial College of Science and Technology, South Kensington.
A. C. D. Crommelin, Esq., B.A., D.Sc.
C. R. Davidson, Esq.
Alfred Fowler, Esq., F.R.S., Professor of Astrophysics, Imperial College of Science and Technology, South Kensington.
J. W. L. Glaisher, Esq., Sc.D., F.R.S.
P. H. Hepburn, Esq., LL.B.
John Jackson, Esq., M.A., B.Sc.
Harold Jeffreys, Esq., M.A., D.Sc.
F. A. Lindemann, Esq., M.A., Ph.D., F.R.S., Professor of Experimental Philosophy, Oxford.
E. A. Milne, Esq., M.A.
J. W. Nicholson, Esq., M.A., D.Sc., F.R.S.
John H. Reynolds, Esq.